5G RAN

IP NR Engineering Guide Feature Parameter Description

Issue Draft A

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Contents

1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes

  Changes in functions and their corresponding parameters
- Editorial changes

  Improvements or revisions to the documentation

1.1 5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 02 (2025-06-28).

Technical Changes

Change Description	Parameter Change	RAT	Base Station Model
Added support for the base station to transmit its backup power saving data to the microwave device. For details, see: • **6.3.3.1.1 Data Preparation (in the New Model)** • **6.3.3.1.2 Data Preparation (in the Old Model)** • **6.3.3.1.3 Using MML Commands (in the New Model)** • **6.3.3.1.4 Using MML Commands (in the Old Model)**	Modified parameter: Added the **BKP_PSAV_EVALUATE_DATA_SW** option to the **LLDPGLOBAL**.***BTSMWCOSW*** parameter.	Low-frequency TDD High-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added support for X2 interfaces in SA networking. For details, see: • **4.2.1.4 Networking** • **4.3.2 Deployment of Service Interfaces**	None	Low-frequency TDD High-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Editorial Changes

None

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FBFD-010016	Transmission Networking	4 Transmission Networking
FBFD-010022	Active/Standby IP Routes	5.2 IP Route Backup
FOFD-010060	Transmission Network Detection and Reliability Improvement	5 Transmission Reliability
		6 Transmission Maintenance and Detection
FOFD-021212	IPv6	4 Transmission Networking
FBFD-021101	IPv4/IPv6 Dual Stack	4 Transmission Networking

2.3 Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Transmission Networking	None	4 Transmission Networking
IP Route Backup	None	5.2 IP Route Backup
Transmission Reliability	None	5 Transmission Reliability
Transmission Maintenance and Detection	None	6 Transmission Maintenance and Detection

2.4 Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Transmission Networking	None	4 Transmission Networking

Function Name	Difference	Chapter/Section
IP Route Backup	None	**5.2 IP Route Backup**
Transmission Reliability	None	**5 Transmission Reliability**
Transmission Maintenance and Detection	None	**6 Transmission Maintenance and Detection**

3 Principles

This document describes only the engineering guidelines for transmission networking, transmission reliability, and transmission maintenance and detection. For details about the principles, see *IPv4 Transmission* and *IPv6 Transmission*.

4 Transmission Networking

4.1 Network Analysis

4.1.1 Deployment of Common Transmission Data

4.1.1.1 Benefits

None

4.1.1.2 Impacts

The impacts between this function and other functions are described in **4.2.1.2.3 Function Impacts**.

The basic IPv6 header is 40 bytes long, which is 20 bytes longer than the basic IPv4 header. This increases the transmission bandwidth overhead. Therefore, the transmission efficiency of IPv6 is slightly lower than that of IPv4. For example, the length of an IPv4 packet is 800 bytes and the length of an IPv6 packet is 820 bytes. The IPv6 transmission efficiency is 2.5% lower than the IPv4 transmission efficiency.

4.1.2 Deployment of Service Interfaces

4.1.2.1 Benefits

None

4.1.2.2 Impacts

The impacts between this function and other functions are described in **4.2.4.2.3 Function Impacts**.

The basic IPv6 header is 40 bytes long, which is 20 bytes longer than the basic IPv4 header. This increases the transmission bandwidth overhead. Therefore, the transmission efficiency of IPv6 is slightly lower than that of IPv4. For example, the length of an IPv4 packet is 800 bytes, and the length of an IPv6 packet is 820 bytes. The IPv6 transmission efficiency is 2.5% lower than the IPv4 transmission efficiency.

4.1.3 Deployment of OM Channels

4.1.3.1 Benefits

None

4.1.3.2 Impacts

The impacts between this function and other functions are described in **4.2.5.2.3 Function Impacts**.

This function has no impact on the network.

4.2 Requirements

4.2.1 Deployment of Common IPv4 Transmission Data

4.2.1.1 Licenses

None

4.2.1.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.2.1.2.1 Prerequisite Functions

None

4.2.1.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	OMCHs binding to routes	None	*IP NR Engineering Guide*	An OMCH cannot be bound to a source-based IPv4 route.

4.2.1.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	IP-Based Multi-mode Co-Transmission on BS side(NR)	None	*Common Transmission*	None

4.2.1.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards support this function. To learn which main control boards are NR-capable, see technical specifications of the BBU in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

This function does not depend on RF modules.

4.2.1.4 Networking

Operators are advised to plan a networking mode for base stations based on their requirements and network conditions. The requirements for IPv4 transmission networking are as follows:

On the base station side

The base station side has different service interfaces, including the S1, X2, NG, Xn, and interfaces of the IP clock link and OM channel. These service interfaces can have identical or different IP addresses. Make an overall plan on base station IP addresses according to the service IP address usage.

- If OM channels and service interfaces require different IP addresses for isolation, plan different IP addresses.
- If OM channels and service interfaces do not need to be isolated, plan identical IP addresses.
- If NR supports both NSA and SA or if SA X2 interfaces need to be configured:
  - It is recommended that X2 and Xn interfaces use different IP addresses.
  - If the X2 and Xn interfaces need to share an IP address, the X2-C and Xn-C interfaces must have different SCTP port numbers, and the X2-U and Xn-U interfaces must share the same **EPGROUP**.
- For multimode base stations including LNR co-MPT base stations:

  LTE and NR must be configured with different IP addresses. That is, the IP addresses of LTE X2/S1 and NR X2/Xn/NG must be different to avoid coupling between RATs. This is because the X2/Xn self-setup message does not include the SCTP port number. If LTE and NR share an IP address, the interface to which the SCTP link belongs cannot be identified.

For IP clock links, if the IP clock link configured in an **IPCLKLINK** MO is compliant with

- Huawei proprietary protocol

  The value of **Port Type** in the **INTERFACE** MO that is referenced by the **DEVIP** (old model)/**IPADDR4** (new model) MO must be **ETH** or **ETHTRK**.
- 1588 protocol

  The value of **Port Type** in the **INTERFACE** MO that is referenced by the **DEVIP** (old model)/**IPADDR4** (new model) MO must be **ETH**, **ETHTRK**, or **LOOPINT**.

The base station uses route forwarding in the panel cascading scenario. An interface IP address must be configured on the cascading interface, which serves as the next-hop IP address of the next-hop leaf node.

Considering transmission reliability, plan multiple physical links or IP addresses and routes for the base station during network planning. Before using a specific transmission maintenance and detection feature, ensure that devices on the transmission network support this feature.

In addition to routes of the core network, MAE, and base station, consider the following requirements during IP route planning:

- If an IP clock server is used, a corresponding route must be planned.
- If a security gateway (SeGW), public key infrastructure (PKI), File Transfer Protocol (FTP) server, or NTP server is used, a corresponding route must be

planned. If multiple devices are on the same network segment, the route to these devices can be configured as a network segment route. It is unnecessary to independently configure a route for each device.

- If an IPsec tunnel is enabled, the interface IP address is used as the external IP address of the IPsec tunnel and the loopback IP address is used as the internal IP address of the IPsec tunnel. The external IP address and internal IP address are encapsulated in the outer IP header and inner IP header of the IPsec tunnel, respectively. For details, see *IPsec*.
- Whether to use source-based IP routing or destination-based IP routing must be planned. Destination-based IP routing is recommended.
  - For details about the application scenarios of source-based IP routing, see *IPv4 Transmission*.
  - The base station supports a maximum of 32 source-based IP routes. If a base station requires more than 32 IP routes, it is recommended that it be configured with only destination-based IP routes.
  - Source-based IP routing and destination-based IP routing are mutually exclusive. Therefore, the control plane and user plane of a base station must use the same routing policy and the active and standby routes must also use the same routing policy.

4.2.1.5 Others

None

4.2.2 Deployment of Common IPv6 Transmission Data

4.2.2.1 Licenses

None

4.2.2.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.2.2.2.1 Prerequisite Functions

None

4.2.2.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	OMCHs binding to routes	None	*IP NR Engineering Guide*	An OMCH cannot be bound to a source-based IPv6 route.

4.2.2.2.3 Function Impacts

The impacts between this function and other functions are described in **4.2.1.2.3 Function Impacts**.

4.2.2.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards support this function. To learn which main control boards are NR-capable, see technical specifications of the BBU in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

This function does not depend on RF modules.

4.2.2.4 Networking

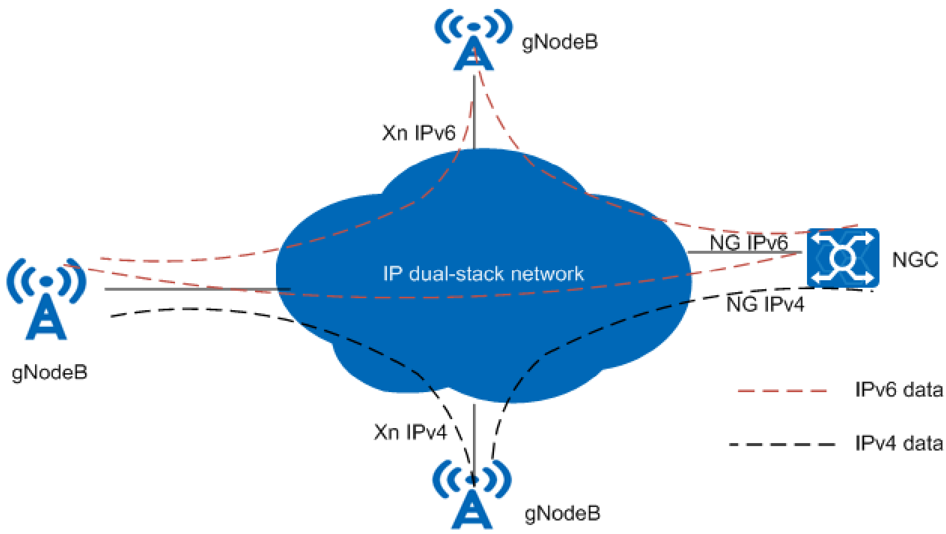
Operators are advised to plan a networking mode for base stations based on their requirements and network conditions. The requirements for IPv6 transmission networking are as follows:

- On the base station side
  - The base station side has different service interfaces, including the S1, X2, NG, Xn, and interfaces of the IP clock link and OM channel. All these service interfaces support IPv6 transmission and have the same

requirements and IP address networking constraints in co-MPT scenarios as those in IPv4 transmission. For details, see **4.2.1.4 Networking**. It is recommended that base station IP addresses be planned according to whether logical IP addresses and interface IP addresses require VLAN isolation. Independent VLAN ID and IP address are recommended for the OM channel to isolate the OM channel from other service VLANs.

- For IPv6 transmission, information about the IPv6 address and IPv6 route of the peer equipment must be collected. When planning IPv6 routes, you need to plan routes for the core network, MAE, FTP server, NTP server, IP clock server, and peer base station (eNodeB and gNodeB). In transmission security scenarios, you also need to plan routes to the SeGW and PKI equipment. If multiple pieces of equipment are on the same network segment, a network segment route can be configured.
- In IPv6 transmission, the main control board of the local BBU can provide a port for connecting to the IPv6 transmission network, and co-transmission through panel interconnection and cascading networking are supported. In addition, IPv6 co-transmission through backplane interconnection is supported.
- The base station uses route forwarding in the panel cascading scenario. An interface IP address must be configured on the cascading interface, which serves as the next-hop IP address of the next-hop leaf node.
- In IPv6 transmission, the MTU of the backhaul network must be collected. It is recommended that the interface MTU of the base station be set to the minimum MTU on all backhaul networks of the interface. Ensure that all the routers support the sending of ICMPv6 Packet Too Big messages and the network firewall allows the packets to pass through.
- If both IPv4 and IPv6 base stations are deployed on the network, Xn-based dual-stack transmission must be configured for the base stations in the border areas of IPv6 transmission and IPv4 transmission. In addition, the bearer network must also support dual-stack transmission. **Figure 4-1** shows the Xn-based dual-stack transmission networking.

Figure 4-1 Xn-based dual-stack transmission



- IPv6 routing supports destination-based routes, source-based routes, and active/standby routes. It also supports Bidirectional Forwarding Detection (BFD)-based active/standby routes.
  - Source-based IPv6 routes can be configured only on the same main control board.
  - If the Xn interface uses Layer 2 networking and does not support source-based routing on the base station but the NG interface uses Layer 3 networking and requires source-based routing on the base station, two IP addresses must be configured for the Xn and NG interfaces, respectively, on the base station.
  - Whether to use source- or destination-based IP routing must be planned on the base station. Destination-based routing is recommended.
    - For details about the application scenarios of source-based IP routing, see *IPv6 Transmission*.
    - The base station supports a maximum of 32 source-based IP routes. If a base station requires more than 32 IP routes, it is recommended that it be configured with only destination-based IP routes.
    - Source-based IP routing and destination-based IP routing are mutually exclusive. Therefore, the control plane and user plane of a base station must use the same routing policy and the active and standby routes must also use the same routing policy. Source-based routes take precedence over destination-based routes.
- On the transmission network side
  - IPv6 transmission has been deployed on the transmission bearer network, and IPv6 routes have been configured between the base station and the peer equipment.
  - It is recommended that the transmission link MTU size of the OM channel be set to 1500 bytes. The default MTU of the MAE is 1500 bytes.
  - Plan a unified MTU size for other service interfaces based on the MTU capabilities of equipment (including base stations, routers, and peer equipment) on end-to-end transmission links. The planned MTU size must be the minimum MTU size supported by the equipment.
  - If the base station uses L2 or L2+L3 networking, the L2 switch must support all multicast packets or support the MLD-Snooping function (allowing the multicast packets of the neighbor discovery protocol to pass).
  - In secure networking, the MTU size of the router between the SeGW and the base station must be the planned size plus the length of the IPsec header.
  - The IPv6 extension headers are optional and used for specific functions. The base station can only receive or send the following IPv6 extension headers. IPv6 packets sent by the peer equipment (router, core network, base station, or clock server) should not carry other extension headers. Otherwise, the IPv6 packets will be discarded by the base station.

  - Fragment header: used for packet fragmentation
  - AH and ESP: supporting encryption
  - HOP by HOP: This field can be carried only in MLD snooping discovery protocol packets, but not other protocol packets.
- On the peer equipment side of the service interface (core network and peer base station)

  Before deploying dual-stack transmission on the peer equipment, ensure that the gNodeB connected to the peer equipment runs SRAN15.1 or a later version, including IPv4-only gNodeBs.

4.2.2.5 Others

None

4.2.3 Deployment of Common IPv4/IPv6 Dual-Stack Transmission Data

4.2.3.1 Licenses

None

4.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.2.3.2.1 Prerequisite Functions

None

4.2.3.2.2 Mutually Exclusive Functions

None

4.2.3.2.3 Function Impacts

The impacts between this function and other functions are described in **4.2.1.2.3 Function Impacts**.

4.2.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards support this function. To learn which main control boards are NR-capable, see technical specifications of the BBU in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

This function does not depend on RF modules.

4.2.3.4 Networking

Operators are advised to plan a networking mode for base stations based on their requirements and network conditions. The requirements for IPv4/IPv6 dual-stack transmission networking are as follows:

- VLAN planning

  It is recommended that the VLAN IDs for IPv4 transmission and IPv6 transmission be different. The VLAN IDs for both IPv4 and IPv6 transmission must be configured in interface VLAN configuration mode. IPv4 transmission cannot use the single VLAN or VLAN group configuration mode.
- IPv6 address and IPv6 route

  IP addresses and routes required for IPv4 and IPv6 transmission need to be planned. For details, see **4.2.1.4 Networking** and **4.2.2.4 Networking**.
- Transmission bearer network
  - IPv4/IPv6 dual-stack transmission must be deployed on the transmission bearer network, and IPv4/IPv6 dual-stack routes must be configured between the base station and the peer equipment.
  - For other requirements of IPv6 transmission on the bearer network, see **4.2.3.4 Networking**.
- Transmission configuration model

  IPv6 transmission or IPv4/IPv6 dual-stack transmission on a gNodeB supports only the new configuration model. If IPv6 configuration is added to IPv4 configuration in the old model on the live network to form dual-stack transmission, the IPv4 configuration must be first converted into the new model, then the single VLAN or VLAN group configuration mode of IPv4 must be converted to the interface VLAN configuration mode, and at last the IPv6 configuration in the new model is added.

4.2.3.5 Others

None

4.2.4 Deployment of Service Interfaces

4.2.4.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-021212	IPv6	NR0S000PV600	per gNodeB

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.2.4.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.2.4.2.1 Prerequisite Functions

None

4.2.4.2.2 Mutually Exclusive Functions

None

4.2.4.2.3 Function Impacts

None

4.2.4.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards support this function. To learn which main control boards are NR-capable, see technical specifications of the BBU in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

This function does not depend on RF modules.

4.2.4.4 Networking

None

4.2.4.5 Others

None

4.2.5 Deployment of OM Channels

4.2.5.1 Licenses

None

4.2.5.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.2.5.2.1 Prerequisite Functions

None

4.2.5.2.2 Mutually Exclusive Functions

None

4.2.5.2.3 Function Impacts

None

4.2.5.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

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Boards

All NR-capable main control boards support this function. To learn which main control boards are NR-capable, see technical specifications of the BBU in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

This function does not depend on RF modules.

4.2.5.4 Networking

When the OM channel uses IPv4 transmission:

- The transmission bearer network must support IPv4 and IPv4 routes must be planned between the base station and the MAE.
- IPv4 addresses need to be planned on the MAE side for IPv4-based communication between the base station and the MAE. These IPv4 addresses include the IPv4 address of the OM channel, and IPv4 addresses of the upper-layer application services such as the FTP server and NTP server.

When the OM channel uses IPv6 transmission:

- The transmission bearer network must support IPv6 and IPv6 routes must be planned between the base station and the MAE.
- IPv6 addresses need to be planned on the MAE side for IPv6-based communication between the base station and the MAE. These IPv6 addresses include the IPv6 address of the OM channel, and IPv6 addresses of the upper-layer application services such as the FTP server and NTP server.

When some OM channels of base stations managed by the MAE use IPv4 transmission and the other OM channels use IPv6 transmission:

Both IPv4 and IPv6 addresses must be planned on the MAE to manage both base stations using IPv4 transmission and base stations using IPv6 transmission.

4.2.5.5 Others

None

4.3 Operation and Maintenance

4.3.1 Deployment of Common Transmission Data

4.3.1.1 IPv4 Data Configuration (in the New Model)

This section describes how to configure IPv4 data on the base station side when the new transmission configuration model is used (**GTRANSPARA**.***TRANSCFGMODE*** is set to **NEW**).

4.3.1.1.1 Data Preparation

Configuring an Ethernet Port

The following table describes the key parameters that must be set in an **ETHPORT** MO to configure an Ethernet port.

Parameter Name	Parameter ID	Setting Notes
Port ID	**ETHPORT**.***PORTID***	Set this parameter based on the network plan.
Subboard Type	**ETHPORT**.***SBT***	Set this parameter based on the network plan.
Port Attribute	**ETHPORT**.***PA***	The port attribute must be the same as that of the peer end. When a UMPTe/UMPTg board is used, automatic port attribute detection is not supported. You need to configure the port attribute according to actual situations. If an optical module with a rate of 50 Gbit/s or higher is used at the local device, automatic detection is not supported. Automatic detection is enabled when the **Port Attribute**, **Speed**, and **Duplex** parameters are set to **AUTO(Automatic Detection)**, **AUTO(Auto Negotiation)**, and **AUTO(Auto Negotiation)**, respectively. When the system detects these parameter settings, it reports ALM-26247 Configuration Failure.
Speed	**ETHPORT**.***SPEED***	Set this parameter to the same value as that of the peer port.
Duplex	**ETHPORT**.***DUPLEX***	Set this parameter based on the values of **ETHPORT**.***PA*** and **ETHPORT**.***SPEED***. For details about the setting suggestions, see **Table 4-1**.
FEC Mode	**ETHPORT**.***FECMODE***	This parameter specifies the FEC mode used when the Ethernet port rate is 25 Gbit/s. It must be set to the same value for the local and peer ports. Set this parameter based on the network plan.

Table 4-1 Suggestions for parameter settings

Port Attribute	Speed	Duplex
COPPER(Copper)	10M(10M) or 100M(100M)	This parameter can only be set to **FULL(Full Duplex)**.
COPPER(Copper)	If 1000 Mbit/s is required, this parameter must be set to **AUTO(Automatic Negotiation)**.	This parameter can only be set to **AUTO(Automatic Negotiation)**.
FIBER(Fiber)	100M(100M)	This parameter can only be set to **FULL(Full Duplex)**.
FIBER(Fiber)	If 1000 Mbit/s is required, set this parameter to **1000M(1000M)** or **AUTO(Automatic Negotiation)**.	● If **Speed** is set to **1000M(1000M)**, set this parameter to **FULL(Full Duplex)**. ● If **Speed** is set to **AUTO(Automatic Negotiation)**, set this parameter to **AUTO(Automatic Negotiation)**.
FIBER(Fiber)	10G/25G	This parameter can only be set to **FULL(Full Duplex)**.
AUTO(Automatic Detection)	This parameter can only be set to **AUTO(Automatic Negotiation)**.	This parameter can only be set to **AUTO(Automatic Negotiation)**.

Configuring a VLAN Priority Mapping

The following table describes the key parameters that must be set in a **DSCP2PCPMAP** MO and a **DSCP2PCPREF** MO to configure a mapping between traffic types and VLAN priorities.

Parameter Name	Parameter ID	Subparameter	Setting Notes
DSCP-to-PCP Mapping ID	**DSCP2PCPMAP**.***DSCP2PCPMAPID***	-	This parameter specifies the ID of the mapping table between DSCPs and VLAN priorities.
Default PCP	**DSCP2PCPMAP**.***DEFAULTPCP***	-	This parameter specifies the default number of the queue used for non-mapped DSCPs.

Parameter Name	Parameter ID	Subparameter	Setting Notes
Differentiated Services Code Point	***DSCP2PCPREF.DSCP***	-	This subparameter specifies the DSCP. A larger value indicates a higher priority.
Priority Code Point	***DSCP2PCPREF.PCP***	-	This subparameter specifies the VLAN priority. A larger value indicates a higher priority.

Setting a VLAN Based on the Interface

The following table describes the key parameters that must be set in an **INTERFACE** MO to configure VLAN interfaces and associated VLANs.

Parameter Name	Parameter ID	Setting Notes
Interface ID	**INTERFACE.*ITFID***	Set this parameter based on the network plan.
Interface Type	**INTERFACE.*ITFTYPE***	For a VLAN interface, set this parameter to **VLAN**.
Port Type	**INTERFACE.*PT***	Set this parameter based on the network plan.
Port ID	**INTERFACE.*PORTID***	Set this parameter based on the network plan.
VLAN ID	**INTERFACE.*VLANID***	Set this parameter based on the network plan.
DSCP-to-PCP Mapping ID	**INTERFACE.*DSCP2PCPMAPID***	Set this parameter based on the network plan.
VRF Index	**INTERFACE.*VRFIDX***	Set this parameter based on the network plan.
IPv4 Maximum Transmission Unit	**INTERFACE.*MTU4***	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
ARP Proxy	**INTERFACE**.***ARPPROXY***	Set this parameter based on the network plan.

Setting a VLAN Based on the VLAN Mapping

NOTE

Setting a VLAN based on the interface is mutually exclusive with setting a VLAN based on the VLAN mapping (only one method can be used at any one time).

The following table describes the key parameters that must be set in a **VLANMAP** MO to configure a VLAN mapping.

Parameter Name	Parameter ID	Setting Notes
VRF Index	**VLANMAP**.***VRFIDX***	Set this parameter based on the network plan.
Next Hop IP	**VLANMAP**.***NEXTHOPIP***	This IP address must be on the same network segment as the IP address of the gateway connected to the physical port of the base station as well as the interface IP address of the base station.
Mask	**VLANMAP**.***MASK***	Set this parameter based on the network plan. A 31-bit mask can be configured for the IP address of an Ethernet port. During transmission network planning, only the directly connected 31-bit network segment can be set to all 0s or all 1s.
VLAN Mode	**VLANMAP**.***VLANMODE***	● If the operator plans a one-to-one mapping between base stations and VLANs, set this parameter to **SINGLEVLAN**. ● If the operator plans a one-to-multiple mapping between base stations and VLANs based on traffic types, set this parameter to **VLANGROUP**.

Parameter Name	Parameter ID	Setting Notes
VLAN Group No.	**VLANMAP**.***VLANGROUPNO***	This parameter is valid only when **VLANMAP**.***VLANMODE*** is set to **VLANGROUP**. It is recommended that VLAN groups be numbered from 0. Generally, each base station has only one VLAN group number. ● Set this parameter if the value of the **VLANMAP**.***VLANMODE*** parameter is changed from **SINGLEVLAN** to **VLANGROUP** during modification of the **VLANMAP** MO. ● Ensure that the value of this parameter is the same as the VLAN group number in the **VLANCLASS** MO.
VLAN ID	**VLANMAP**.***VLANID***	This parameter is valid only when **VLANMAP**.***VLANMODE*** is set to **SINGLEVLAN**.
Set VLAN Priority	**VLANMAP**.***SETPRIO***	This parameter is valid only when **VLANMAP**.***VLANMODE*** is set to **SINGLEVLAN** and one VLAN has only one priority. [a]
a: ● To use the default mapping between user data types and VLAN priorities, set this parameter to **DISABLE**. ● To specify the mapping between user data types and VLAN priorities, set this parameter to **ENABLE**. ● The default mapping is recommended. That is, set **VLANMAP**.***SETPRIO*** to **DISABLE**. To specify the mapping between DSCPs and VLAN priorities when the single VLAN mode is used, configure parameters in the **DSCPMAP** MO.		

Configuring a VLAN Priority Mapping

If the VLAN group mode is used, it is recommended that both the **VLANCLASS** and **VLANMAP** MOs be configured to set the VLAN group mapping. The following table describes the key parameters that must be set in a **VLANCLASS** MO.

Parameter Name	Parameter ID	Setting Notes
VLAN Group No.	**VLANCLASS**.***VLANGROUPNO***	Set this parameter based on the network plan.
Traffic Type	**VLANCLASS**.***TRAFFIC***	There are the following traffic types: **USERDATA**, **SIG**, **OM_HIGH**, **OM_LOW**, and **OTHER**. A VLAN ID must be specified for each traffic type to prevent service exceptions.

Parameter Name	Parameter ID	Setting Notes
User Data Service Priority	**VLANCLASS**.***SRVPRIO***	• This parameter is valid only when the **VLANCLASS**.***TRAFFIC*** parameter is set to **USERDATA**. • It is recommended that the setting of this parameter be consistent with the configuration in the **DSCPMAP** MO.
VLAN ID	**VLANCLASS**.***VLANID***	Set this parameter based on the network plan.
VLAN Priority	**VLANCLASS**.***VLANPRIO***	Set this parameter based on the network plan.

Configuring an Interface

The following table describes the key parameters that must be set in an **INTERFACE** MO to configure an interface.

Parameter Name	Parameter ID	Setting Notes
Interface ID	**INTERFACE**.***ITFID***	Set this parameter based on the network plan.
Interface Type	**INTERFACE**.***ITFTYPE***	For a normal interface, set this parameter to **NORMAL**.
Port Type	**INTERFACE**.***PT***	Set this parameter based on the network plan.
Port ID	**INTERFACE**.***PORTID***	Set this parameter based on the network plan.
VLAN Tagged Switch	**INTERFACE**.***TAGGED***	Set this parameter based on the network plan.
DSCP-to-PCP Mapping ID	**INTERFACE**.***DSCP2PCPMAPID***	Set this parameter based on the network plan.
VRF Index	**INTERFACE**.***VRFIDX***	Set this parameter based on the network plan.
IPv4 Maximum Transmission Unit	**INTERFACE**.***MTU4***	Set this parameter based on the network plan.
ARP Proxy	**INTERFACE**.***ARPPROXY***	Set this parameter based on the network plan.

Configuring a Loopback Interface

The following table describes the key parameters that must be set in a **LOOPBACK** MO to configure a loopback interface.

Parameter Name	Parameter ID	Setting Notes
Port ID	**LOOPBACK.*PORTID***	This parameter specifies the ID of a loopback interface and is unique in the system.
Cabinet No.	**LOOPBACK.*CN***	This parameter specifies the cabinet number of the board to which the loopback interface belongs.
Subrack No.	**LOOPBACK.*SRN***	This parameter specifies the subrack number of the board to which the loopback interface belongs.
Slot No.	**LOOPBACK.*SN***	This parameter specifies the slot number of the board to which the loopback interface belongs.

The following table describes the key parameters that must be set in an **INTERFACE** MO to configure an interface on the loopback interface.

Parameter Name	Parameter ID	Setting Notes
Interface ID	**INTERFACE.*ITFID***	Set this parameter based on the network plan.
Interface Type	**INTERFACE.*ITFTYPE***	This parameter specifies the type of an interface, which can be set to **NORMAL** or **VLAN**. For a loopback interface, set this parameter to **NORMAL**.
Port Type	**INTERFACE.*PT***	For a loopback interface, set this parameter to **LOOPINT**.
Port ID	**INTERFACE.*PORTID***	For a loopback interface, set this parameter to the same value as **LOOPBACK.*PORTID***.
VRF Index	**INTERFACE.*VRFIDX***	Set this parameter based on the network plan.

Configuring an IP Address

Configure an IP address for a VLAN or normal interface.

The following table describes the key parameters that must be set in an **IPADDR4** MO to configure an IPv4 address.

Parameter Name	Parameter ID	Setting Notes
Interface ID	**IPADDR4.*ITFID***	When **INTERFACE.*PT*** corresponding to the interface ID is set to: ● **ETH**, **ETHTRK**, **PPP**, **MPGRP**, or **ETHCI** Configure an IP address, which is called an interface IP address. ● **LOOPINT** Configure an IP address, which is called a logical IP address.
IP Address	**IPADDR4.*IP***	If NR and LTE share the main control board, plan different local IP addresses for NR and LTE.
Mask	**IPADDR4.*MASK***	Set this parameter based on the network plan. A 31-bit mask can be configured for the IP address of an Ethernet port. During transmission network planning, only the directly connected 31-bit network segment can be set to all 0s or all 1s.
VRF Index	**IPADDR4.*VRFIDX***	Set this parameter based on the network plan.

Configuring an IP Route

The following table describes the key parameters that must be set in an **IPROUTE4** MO to configure a destination-based IPv4 route. In Layer 2 networking, these parameters are not required.

Parameter Name	Parameter ID	Setting Notes
Route Index	**IPROUTE4.*RTIDX***	Set this parameter based on the network plan.
VRF Index	**IPROUTE4.*VRFIDX***	Set this parameter based on the network plan.
Destination IP	**IPROUTE4.*DSTIP***	The default route, with the destination IP address and subnet mask being 0.0.0.0, is not recommended. If the peer NE such as the S-GW, MME, or peer eNodeB is located on the same network segment, an IP route to this network segment is recommended, with a destination IP address and a subnet mask for this segment.

Parameter Name	Parameter ID	Setting Notes
Mask	**IPROUTE4**.***DSTMASK***	The default route, with the destination IP address and subnet mask being 0.0.0.0, is not recommended. If the peer NE such as the S-GW, MME, or peer eNodeB is located on the same network segment, an IP route to this network segment is recommended, with a destination IP address and a subnet mask for this segment.
Route Type	**IPROUTE4**.***RTTYPE***	Set this parameter to **NEXTHOP** in Ethernet scenarios.
Port Type	**IPROUTE4**.***PT***	Set this parameter based on the network plan.
Port ID	**IPROUTE4**.***PORTID***	Set this parameter based on the network plan.
Next Hop IP	**IPROUTE4**.***NEXTHOP***	This parameter is valid only when the **IPROUTE4**.***RTTYPE*** parameter is set to **NEXTHOP** for the IPv4 route. Generally, this parameter is set to the IP address of the gateway on the transmission network connecting the base station.
Preference	**IPROUTE4**.***PREF***	This parameter is required for IP route backup. A smaller parameter value indicates a higher priority. The route with a higher priority is activated. The base station does not support route load sharing. Routes to the same destination network segment must have different priorities.

The following table describes the key parameters that must be set in an **SRCIPROUTE4** MO to configure a source-based IPv4 route. In Layer 2 networking, these parameters are not required.

Parameter Name	Parameter ID	Setting Notes
Source Route Index	**SRCIPROUTE4**.***SRCRTIDX***	Set this parameter based on the network plan.
Source IP Address	**SRCIPROUTE4**.***SRCIP***	Set this parameter based on the network plan.
Route Type	**SRCIPROUTE4**.***RTTYPE***	Set this parameter to **NEXTHOP** in Ethernet scenarios.

Parameter Name	Parameter ID	Setting Notes
Port Type	**SRCIPROUTE4.*PT***	Set this parameter based on the network plan.
Port ID	**SRCIPROUTE4.*PORTID***	Set this parameter based on the network plan.
Next Hop IP	**SRCIPROUTE4.*NEXTHOP***	This parameter is valid only when the **SRCIPROUTE4.*RTTYPE*** parameter is set to **NEXTHOP** for the source IPv4 route. Generally, this parameter is set to the IP address of the gateway on the transmission network connecting the base station.
Priority	**SRCIPROUTE4.*PREF***	This parameter is required for IP route backup. A smaller parameter value indicates a higher priority. The route with a higher priority is activated. The base station does not support route load sharing. Routes to the same destination network segment must have different priorities.

4.3.1.1.2 Using MML Commands

Before using MML commands, refer to **4.2 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Step 1 Run the **SET ETHPORT** command to set the attributes of an Ethernet port.

Step 2 (Optional) Run the **ADD DSCP2PCPMAP** command to configure a VLAN priority mapping.

Step 3 Run the following commands to set a VLAN:

- To set a VLAN based on the interface:

  Run the **ADD INTERFACE** command to configure the interface attribute.
- To set a VLAN based on the VLAN mapping:
  a. Run the **ADD VLANMAP** command to add a VLAN mapping.

     If **VLAN Mode** is set to **SINGLEVLAN**, go to **Step 3.c**.

     If **VLAN Mode** is set to **VLANGROUP**, go to **Step 3.b**.
  b. Run the **ADD VLANCLASS** command to add a mapping between traffic types and the VLAN group.

c. Run the **ADD INTERFACE** command to add a normal interface.

Step 4 (Optional) Configure a loopback interface.

1. Run the **ADD LOOPBACK** command to add a loopback interface.
2. Run the **ADD INTERFACE** command to add a normal interface.

Step 5 Run the **ADD IPADDR4** command to add an IP address.

Step 6 (Optional: to configure a destination-based IP route for the base station) Run the **ADD IPROUTE4** command to configure an IP route from the base station to the peer device.

Step 7 (Optional: to configure a source-based IP route for the base station) Run the **ADD SRCIPROUTE4** command to add a source-based IP route with the local IP address of the base station as the source IP address.

NOTE

Before performing this step, configure a DSCP for each traffic type. For details about the principle and MML commands for configuring DSCPs, see *Transmission Resource Management*. This section provides the MML command examples of configuring DSCPs for different types of traffic.

----End

Activation Command Examples

```
//Setting Ethernet port attributes for the base station
SET ETHPORT: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, PN=1, PORTID=1, PA=FIBER, SPEED=10G,
DUPLEX=FULL, FC=OPEN, FERAT=10, FERDT=10, RXBCPKTALMOCRTHD=322, RXBCPKTALMCLRTHD=290;

//(Optional) Configuring a VLAN priority mapping
ADD DSCP2PCPMAP: DSCP2PCPMAPID=0, DEFAULTPCP=0;
ADD DSCP2PCPREF: DSCP2PCPMAPID=0, DSCP=48, PCP=6;
ADD DSCP2PCPREF: DSCP2PCPMAPID=0, DSCP=46, PCP=5;
ADD DSCP2PCPREF: DSCP2PCPMAPID=0, DSCP=34, PCP=4;
ADD DSCP2PCPREF: DSCP2PCPMAPID=0, DSCP=26, PCP=3;
ADD DSCP2PCPREF: DSCP2PCPMAPID=0, DSCP=10, PCP=1;
ADD DSCP2PCPREF: DSCP2PCPMAPID=0, DSCP=0, PCP=0;

//Configuring the VLAN
//Configuring the VLAN based on the interface
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=1, VLANID=100, VRFIDX=0, MTU4=1500,
ARPPROXY=ENABLE;

//Configuring the VLAN based on the VLAN mapping
//Adding a mapping between the VLAN IDs and next-hop IP addresses
ADD VLANMAP: NEXTHOPIP="10.1.1.2", MASK="255.255.255.0", VLANMODE=SINGLEVLAN, VLANID=1,
SETPRIO=DISABLE;

//Adding a mapping among traffic types, DSCPs, and VLAN IDs
ADD VLANCLASS: VLANGROUPNO=0, TRAFFIC=SIG, VLANID=1, VLANPRIO=6;
ADD VLANCLASS: VLANGROUPNO=0, TRAFFIC=OM_HIGH, VLANID=1, VLANPRIO=5;
ADD VLANCLASS: VLANGROUPNO=0, TRAFFIC=OM_LOW, VLANID=1, VLANPRIO=2;
ADD VLANCLASS: VLANGROUPNO=0, TRAFFIC=USERDATA, SRVPRIO=46, VLANID=1, VLANPRIO=5;
ADD VLANCLASS: VLANGROUPNO=0, TRAFFIC=USERDATA, SRVPRIO=26, VLANID=1, VLANPRIO=3;
ADD VLANCLASS: VLANGROUPNO=0, TRAFFIC=USERDATA, SRVPRIO=34, VLANID=1, VLANPRIO=4;
ADD VLANCLASS: VLANGROUPNO=0, TRAFFIC=USERDATA, SRVPRIO=18, VLANID=1, VLANPRIO=2;
ADD VLANCLASS: VLANGROUPNO=0, TRAFFIC=USERDATA, SRVPRIO=10, VLANID=1, VLANPRIO=1;
ADD VLANCLASS: VLANGROUPNO=0, TRAFFIC=USERDATA, SRVPRIO=0, VLANID=1, VLANPRIO=0;
ADD VLANCLASS: VLANGROUPNO=0, TRAFFIC=OTHER, VLANID=1, VLANPRIO=5;
ADD INTERFACE: ITFID=0, ITFTYPE=NORMAL, PT=ETH, PORTID=1, VRFIDX=0, MTU4=1500,
ARPPROXY=ENABLE;

//Adding an interface IP address for the base station
```



```
ADD IPADDR4: ITFID=0, IP="10.1.1.1", MASK="255.255.255.0", VRFIDX=0;

//(Optional) Adding a loopback IP address for the base station
ADD LOOPBACK: PORTID=0, CN=0, SRN=0, SN=7;
ADD INTERFACE: ITFID=1, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=0, VRFIDX=0;
ADD IPADDR4: ITFID=1, IP="20.1.1.1", MASK="255.255.255.0", VRFIDX=0;

//(Optional) Adding a destination-based IP route from the base station to the peer device
ADD IPROUTE4: RTIDX=0, DSTIP="100.1.1.1", DSTMASK="255.255.255.0", RTTYPE=NEXTHOP,
NEXTHOP="10.1.1.2", PREF=60;
//(Optional) Adding a source-based IP route with the local IP address of the base station being the source
IP address
//When an interface IP address is configured
ADD SRCIPROUTE4: SRCRTIDX=0, SRCIP="10.1.1.1", RTTYPE=NEXTHOP, NEXTHOP="10.1.1.2", PREF=60;
//When a loopback IP address is configured
ADD SRCIPROUTE4: SRCRTIDX=0, SRCIP="20.1.1.1", RTTYPE=NEXTHOP, NEXTHOP="10.1.1.2", PREF=60;
//Setting the DSCP values for signaling data, OM data, and IP clock data
SET DIFPRI: PRIRULE=DSCP, SIGPRI=48, OMHIGHPRI=46, OMLOWPRI=18, IPCLKPRI=46;
//Setting the DSCP for BFD packets when adding the BFD session
ADD BFD: BFDSN=100, SRCIP="10.1.1.1", DSTIP="10.1.1.2", MYDISCREAMINATOR=1, HT=SINGLE_HOP,
CATLOG=RELIABILITY, DSCP=46, VER=STANDARD, BFDAUTHSW=OFF;
```

4.3.1.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.3.1.2 IPv4 Data Configuration (in the Old Model)

This section describes how to configure IPv4 data on the base station side when the old transmission configuration model is used (**GTRANSPARA**.***TRANSCFGMODE*** is set to **OLD**).

4.3.1.2.1 Data Preparation

Deploying common data involves setting up the transmission paths for the bottom layers. Common data includes data at the physical layer, data link layer, and network layer.

- Physical layer data

  Operator-planned information about the cabinet, subrack, and slot housing the transmission port, optical/electrical port attribute, transmission rate, and duplex mode
- Data link layer data
  - Information about the Ethernet MAC layer, which includes the port's flow control frame and ARP proxy information
  - Operator-planned VLAN ID and VLAN priority of each service flow
- Network layer data
  - Local IP address information about the base station, including the interface IP addresses and logical IP addresses
  - Operator-planned IP addresses for the S1-U, X2-C, X2-U, NG-C, NG-U, Xn-C, Xn-U, IP clock link, IPsec tunnel, and OM channel

– Information about routes from the base station to a peer NE when there are routers between them. The peer NE can be the core network, eNodeB, MAE, or IP clock server.

NOTE

If the processing capability of the base station is insufficient, pause frames are sent to the transmission network. It is recommended that Pause frame flow control be enabled on the transmission equipment.

Configuring an Ethernet Port

The following table describes the key parameters that must be set in an **ETHPORT** MO to configure an Ethernet port.

Parameter Name	Parameter ID	Setting Notes
Subboard Type	**ETHPORT**.***SBT***	Set this parameter based on the network plan.
Port Attribute	**ETHPORT**.***PA***	The port attribute must be the same as that of the peer end. When a UMPTe/UMPTg board is used, automatic port attribute detection is not supported. You need to configure the port attribute according to actual situations. If an optical module with a rate of 50 Gbit/s or higher is used at the local device, automatic detection is not supported. Automatic detection is enabled when the **Port Attribute**, **Speed**, and **Duplex** parameters are set to **AUTO(Automatic Detection)**, **AUTO(Auto Negotiation)**, and **AUTO(Auto Negotiation)**, respectively. When the system detects these parameter settings, it reports ALM-26247 Configuration Failure.
Maximum Transmission Unit	**ETHPORT**.***MTU***	Set this parameter based on the network plan.
Speed	**ETHPORT**.***SPEED***	Set this parameter to the same value as that of the peer port.
Duplex	**ETHPORT**.***DUPLEX***	Set this parameter based on the values of **ETHPORT**.***PA*** and **ETHPORT**.***SPEED***. For details about the setting suggestions, see **Table 4-2**.

Parameter Name	Parameter ID	Setting Notes
FEC Mode	**ETHPORT**.***FECMODE***	This parameter specifies the FEC mode used when the Ethernet port rate is 25 Gbit/s. It must be set to the same value for the local and peer ports. Set this parameter based on the network plan.

Table 4-2 Suggestions for parameter settings

Port Attribute	Speed	Duplex
COPPER(Copper)	**10M(10M)** or **100M(100M)**	This parameter can only be set to **FULL(Full Duplex)**.
COPPER(Copper)	If 1000 Mbit/s is required, this parameter must be set to **AUTO(Automatic Negotiation)**.	This parameter can only be set to **AUTO(Automatic Negotiation)**.
FIBER(Fiber)	**100M(100M)**	This parameter can only be set to **FULL(Full Duplex)**.
FIBER(Fiber)	If 1000 Mbit/s is required, set this parameter to **1000M(1000M)** or **AUTO(Automatic Negotiation)**.	● If **Speed** is set to **1000M(1000M)**, set this parameter to **FULL(Full Duplex)**. ● If **Speed** is set to **AUTO(Automatic Negotiation)**, set this parameter to **AUTO(Automatic Negotiation)**.
FIBER(Fiber)	**10G** or **25G**	This parameter can only be set to **FULL(Full Duplex)**.
AUTO(Automatic Detection)	This parameter can only be set to **AUTO(Automatic Negotiation)**.	This parameter can only be set to **AUTO(Automatic Negotiation)**.

Configuring a VLAN Mapping

The following table describes the key parameters that must be set in a **VLANMAP** MO to configure a VLAN mapping.

Parameter Name	Parameter ID	Setting Notes
VRF Index	**VLANMAP**.***VRFIDX***	Set this parameter based on the network plan.
Next Hop IP	**VLANMAP**.***NEXTHOPIP***	This IP address must be on the same network segment as the IP address of the gateway connected to the physical port of the base station as well as the interface IP address of the base station.
Mask	**VLANMAP**.***MASK***	The interface IP address of the base station must be on the network segment determined by the next-hop IP address and subnet mask. A 31-bit mask can be configured for the IP address of an Ethernet port. During transmission network planning, only the directly connected 31-bit network segment can be set to all 0s or all 1s.
VLAN Mode	**VLANMAP**.***VLANMODE***	● If the operator plans a one-to-one mapping between base stations and VLANs, set this parameter to **SINGLEVLAN**. ● If the operator plans a one-to-multiple mapping between base stations and VLANs based on traffic types, set this parameter to **VLANGROUP**.
VLAN Group No.	**VLANMAP**.***VLANGROUPNO***	This parameter is valid only when the **VLANMAP**.***VLANMODE*** parameter is set to **VLANGROUP**. It is recommended that VLAN groups be numbered from 0. Generally, each base station has only one VLAN group number. ● Set this parameter if the value of the **VLANMAP**.***VLANMODE*** parameter is changed from **SINGLEVLAN** to **VLANGROUP** during modification of the **VLANMAP** MO. ● Ensure that the value of this parameter is the same as the VLAN group number in the **VLANCLASS** MO.
VLAN ID	**VLANMAP**.***VLANID***	This parameter is valid only when the **VLANMAP**.***VLANMODE*** parameter is set to **SINGLEVLAN**.

Parameter Name	Parameter ID	Setting Notes
Set VLAN Priority	**VLANMAP**.***SETPRIO***	This parameter is valid only when the **VLANMAP**.***VLANMODE*** parameter is set to **SINGLEVLAN** and one VLAN has only one priority. [a]
a: ● To use the default mapping between user data types and VLAN priorities, set this parameter to **DISABLE**. ● To specify the mapping between user data types and VLAN priorities, set this parameter to **ENABLE**. ● The default mapping is recommended. That is, set **VLANMAP**.***SETPRIO*** to **DISABLE**. To specify the mapping between DSCPs and VLAN priorities when the single VLAN mode is used, configure parameters in the **DSCPMAP** MO.		

Configuring a VLAN Priority Mapping

If the single VLAN mode is used, it is recommended that the **DSCPMAP** MO be configured to set up a mapping between traffic types and VLAN priorities. The following table describes the key parameters.

Parameter Name	Parameter ID	Setting Notes
VRF Index	**DSCPMAP**.***VRFIDX***	Set this parameter based on the network plan.
Differentiated Service Codepoint	**DSCPMAP**.***DSCP***	Set this parameter based on the network plan.
VLAN Priority	**DSCPMAP**.***VLANPRIO***	Set this parameter based on the network plan.

If the VLAN group mode is used, it is recommended that both the **VLANCLASS** and **VLANMAP** MOs be configured to set the VLAN group mapping. The following table describes the key parameters.

Parameter Name	Parameter ID	Setting Notes
VLAN Group No.	**VLANCLASS**.***VLANGROUPNO***	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Traffic Type	**VLANCLASS**.***TRAFFIC***	There are the following traffic types: **USERDATA**, **SIG**, **OM_HIGH**, **OM_LOW**, and **OTHER**. A VLAN ID must be specified for each traffic type to prevent service exceptions.
User Data Service Priority	**VLANCLASS**.***SRVPRIO***	This parameter is valid only when the **VLANCLASS**.***TRAFFIC*** parameter is set to **USERDATA**. It is recommended that the setting of this parameter be consistent with the configuration in the **DSCPMAP** MO.
VLAN ID	**VLANCLASS**.***VLANID***	Set this parameter based on the network plan.
VLAN Priority	**VLANCLASS**.***VLANPRIO***	Set this parameter based on the network plan.

Configuring a Device IP Address

The following table describes the key parameters that must be set in a **DEVIP** MO to configure a device IP address.

Parameter Name	Parameter ID	Setting Notes
Subboard Type	**DEVIP**.***SBT***	It is recommended that this parameter be set to **BASE_BOARD**.
IP Address	**DEVIP**.***IP***	The control plane and the user plane of the S1/X2 interface use this IP address. If NR and LTE share the main control board, plan different local IP addresses for NR and LTE.
Mask	**DEVIP**.***MASK***	Set this parameter based on the network plan. A 31-bit mask can be configured for the IP address of an Ethernet port. During transmission network planning, only the directly connected 31-bit network segment can be set to all 0s or all 1s.

Parameter Name	Parameter ID	Setting Notes
Port Type	**DEVIP**.*PT*	● In an Ethernet transmission scenario, configure a device IP address with this parameter set to **ETH**. This IP address is also called interface IP address. ● In an IPsec-enabled scenario, a loopback IP address is required in addition to the interface IP address. A device IP address is a loopback IP address when this parameter is set to **LOOPINT**. ● When the base station uses an interface IP address to set up an IPsec tunnel with the SeGW, the base station uses the loopback IP address to communicate with the MAE, S-GW, or eNodeB. ● If a device IP address is to be configured to serve as the UMPTe/UMPTg CI port address, set this parameter to **ETHCI**.

Configuring an IP Route

The following table describes the key parameters that must be set in an **IPRT** MO to configure a destination-based IP route. In Layer 2 networking, these parameters are not required.

Parameter Name	Parameter ID	Setting Notes
Route Index	**IPRT**.*RTIDX*	Set this parameter based on the network plan.
Destination IP	**IPRT**.*DSTIP*	The default route, with the destination IP address and subnet mask being 0.0.0.0, is not recommended. If the peer NE is located on the same network segment, an IP route to this network segment (with a destination IP address and a subnet mask) is recommended.

Parameter Name	Parameter ID	Setting Notes
Mask	**IPRT**.***DSTMASK***	The default route, with the destination IP address and subnet mask being 0.0.0.0, is not recommended. If the S-GW or the peer base station is on the same network segment, an IP route to this network segment is recommended.
Subboard Type	**IPRT**.***SBT***	It is recommended that this parameter be set to **BASE_BOARD**.
Route Type	**IPRT**.***RTTYPE***	Set this parameter based on the network plan.
Port Type	**IPRT**.***IFT***	Set this parameter based on the network plan.
Next Hop IP	**IPRT**.***NEXTHOP***	This parameter is valid only when the **IPRT**.***RTTYPE*** parameter is set to **NEXTHOP**. Generally, this parameter is set to the IP address of the gateway on the transmission network connecting the base station.
Preference	**IPRT**.***PREF***	This parameter is required for IP route backup. A smaller parameter value indicates a higher priority. The route with a higher priority is activated. The base station does not support route load sharing. Routes to the same destination network segment must have different priorities.

The following table describes the key parameters that must be set in an **SRCIPRT** MO to configure a source-based IP route. In Layer 2 networking, these parameters are not required.

Parameter Name	Parameter ID	Setting Notes
Source Route Index	**SRCIPRT**.***SRCRTIDX***	Set this parameter based on the network plan.
Source IP Address	**SRCIPRT**.***SRCIP***	Set this parameter based on the network plan.
Subboard Type	**SRCIPRT**.***SBT***	It is recommended that this parameter be set to **BASE_BOARD**.

Parameter Name	Parameter ID	Setting Notes
Route Type	**SRCIPRT**.*RTTYPE*	Set this parameter based on the network plan.
Interface Type	**SRCIPRT**.*IFT*	Set this parameter based on the network plan.
Next Hop IP	**SRCIPRT**.*NEXTHOP*	This parameter is valid only when the **SRCIPRT**.*RTTYPE* parameter is set to **NEXTHOP**. Generally, this parameter is set to the IP address of the gateway on the transmission network connecting the base station.
Priority	**SRCIPRT**.*PREF*	This parameter is required for IP route backup. A smaller parameter value indicates a higher priority. The route with a higher priority is activated. The base station does not support route load sharing. Routes to the same destination network segment must have different priorities.

4.3.1.2.2 Using MML Commands

Before using MML commands, refer to **4.2 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Step 1 Run the **SET ETHPORT** command to set the attributes of an Ethernet port.

Step 2 Run the **ADD DEVIP** command to add a device IP address.

Step 3 Run the **ADD IPRT** command to configure an IP route from the base station to the peer device.

Step 4 (Optional) Run the **ADD SRCIPRT** command to configure a source-based IP route with the local IP address of the base station being the source IP address.

NOTE

Before performing **Step 4**, configure a DSCP for each traffic type. For details about the principle and MML commands for configuring DSCPs, see *Transmission Resource Management*. This section provides the MML command examples of configuring DSCPs for different types of traffic.

Step 5 Run the **ADD VLANMAP** command to add a VLAN mapping.

- If **VLAN Mode** is set to **SINGLEVLAN**, go to **Step 6**.
- If **VLAN Mode** is set to **VLANGROUP** go to **Step 7**.

Step 6 Run the **SET DSCPMAP** command to set a mapping between DSCPs and VLAN priorities.

Step 7 For a macro base station, run the **ADD VLANCLASS** command to add a mapping between traffic types and VLANs.

----End

Activation Command Examples

```
//Setting Ethernet port attributes for the base station
SET ETHPORT: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, PN=1, PA=FIBER, MTU=1500, SPEED=10G,
DUPLEX=FULL, ARPPROXY=ENABLE, FC=OPEN, FERAT=10, FERDT=10, RXBCPKTALMOCRTHD=322,
RXBCPKTALMCLRTHD=290;
//Adding a device IP address for the base station
ADD DEVIP: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, PT=ETH, PN=1, IP="10.1.1.1", MASK="255.255.255.0";
//Adding a destination-based IP route from the base station to the peer device
ADD IPRT: RTIDX=0, SN=7, SBT=BASE_BOARD, DSTIP="100.1.1.1", DSTMASK="255.255.255.0",
RTTYPE=NEXTHOP, NEXTHOP="10.1.1.2", PREF=60;
//(Optional) Adding a source-based IP route with the local IP address of the base station being the source
IP address
ADD SRCIPRT: SRCRTIDX=0, SN=7, SBT=BASE_BOARD, SRCIP="10.1.1.1", RTTYPE=NEXTHOP,
NEXTHOP="10.1.1.2", PREF=60;
//Setting the DSCP values for signaling data, OM data, and IP clock data
SET DIFPRI: PRIRULE=DSCP, SIGPRI=48, OMHIGHPRI=46, OMLOWPRI=18, IPCLKPRI=46;
//Setting the DSCP for BFD packets when adding the BFD session
ADD BFDSESSION: SN=7, BFDSN=0, SRCIP="10.1.1.1", DSTIP="10.1.1.2", HT=SINGLE_HOP,
CATLOG=RELIABILITY, DSCP=46, VER=STANDARD, BFDAUTHSW=OFF;
//Setting VLAN information in single VLAN mode
//Adding a mapping between the VLAN IDs and next-hop IP addresses
ADD VLANMAP: NEXTHOPIP="10.1.1.2", MASK="255.255.255.0", VLANMODE=SINGLEVLAN, VLANID=1,
SETPRIO=DISABLE;
//Setting a mapping between DSCPs and VLAN priorities
SET DSCPMAP: DSCP=48, VLANPRIO=6;
SET DSCPMAP: DSCP=46, VLANPRIO=5;
SET DSCPMAP: DSCP=34, VLANPRIO=4;
SET DSCPMAP: DSCP=26, VLANPRIO=3;
SET DSCPMAP: DSCP=18, VLANPRIO=2;
SET DSCPMAP: DSCP=10, VLANPRIO=1;
SET DSCPMAP: DSCP=0, VLANPRIO=0;
//Setting VLAN information in VLAN group mode
//Adding a mapping between next-hop IP addresses and VLAN groups
ADD VLANMAP:NEXTHOPIP="10.1.1.2", MASK="255.255.255.0", VLANMODE=VLANGROUP,
VLANGROUPNO=0;
//Adding a mapping among traffic types, DSCPs, and VLAN IDs
ADD VLANCLASS: VLANGROUPNO=0, TRAFFIC=SIG, VLANID=1, VLANPRIO=6;
ADD VLANCLASS: VLANGROUPNO=0, TRAFFIC=OM_HIGH, VLANID=1, VLANPRIO=5;
ADD VLANCLASS: VLANGROUPNO=0, TRAFFIC=OM_LOW, VLANID=1, VLANPRIO=2;
ADD VLANCLASS: VLANGROUPNO=0, TRAFFIC=USERDATA, SRVPRIO=46, VLANID=1, VLANPRIO=5;
ADD VLANCLASS: VLANGROUPNO=0, TRAFFIC=USERDATA, SRVPRIO=26, VLANID=1, VLANPRIO=3;
ADD VLANCLASS: VLANGROUPNO=0, TRAFFIC=USERDATA, SRVPRIO=34, VLANID=1, VLANPRIO=4;
ADD VLANCLASS: VLANGROUPNO=0, TRAFFIC=USERDATA, SRVPRIO=18, VLANID=1, VLANPRIO=2;
ADD VLANCLASS: VLANGROUPNO=0, TRAFFIC=USERDATA, SRVPRIO=10, VLANID=1, VLANPRIO=1;
ADD VLANCLASS: VLANGROUPNO=0, TRAFFIC=USERDATA, SRVPRIO=0, VLANID=1, VLANPRIO=0;
ADD VLANCLASS: VLANGROUPNO=0, TRAFFIC=OTHER, VLANID=1, VLANPRIO=5;
```

4.3.1.2.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.3.1.3 IPv6 Data Configuration (in the New Model)

This section describes how to configure IPv6 transmission. IPv6 transmission supports only the new configuration model (**GTRANSPARA**.***TRANSCFGMODE*** is set to **NEW**).

4.3.1.3.1 Data Preparation

Configuring an Ethernet Port

The following table describes the key parameters that must be set in an **ETHPORT** MO to configure an Ethernet port.

Parameter Name	Parameter ID	Setting Notes
Port ID	**ETHPORT**.***PORTID***	Set this parameter based on the network plan.
Subboard Type	**ETHPORT**.***SBT***	Set this parameter based on the network plan.
Port Attribute	**ETHPORT**.***PA***	The port attribute must be the same as that of the peer end. When a UMPTe/UMPTg board is used, automatic port attribute detection is not supported. You need to configure the port attribute according to actual situations. If an optical module with a rate of 50 Gbit/s or higher is used at the local device, automatic detection is not supported. Automatic detection is enabled when the **Port Attribute**, **Speed**, and **Duplex** parameters are set to **AUTO(Automatic Detection)**, **AUTO(Auto Negotiation)**, and **AUTO(Auto Negotiation)**, respectively. When the system detects these parameter settings, it reports ALM-26247 Configuration Failure.
Speed	**ETHPORT**.***SPEED***	Set this parameter to the same value as that of the peer port.
Duplex	**ETHPORT**.***DUPLEX***	Set this parameter based on the values of **ETHPORT**.***PA*** and **ETHPORT**.***SPEED***. For details about the setting suggestions, see Table 4-3.

Parameter Name	Parameter ID	Setting Notes
FEC Mode	**ETHPORT**.***FECMODE***	This parameter specifies the FEC mode used when the Ethernet port rate is 25 Gbit/s. It must be set to the same value for the local and peer ports. Set this parameter based on the network plan.

Table 4-3 Suggestions for parameter settings

Port Attribute	Speed	Duplex
COPPER(Copper)	10M(10M) or 100M(100M)	This parameter can only be set to **FULL(Full Duplex)**.
COPPER(Copper)	If 1000 Mbit/s is required, this parameter must be set to **AUTO(Automatic Negotiation)**.	This parameter can only be set to **AUTO(Automatic Negotiation)**.
FIBER(Fiber)	100M(100M)	This parameter can only be set to **FULL(Full Duplex)**.
FIBER(Fiber)	If 1000 Mbit/s is required, set this parameter to **1000M(1000M)** or **AUTO(Automatic Negotiation)**.	• If **Speed** is set to **1000M(1000M)**, set this parameter to **FULL(Full Duplex)**. • If **Speed** is set to **AUTO(Automatic Negotiation)**, set this parameter to **AUTO(Automatic Negotiation)**.
FIBER(Fiber)	**10G(10G)** or **25G(25G)**	This parameter can only be set to **FULL(Full Duplex)**.
AUTO(Automatic Detection)	This parameter can only be set to **AUTO(Automatic Negotiation)**.	This parameter can only be set to **AUTO(Automatic Negotiation)**.

Configuring a VLAN Priority Mapping

The following table describes the key parameters that must be set in a **DSCP2PCPMAP** MO and a **DSCP2PCPREF** MO to configure a mapping between traffic types and VLAN priorities.

Parameter Name	Parameter ID	Subparameter	Setting Notes
DSCP-to-PCP Mapping ID	**DSCP2PCPMAP**.***DSCP2PCPMAPID***	-	Set this parameter based on the network plan.
Default PCP	**DSCP2PCPMAP**.***DEFAULTPCP***	-	Set this parameter based on the network plan.
Differentiated Services Code Point	***DSCP2PCPREF.DSCP***	-	Set this parameter based on the network plan.
Priority Code Point	***DSCP2PCPREF.PCP***	-	Set this parameter based on the network plan.

NOTE

VLAN priority mapping applies to both IPv4 and IPv6 transmission. In IPv4/IPv6 dual-stack transmission, this mapping only needs to be configured once.

Configuring a Loopback Interface

The following table describes the key parameters that must be set in a **LOOPBACK** MO to configure a loopback interface.

Parameter Name	Parameter ID	Setting Notes
Port ID	**LOOPBACK**.***PORTID***	This parameter specifies the ID of a loopback interface and is unique in the system.
Cabinet No.	**LOOPBACK**.***CN***	This parameter specifies the cabinet number of the board to which the loopback interface belongs.
Subrack No.	**LOOPBACK**.***SRN***	This parameter specifies the subrack number of the board to which the loopback interface belongs.
Slot No.	**LOOPBACK**.***SN***	This parameter specifies the slot number of the board to which the loopback interface belongs.

Configuring a VLAN Interface

The following table describes the key parameters that must be set in an **INTERFACE** MO to configure a VLAN interface and VLAN.

Parameter Name	Parameter ID	Setting Notes
Interface ID	**INTERFACE.*ITFID***	Set this parameter based on the network plan.
Interface Type	**INTERFACE.*ITFTYPE***	This parameter specifies the type of an interface, which can be set to **NORMAL** or **VLAN**. For a VLAN interface, set this parameter to **VLAN**.
Port Type	**INTERFACE.*PT***	This parameter can only be set to **ETH** for a VLAN interface in IPv6 transmission.
Port ID	**INTERFACE.*PORTID***	Set this parameter based on the network plan.
VLAN ID	**INTERFACE.*VLANID***	This parameter must be set when a VLAN interface is configured.
DSCP-to-PCP Mapping ID	**INTERFACE.*DSCP2PCPMAPID***	Set this parameter based on the network plan.
IPv6 Maximum Transmission Unit	**INTERFACE.*MTU6***	Use the minimum MTU on all backhaul networks of the interface as the interface MTU on the base station side. For details about the planning requirements, see **4.2.2.4 Networking**.
IPv6 Switch	**INTERFACE.*IPV6SW***	Set this parameter to **ENABLE** for an interface on which IPv6 transmission or dual-stack transmission is used.

Configuring a Normal Interface

The following table describes the key parameters that must be set in an **INTERFACE** MO to configure a normal interface.

Parameter Name	Parameter ID	Setting Notes
Interface ID	**INTERFACE.*ITFID***	Set this parameter based on the network plan.
Interface Type	**INTERFACE.*ITFTYPE***	This parameter specifies the type of an interface, which can be set to **NORMAL** or **VLAN**. For a normal interface, set this parameter to **NORMAL**.

Parameter Name	Parameter ID	Setting Notes
Port Type	**INTERFACE**.***PT***	For an IPv6 normal interface, set this parameter to **ETH** or **LOOPINT**. ● If packets that do not carry VLAN tags need to be received or sent, set this parameter to **ETH**. ● If the service IP address is a logical IP address, set this parameter to **LOOPINT**, for example, in security scenarios.
Port ID	**INTERFACE**.***PORTID***	Set this parameter based on the network plan.
VLAN Tagged Switch	**INTERFACE**.***TAGGED***	Set this parameter based on the network plan.
DSCP-to-PCP Mapping ID	**INTERFACE**.***DSCP2PCPMAPID***	Set this parameter based on the network plan.
IPv6 Maximum Transmission Unit	**INTERFACE**.***MTU6***	Use the minimum MTU on all backhaul networks of the interface as the interface MTU on the base station side.
IPv6 Switch	**INTERFACE**.***IPV6SW***	Set this parameter to **ENABLE** for an interface on which IPv6 transmission or dual-stack transmission is used.

NOTE

If an interface has multiple transmission links, for example, links for the OM channel, control plane, and user plane, the MTU size of the OM channel transmission link is 1500, the MTU size of the control plane link is 1300, and the MTU size of the user plane path is 1800, then set the MTU size of IPv6 transmission to 1300.

The following table describes the key parameters that must be set in an **INTERFACE** MO to configure an interface on the loopback interface.

Parameter Name	Parameter ID	Setting Notes
Interface ID	**INTERFACE**.***ITFID***	Set this parameter based on the network plan.
Interface Type	**INTERFACE**.***ITFTYPE***	This parameter specifies the type of an interface, which can be set to **NORMAL** or **VLAN**. For a loopback interface, set this parameter to **NORMAL**.

Parameter Name	Parameter ID	Setting Notes
Port Type	**INTERFACE**.***PT***	For a loopback interface, set this parameter to **LOOPINT**.
Port ID	**INTERFACE**.***PORTID***	For a loopback interface, set this parameter to the same value as **LOOPBACK**.***PORTID***.

Configuring an IPv6 Address

Configure an IPv6 address for a VLAN or normal interface.

The following table describes the key parameters that must be set in an **IPADDR6** MO to configure an IPv6 address.

Parameter Name	Parameter ID	Setting Notes
IPv6 Address ID	**IPADDR6**.***IPADDR6ID***	Set this parameter based on the network plan.
Interface ID	**IPADDR6**.***ITFID***	Set this parameter based on the network plan.
IPv6 Address	**IPADDR6**.***IPV6***	If NR and LTE share the main control board, plan different local IPv6 addresses for NR and LTE.
Prefix Length	**IPADDR6**.***PFXLEN***	Set this parameter based on the network plan.

Configuring an IPv6 Route

The following table describes the key parameters that must be set in an **IPROUTE6** MO to configure an IPv6 route. In Layer 2 networking, these parameters are not required.

Parameter Name	Parameter ID	Setting Notes
Route Index	**IPROUTE6**.***RTIDX***	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Destination IPv6 Address	**IPROUTE6**.***DSTIP***	The default route, of which the destination IPv6 address is 0::0 and the prefix length is 0, is not recommended. If the peer NE such as the core network equipment or peer eNodeB/gNodeB is located on the same network segment, an IP route to this network segment is recommended, with the destination IPv6 address being the network segment address and the prefix length being less than 128.
Prefix Length	**IPROUTE6**.***PFXLEN***	Set this parameter based on the network plan.
Route Type	**IPROUTE6**.***RTTYPE***	Set this parameter to **NEXTHOP**.
Next-Hop IPv6 Address	**IPROUTE6**.***NEXTHOP***	Set this parameter based on the network plan.
Preference	**IPROUTE6**.***PREF***	Set this parameter based on the network plan.

Configuring a Source-based IPv6 Route

The following table describes the key parameters that must be set in an **SRCIPROUTE6** MO to configure a source-based IPv6 route. IPv6 routes do not need to be configured in Layer 2 networking.

Parameter Name	Parameter ID	Setting Notes
IPv6 Source Route Index	**SRCIPROUTE6**.***SRCRTIDX***	Set this parameter based on the network plan.
Source IPv6 Address	**SRCIPROUTE6**.***SRCIP***	Set this parameter to the value of **IPADDR6**.***IPV6***.
Route Type	**SRCIPROUTE6**.***RTTYPE***	Set this parameter based on the network plan.
Next-Hop IPv6 Address	**SRCIPROUTE6**.***NEXTHOP***	Set this parameter based on the network plan.
Preference	**SRCIPROUTE6**.***PREF***	Set this parameter based on the network plan.

4.3.1.3.2 Using MML Commands

Before using MML commands, refer to **4.2 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Step 1 Run the **SET ETHPORT** command to set the attributes of an Ethernet port.

Step 2 Configure the VLAN and VLAN interface.

1. (Optional) Run the **ADD DSCP2PCPMAP** command to configure a VLAN priority mapping.
2. Run the **ADD INTERFACE** command to set **VLAN ID**, **IPv6 Maximum Transmission Unit**, and **IPv6 Switch**.

Step 3 Run the **ADD INTERFACE** command to configure a normal interface. In this step, set **IPv6 Maximum Transmission Unit** and **IPv6 Switch** to appropriate values.

Step 4 (Optional) Configure a loopback interface.

1. Run the **ADD LOOPBACK** command to add a loopback interface.
2. Run the **ADD INTERFACE** command to add a normal interface.

Step 5 Run the **ADD IPADDR6** command to add an IPv6 address.

Step 6 (Optional) Run the **ADD IPROUTE6** or **ADD SRCIPROUTE6** command to configure an IPv6 route from the gNodeB to the peer equipment.

NOTE

The DSCP values must be set for each traffic type. For details about the principle and MML commands for configuring DSCPs, see *Transmission Resource Management*. This section provides the MML command examples of configuring DSCPs for different types of traffic.

----End

Activation Command Examples

```
//Setting the Ethernet port attributes
SET ETHPORT: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, PN=1, PORTID=1, PA=FIBER, SPEED=1000M,
DUPLEX=FULL, FC=OPEN, FERAT=10, FERDT=10, RXBCPKTALMOCRTHD=322, RXBCPKTALMCLRTHD=290;
//(Optional) Configuring a VLAN priority mapping
ADD DSCP2PCPMAP: DSCP2PCPMAPID=0, DEFAULTPCP=0;
ADD DSCP2PCPREF: DSCP2PCPMAPID=0, DSCP=48, PCP=6;
ADD DSCP2PCPREF: DSCP2PCPMAPID=0, DSCP=46, PCP=5;
ADD DSCP2PCPREF: DSCP2PCPMAPID=0, DSCP=34, PCP=4;
ADD DSCP2PCPREF: DSCP2PCPMAPID=0, DSCP=26, PCP=3;
ADD DSCP2PCPREF: DSCP2PCPMAPID=0, DSCP=18, PCP=2;
ADD DSCP2PCPREF: DSCP2PCPMAPID=0, DSCP=10, PCP=1;
ADD DSCP2PCPREF: DSCP2PCPMAPID=0, DSCP=0, PCP=0;
//Configuring a VLAN interface with the VLAN ID, maximum IPv6 transmission unit, and IPv6 switch set to
appropriate values
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=1, VLANID=100, DSCP2PCPMAPID=0,
MTU6=1500, IPV6SW=ENABLE;
ADD INTERFACE: ITFID=1, ITFTYPE=VLAN, PT=ETH, PORTID=1, VLANID=200, DSCP2PCPMAPID=0,
```

```
MTU6=1500, IPV6SW=ENABLE;
//Configuring a normal interface with the maximum IPv6 transmission unit and IPv6 switch specified
ADD INTERFACE: ITFID=1, ITFTYPE=NORMAL, PT=ETH, PORTID=1, VRFIDX=0, MTU6=1500,
IPV6SW=ENABLE;
//Adding a loopback IP address for the gNodeB
ADD LOOPBACK: PORTID=8, CN=0, SRN=0, SN=7;
ADD INTERFACE: ITFID=10, ITFTYPE=NORMAL, PT=LOOPINT, PORTID=8, VRFIDX=0, IPV6SW=ENABLE;
ADD IPADDR6: IPADDR6ID="NRLOOPBACK", ITFID=10, IPV6="2001:db8:100:ad1:200:100:200:100",
PFXLEN=128;
//Adding an IPv6 address for the gNodeB
ADD IPADDR6: IPADDR6ID="NR", ITFID=0, IPV6="2001:db8:100:ad1:200:100:100:2", PFXLEN=126;
//(Optional) Adding a destination-based IP route from the gNodeB to the peer equipment
ADD IPROUTE6: RTIDX=0, DSTIP="2001:db8:100:ad1:200:100:3001:0", PFXLEN=112, RTTYPE=NEXTHOP,
NEXTHOP="2001:db8:100:ad1:200:100:100:1", PREF=60;
//(Optional) Adding a source-based IP route from the gNodeB to the peer equipment
ADD SRCIPROUTE6: SRCRTIDX=0, SRCIP="2001:db8:100:ad1:200:100:200:100", RTTYPE=NEXTHOP,
NEXTHOP="2001:db8:100:ad1:200:100:100:1", PREF=60;
//Setting the DSCP values for signaling data, OM data, and IP clock data
SET DIFPRI: PRIRULE=DSCP, SIGPRI=48, OMHIGHPRI=46, OMLOWPRI=18, IPCLKPRI=46;
//Setting the DSCP values for user data
ADD UDT: UDTNO=9, UDTPARAGRPID=48;
ADD UDT: UDTNO=8, UDTPARAGRPID=47;
ADD UDT: UDTNO=7, UDTPARAGRPID=46;
ADD UDT: UDTNO=6, UDTPARAGRPID=45;
ADD UDT: UDTNO=5, UDTPARAGRPID=44;
ADD UDT: UDTNO=4, UDTPARAGRPID=43;
ADD UDT: UDTNO=3, UDTPARAGRPID=42;
ADD UDT: UDTNO=2, UDTPARAGRPID=41;
ADD UDT: UDTNO=1, UDTPARAGRPID=40;
ADD UDTPARAGRP: UDTPARAGRPID=40, PRIRULE=DSCP, PRI=46, PRIMTRANRSCTYPE=HQ,
PRIMPTLOADTH=100, PRIM2SECPTLOADRATH=0;
ADD UDTPARAGRP: UDTPARAGRPID=41, PRIRULE=DSCP, PRI=34, PRIMTRANRSCTYPE=HQ,
PRIMPTLOADTH=100, PRIM2SECPTLOADRATH=1000;
ADD UDTPARAGRP: UDTPARAGRPID=42, PRIRULE=DSCP, PRI=34, PRIMTRANRSCTYPE=HQ,
PRIMPTLOADTH=100, PRIM2SECPTLOADRATH=1000;
ADD UDTPARAGRP: UDTPARAGRPID=43, PRIRULE=DSCP, PRI=34, PRIMTRANRSCTYPE=HQ,
PRIMPTLOADTH=100, PRIM2SECPTLOADRATH=1000;
ADD UDTPARAGRP: UDTPARAGRPID=44, PRIRULE=DSCP, PRI=46, PRIMTRANRSCTYPE=HQ,
PRIMPTLOADTH=100, PRIM2SECPTLOADRATH=1000;
ADD UDTPARAGRP: UDTPARAGRPID=45, PRIRULE=DSCP, PRI=18, PRIMTRANRSCTYPE=HQ,
PRIMPTLOADTH=100, PRIM2SECPTLOADRATH=1000;
ADD UDTPARAGRP: UDTPARAGRPID=46, PRIRULE=DSCP, PRI=18, PRIMTRANRSCTYPE=HQ,
PRIMPTLOADTH=100, PRIM2SECPTLOADRATH=1000;
ADD UDTPARAGRP: UDTPARAGRPID=47, PRIRULE=DSCP, PRI=18, PRIMTRANRSCTYPE=HQ,
PRIMPTLOADTH=100, PRIM2SECPTLOADRATH=1000;
ADD UDTPARAGRP: UDTPARAGRPID=48, PRIRULE=DSCP, PRI=0, PRIMTRANRSCTYPE=HQ,
PRIMPTLOADTH=100, PRIM2SECPTLOADRATH=1000;
//Modifying the DSCP value for GTP-U echo packets
MOD GTPU: TIMEOUTTH=5000, TIMEOUTCNT=3, DSCP=0, STATICCHK=ENABLE;
//Setting the DSCP value for IKE packets
SET IKECFG:IKELNM="IKE", IKEKLI=20, IKEKLT=60, DSCP=46;
```

4.3.1.3.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.3.1.4 IPv4/IPv6 Dual-Stack Data Configuration

- For details about how to prepare IPv4 transmission data (new model), see **4.3.1.1.1 Data Preparation**.
- For details about how to prepare IPv6 transmission data (new model), see **4.3.1.3.1 Data Preparation**.

4.3.1.5 IPv4 Activation Verification

The verification procedure is as follows:

Step 1 Run the **DSP ETHPORT** command to query the status of the Ethernet port.

Expected result:

- The values of **Port Status** and **Physical Layer Status** are **UP**, indicating that the physical port has been successfully activated.
- The values of **Local Speed** and **Peer Speed** are the same.
- The values of **Local Duplex** and **Peer Duplex** are the same.

Step 2 Query the route status.

- If a destination IP route is configured for the base station, run the **DSP IPRT** (old model)/**DSP IPROUTE4** (new model) command to check the IP address specified in the **DEVIP** (in the old model)/**IPADDR4** (new model) MO and route status.

  Expected result: The state of IP route is **Valid**, indicating that the route has taken effect.
- If a source IP route is configured for the base station, run the **DSP SRCIPRT** (old model)/**DSP SRCIPROUTE4** (new model) command to check the route status.

  Expected result: The value of **Valid State** is **Valid**, indicating that the route has taken effect.

Step 3 Run the **PING** command to ping the peer IP address, with the local IP address set to the corresponding IP address specified in the **DEVIP** (old model)/**IPADDR4** (new model) MO.

Expected result: The peer IP address can be pinged, indicating that the route and the IP address specified in the **DEVIP** (old model)/**IPADDR4** (new model) MO have taken effect.

----End

4.3.1.6 IPv6 Activation Verification

The verification procedure is as follows:

Step 1 Run the **DSP ETHPORT** command to query the status of the Ethernet port.

Expected result:

- The values of **Port Status** and **Physical Layer Status** are **UP**, indicating that the physical port has been successfully activated.
- The values of **Local Speed** and **Peer Speed** are the same.
- The values of **Local Duplex** and **Peer Duplex** are the same.

Step 2 Run the **DSP IPADDR6** command to check whether the IPv6 address takes effect.

Expected result: The configured IPv6 address has taken effect.

Step 3 (Optional) Run the **DSP IPROUTE6** or **DSP SRCIPROUTE6** command to check the route status.

Expected result: The value of **Route Status** is **Valid**, indicating that the route has taken effect.

Step 4 Run the **PING6** command to ping the peer IP address, with the local IP address set to the corresponding IPv6 address (**IPADDR6**).

Expected result: The peer IP address is successfully pinged, indicating that the IPv6 route and IPv6 address (**IPADDR6**) have taken effect.

----End

4.3.1.7 Network Monitoring

After common transmission data is deployed, you can monitor the performance by using the counters listed in **Table 4-4**, **Table 4-5**, **Table 4-6**, and **Table 4-7**.

Table 4-4 Ethernet port related counters

Counter ID	Counter Name
1542455297	**VS.FEGE.TxBytes**
1542455298	**VS.FEGE.TxPackets**
1542455299	**VS.FEGE.RxBytes**
1542455300	**VS.FEGE.RxPackets**
1542455301	**VS.FEGE.RxErrPackets**
1542455302	**VS.FEGE.TxMaxSpeed**
1542455303	**VS.FEGE.TxMinSpeed**
1542455304	**VS.FEGE.TxMeanSpeed**
1542455305	**VS.FEGE.RxMaxSpeed**
1542455306	**VS.FEGE.RxMinSpeed**
1542455307	**VS.FEGE.RxMeanSpeed**
1542460296	**VS.FEGE.TxTotalBW**
1542460297	**VS.FEGE.RxTotalBW**

Table 4-5 Interface related counters

Counter ID	Counter Name
1542460734	**VS.INTERFACE.IPv4TxBytes**
1542460735	**VS.INTERFACE.IPv4RxBytes**
1542460736	**VS.INTERFACE.IPv6TxBytes**
1542460737	**VS.INTERFACE.IPv6RxBytes**

Counter ID	Counter Name
1542460738	**VS.INTERFACE.IPv4TxPackets**
1542460739	**VS.INTERFACE.IPv4RxPackets**
1542460740	**VS.INTERFACE.IPv6TxPackets**
1542460741	**VS.INTERFACE.IPv6RxPackets**
1542460742	**VS.INTERFACE.IPv4TxMaxSpeed**
1542460744	**VS.INTERFACE.IPv4TxMeanSpeed**
1542460745	**VS.INTERFACE.IPv4TxMinSpeed**
1542460746	**VS.INTERFACE.IPv4RxMaxSpeed**
1542460747	**VS.INTERFACE.IPv4RxMeanSpeed**
1542460748	**VS.INTERFACE.IPv4RxMinSpeed**
1542460749	**VS.INTERFACE.IPv6TxMaxSpeed**
1542460750	**VS.INTERFACE.IPv6TxMeanSpeed**
1542460751	**VS.INTERFACE.IPv6TxMinSpeed**
1542460752	**VS.INTERFACE.IPv6RxMaxSpeed**
1542460753	**VS.INTERFACE.IPv6RxMeanSpeed**
1542460754	**VS.INTERFACE.IPv6RxMinSpeed**

Table 4-6 IP related counters

Counter ID	Counter Name
1542455326	**VS.IP.TxPackets**
1542455327	**VS.IP.TxBytes**
1542455328	**VS.IP.TxDropPkts**
1542455329	**VS.IP.TxDropBytes**
1542455330	**VS.IP.RxPackets**
1542455331	**VS.IP.RxBytes**
1542455332	**VS.IP.RxDropPkts**
1542455333	**VS.IP.RxDropBytes**
1542460727	**VS.IP.TxFragments**
1542460728	**VS.IP.ReassembleSuccessPkts**

Counter ID	Counter Name
1542460729	**VS.IP.ReassembleFailPkts**

Table 4-7 IPv6 related counters

Counter ID	Counter Name
1542455796	**VS.IPv6.TxPackets**
1542455797	**VS.IPv6.TxBytes**
1542455798	**VS.IPv6.TxDropPkts**
1542455799	**VS.IPv6.TxDropBytes**
1542455800	**VS.IPv6.RxPackets**
1542455801	**VS.IPv6.RxBytes**
1542455802	**VS.IPv6.RxDropPkts**
1542455803	**VS.IPv6.RxDropBytes**
1542460730	**VS.IPv6.TxFragments**
1542460731	**VS.IPv6.ReassembleSuccessPkts**
1542460732	**VS.IPv6.ReassembleFailPkts**

If any of the alarms listed in **Table 4-8** is reported after common transmission parameters are configured, check whether the configurations are correct by referring to the alarm help.

Table 4-8 Alarms related to common transmission parameters

Alarm ID	Alarm Name	Alarm Severity
ALM-25880	Ethernet Link Fault	Major
ALM-25885	IP Address Conflict	Minor
ALM-25947	IPv6 Address Conflict	Minor/Warning

4.3.2 Deployment of Service Interfaces

- For the operations of setting up S1 and X2 interfaces in endpoint configuration mode, see *X2 and S1 Self-Management in NSA Networking*.
- For the operations of setting up SA X2 interfaces in endpoint configuration mode, see *X2 Self-Management in SA Networking*.

- For the operations of setting up NG and Xn interfaces in endpoint configuration mode, see *NG and Xn Self-Management*.

The **RSCGRP** (old model)/**IPRSCGRP** (new model) MO is required for accurate management of base station transmission resources. If the MO is not configured, user-plane data is managed by the default transmission resource group. For details, see *Transmission Resource Management*.

It is recommended that GTP-U detection be enabled on the S1-U/NG-U interface to check the link connectivity over the S1-U/NG-U interface. GTP-U detection can be enabled by configuring the **GTPU** MO. For details about how to configure this MO, see **6.2.3.2.1 Data Preparation**.

4.3.3 Deployment of OM Channels

4.3.3.1 IPv4 Data Configuration

4.3.3.1.1 Data Preparation

After configuring common transmission data, you can deploy OM channels.

If the MAE functions as an NTP server, a separate route to the NTP server is not required. If a standalone NTP server is deployed but no routes are available (default route or network segment routes), a route to this NTP server must be configured. The configuration of a route to an FTP server is the same as that to an NTP server.

The following table describes the parameters that must be set in an **OMCH** MO to configure an OM channel.

Parameter Name	Parameter ID	Setting Notes
Standby Status	**OMCH**.***FLAG***	Set this parameter to **MASTER**. Before setting this parameter to **SLAVE**, enable the OM channel backup feature.
Local IP	**OMCH**.***IP***	Set this parameter based on the network plan.
Local Mask	**OMCH**.***MASK***	Set this parameter based on the network plan.
Peer IP	**OMCH**.***PEERIP***	Set this parameter based on the network plan.
Peer Mask	**OMCH**.***PEERMASK***	Set this parameter based on the network plan.
Binding Route	**OMCH**.***BRT***	• Set this parameter to **YES** when the OM channels use destination-based routes. • Set this parameter to **NO** when the OM channels use source-based routes.

Parameter Name	Parameter ID	Setting Notes
Route Index	**OMCH.*RTIDX***	Set this parameter based on the network plan.
Binding Secondary Route	**OMCH.*BINDSECONDARYRT***	Set this parameter based on the network plan.
Secondary Route Index	**OMCH.*SECONDARYRTIDX***	Set this parameter based on the network plan.

NOTICE

The IP addresses of the local maintenance port and the MAE cannot be on the same network segment. If they are on the same network segment and the local maintenance port and the MAE are connected to the same transmission network, the base station deployment will fail.

4.3.3.1.2 Using MML Commands

Before using MML commands, refer to **4.2 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

In the examples provided in this section, the **DEVIP** (in the old model)/**IPADDR4** (in the new model) MO has been configured for the OM channel. If no route is bound to the OM channel, the route from the base station to the MAE is to be configured. For details on route configuration, see **4.3.1 Deployment of Common Transmission Data**.

Run the **ADD OMCH** command to add an OM channel from the base station to the MAE.

Activation Command Examples

```
//Adding an OM channel from the base station to the MAE
ADD OMCH: FLAG=MASTER, IP="10.10.25.8", MASK="255.255.255.0", PEERIP="10.25.36.9",
PEERMASK="255.255.255.0", BEAR=IPV4, BRT=NO, CHECKTYPE=NONE;
```

4.3.3.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.3.3.2 IPv6 Data Configuration

4.3.3.2.1 Data Preparation

Ensure that the IPv6 route of the upper-layer application transmitted over the OM channel is reachable. For example, if the MAE IP address is used as the NTP server IP address, you do not need to configure a separate route for the NTP server. If a standalone NTP server is deployed but there are no routes (neither the default route nor network segment routes) to this NTP server, an IPv6 route to this NTP server must be configured. The configuration of a route to applications such as an FTP server is the same as that to an NTP server.

The following table describes the key parameters that must be set in an **OMCH** MO to configure an OM channel.

Parameter Name	Parameter ID	Setting Notes
Standby Status	**OMCH**.***FLAG***	Set this parameter to **MASTER**. Before setting this parameter to **SLAVE**, enable the OM channel backup feature.
Bearer Type	**OMCH**.***BEAR***	If the OM channel uses IPv6 transmission, set this parameter to **IPV6**.
Local IPv6 Address	**OMCH**.***IP6***	Set this parameter based on the network plan. This parameter can be set in either of the following ways: ● Set this parameter directly without the need of configuring the **IPADDR6** MO. The IP address takes effect on the loopback interface directly. This configuration method must be used when the main control board backup function is enabled. ● Configure an **IPADDR6** MO before setting this parameter. This method can be used when the IPv6 address of the OM channel must be an interface IP address. In non-IPsec networking, **Port Type** for the local IPv6 address can be set to **ETH**. In IPsec networking, **Port Type** for the local IPv6 address must be set to **LOOPINT** and the local IPv6 address serves as the inner IP address of the IPsec tunnel. The base station OM IPv6 address configured on the MAE topology view must be the same as this local IPv6 address.

Parameter Name	Parameter ID	Setting Notes
Peer IPv6 Address	**OMCH**.***PEERIP6***	Set this parameter based on the network plan. This address can be an MAE address or an MAE network segment address.
Peer IPv6 Address Prefix Length	**OMCH**.***PEERIP6PFXLEN***	Set this parameter based on the network plan.

4.3.3.2.2 Using MML Commands

Before using MML commands, refer to **4.2 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

This section assumes that the **IPADDR6**, and **IPROUTE6** or **SRCIPROUTE6** MOs have been configured for the OM channel.

Run the **ADD OMCH** command to add an OM channel from the base station to the MAE.

Activation Command Examples

Configuration method 1: Setting the **OMCH.IP6** parameter in the **OMCH** MO

```
//Adding an OM channel from the base station to the MAE
ADD OMCH: FLAG=MASTER, BEAR=IPV6, IP6="2001:db8:100:ad1:200:100:100:11",
PEERIP6="2001:db8:100:ad1:200:100:211:10", PEERIP6PFXLEN=124;
```

Configuration method 2: Configuring an **IPADDR6** MO and setting the **OMCH.IP6** parameter to the same value as the IP address specified in the **IPADDR6** MO (assuming that the interface with **ITFID** being **0** has been configured)

```
//Adding an IPv6 address for the base station
ADD IPADDR6: IPADDR6ID="NROM", ITFID=0, IPV6="2001:db8:100:ad1:200:100:100:11", PFXLEN=126;
//Adding an OM channel from the base station to the MAE
ADD OMCH: FLAG=MASTER, BEAR=IPV6, IP6="2001:db8:100:ad1:200:100:100:11",
PEERIP6="2001:db8:100:ad1:200:100:211:10", PEERIP6PFXLEN=124;
```

4.3.3.2.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.3.3.3 Activation Verification

The verification procedure is as follows:

Step 1 Run the **DSP OMCH** command to query the OM channel status.

Expected result: If the value of **OM Channel Status** is **Normal**, the OM channel status is normal.

Step 2 Log in to the MAE-Access. Choose **Topology** > **Main Topology** to check the base station topology.

Expected result: The base station topology is normal.

----End

4.3.3.4 Network Monitoring

After the OM channel is deployed, you can use counters listed in **Table 4-9** to monitor the performance.

Table 4-9 OMCH related counters

Counter ID	Counter Name
1542455826	**VS.OMCH.RxTotalPkts**
1542455827	**VS.OMCH.RxInSeqPkts**
1542455828	**VS.OMCH.RxWndProbePkts**
1542455829	**VS.OMCH.RxWndUpdatePkts**
1542455833	**VS.OMCH.RxDupPkts**
1542455834	**VS.OMCH.RxPartDupPkts**
1542455835	**VS.OMCH.RxOutOfOrderPkts**
1542455836	**VS.OMCH.RxAfterWndPkts**
1542455837	**VS.OMCH.RxAfterClosePkts**
1542455838	**VS.OMCH.RxAckPkts**
1542455839	**VS.OMCH.RxDupAckPkts**
1542455840	**VS.OMCH.RxTooMuchAckPkts**
1542455841	**VS.OMCH.TxTotalPkts**
1542455842	**VS.OMCH.TxUrgentPkts**
1542455843	**VS.OMCH.TxCtrlPkts**

Counter ID	Counter Name
1542455844	**VS.OMCH.TxWndProbePkts**
1542455845	**VS.OMCH.TxWndUpdatePkts**
1542455846	**VS.OMCH.TxDataPkts**
1542455847	**VS.OMCH.TxDataRexntPkts**
1542455848	**VS.OMCH.TxAckOnlyPkts**

If ALM-25901 Remote Maintenance Link Failure is reported after the OM channel is deployed, check whether the deployment is correct according to the alarm help.

5 Transmission Reliability

5.1 Ethernet Link Aggregation

5.1.1 Network Analysis

5.1.1.1 Benefits

This function improves the reliability of an Ethernet link.

5.1.1.2 Impacts

The impacts between this function and other functions are described in **5.1.2.2.3 Function Impacts**.

Link aggregation in load sharing mode improves transmission bandwidth efficiency.

The volumes of traffic transmitted to the bearer network through the member ports in an Ethernet trunk vary with load balancing modes.

5.1.2 Requirements

5.1.2.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-010060	Transmission Network Detection and Reliability Improvement	NR0S0TNDER00	per gNodeB

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.1.2.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.1.2.2.1 Prerequisite Functions

None

5.1.2.2.2 Mutually Exclusive Functions

None

5.1.2.2.3 Function Impacts

None

5.1.2.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

The main control boards UMPTe and UMPTg support this function. For details, see technical specifications of the BBU in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

This function does not depend on RF modules.

5.1.2.4 Networking

The base station is directly connected to a Layer 2/Layer 3 transmission device that supports link aggregation.

The base station provides multiple FE/GE/10GE/25GE ports and these ports are all FE ports or all GE/10GE/25GE ports. In addition, these ports must have the same settings of the following parameters: **ETHPORT**.***SPEED***, **ETHPORT**.***DUPLEX***, and **ETHPORT**.***FECMODE***.

5.1.2.5 Others

None

5.1.3 Operation and Maintenance

5.1.3.1 When to Use

This function is recommended when a base station is connected to Layer 2 or Layer 3 transmission devices that support link aggregation.

5.1.3.2 Data Configuration

5.1.3.2.1 Data Preparation

Table 5-1 Data to be prepared for link aggregation

Parameter Name	Parameter ID	Setting Notes
ETH Trunk Load Balancing Mode	**TRANSFUNCTIONSW**.***ETHTRKLBMODE***	Set this parameter to **LB_MODE_1**.
Trunk Type	**ETHTRK**.***LACP***	Set this parameter to **ENABLE**.

5.1.3.2.2 Using MML Commands (in the New Model)

Before using MML commands, refer to **5.1.2 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation:

Step 1 Run the **SET TRANSFUNCTIONSW** command to set the Ethernet trunk load balancing mode.

Step 2 Run the **ADD ETHTRUNK** command to add an Ethernet link aggregation group.

Step 3 Run the **ADD ETHTRUNKLNK** command with **Master Port Flag** set to **YES** to add a primary port to the Ethernet link aggregation group.

Step 4 Run the **ADD ETHTRUNKLNK** command with **Master Port Flag** set to **NO** to add another port to the Ethernet link aggregation group.

NOTE

The member ports in an Ethernet link aggregation group must have the same settings of the following parameters: **ETHPORT**.***SPEED***, **ETHPORT**.***DUPLEX***, and **ETHPORT**.***FECMODE***.

Step 5 Run the **ADD INTERFACE** command to add an interface to the Ethernet link aggregation group. In this step, set **Port Type** to **ETHTRK** and **Port ID** to the value of **Port ID** in the **ETHTRUNK** MO.

Step 6 Run the **ADD IPADDR4/ADD IPADDR6** command to add an IPv4/IPv6 address for the Ethernet port. In this step, set **Interface ID** to the value of **Interface ID** added in **Step 5**.

----End

Deactivation:

Step 1 Run the **RMV IPADDR4/RMV IPADDR6** command to remove the IPv4/IPv6 address.

Step 2 Run the **RMV INTERFACE** command to remove the interface.

Step 3 Run the **RMV ETHTRUNKLNK** command to remove the ports from the Ethernet link aggregation group. The primary port is the last to be removed.

Step 4 Run the **RMV ETHTRUNK** command to remove the Ethernet link aggregation group.

----End

Activation Command Examples

```
//Adding a link aggregation group
SET TRANSFUNCTIONSW: ETHTRKLBMODE=LB_MODE_1;
ADD ETHTRUNK: PORTID=0, LACP=ENABLE;
ADD ETHTRUNKLNK: ETHTRKPORTID=0, ETHTRUNKLNKID=0, ETHPORTID=1, FLAG=YES;
ADD ETHTRUNKLNK: ETHTRKPORTID=0, ETHTRUNKLNKID=1, ETHPORTID=3, FLAG=NO;
//IPv4 scenario
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETHTRK, PORTID=0, VLANID=100, VRFIDX=0, MTU4=1500,
ARPPROXY=ENABLE;
ADD IPADDR4: ITFID=0, IP="126.126.126.1", MASK="255.255.255.0";
//IPv6 scenario
ADD INTERFACE: ITFID=1, ITFTYPE=VLAN, PT=ETHTRK, PORTID=0, VLANID=200, VRFIDX=0, MTU6=1500,
IPV6SW=ENABLE;
ADD IPADDR6: IPADDR6ID="0", ITFID=1, IPV6="2001:DB8::1", PFXLEN=32;
```

Deactivation Command Examples

```
//Removing the link aggregation group
//IPv4 scenario
RMV IPADDR4: ITFID=0, IP="126.126.126.1";
RMV INTERFACE: ITFID=0;
//IPv6 scenario
RMV IPADDR6: IPADDR6ID="0";
RMV INTERFACE: ITFID=1;
RMV ETHTRUNKLNK: ETHTRKPORTID=0, ETHTRUNKLNKID=1;
RMV ETHTRUNKLNK: ETHTRKPORTID=0, ETHTRUNKLNKID=0;
RMV ETHTRUNK: PORTID=0;
```

5.1.3.2.3 Using MML Commands (in the Old Model)

Before using MML commands, refer to **5.1.2 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation:

Step 1 Run the **SET TRANSFUNCTIONSW** command to set the Ethernet trunk load balancing mode.

Step 2 Run the **ADD ETHTRK** command to add an Ethernet link aggregation group.

Step 3 Run the **ADD ETHTRKLNK** command with **Master Flag** set to **YES** to add a primary port to the Ethernet link aggregation group.

Step 4 Run the **ADD ETHTRKLNK** command with **Master Flag** set to **NO** to add another port to the Ethernet link aggregation group.

NOTE

The member ports in an Ethernet link aggregation group must have the same settings of the following parameters: **ETHPORT**.***SPEED***, **ETHPORT**.***DUPLEX***, and **ETHPORT**.***FECMODE***.

Step 5 Run the **ADD DEVIP** command to add a device IP address for the Ethernet port. In this step, set **Port Type** to **ETHTRK** and **Port No.** to the value of **Trunk No.** added in **Step 2**.

----End

Deactivation:

Step 1 Run the **RMV DEVIP** command to remove the device IP address.

Step 2 Run the **RMV ETHTRKLNK** to remove the ports from the link aggregation group. The primary port is the last to be removed.

Step 3 Run the **RMV ETHTRK** command to remove the link aggregation group.

----End

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Adding a link aggregation group
SET TRANSFUNCTIONSW: ETHTRKLBMODE=LB_MODE_1;
ADD ETHTRK: SN=7, SBT=BASE_BOARD, TN=0, LACP=ENABLE;
ADD ETHTRKLNK: SN=7, SBT=BASE_BOARD, TN=0, PN=1, FLAG=YES;
ADD ETHTRKLNK: SN=7, SBT=BASE_BOARD, TN=0, PN=3, FLAG=NO;
```



```
ADD DEVIP: VRFIDX=0, SN=7, SBT=BASE_BOARD, PT=ETHTRK, PN=0, IP="126.126.126.1",
MASK="255.255.255.0";
```

Deactivation Command Examples

```
//Removing the link aggregation group
RMV DEVIP: VRFIDX=0, SN=7, SBT=BASE_BOARD, PT=ETHTRK, PN=0, IP="126.126.126.1";
RMV ETHTRKLNK: SN=7, SBT=BASE_BOARD, TN=0, PN=3;
RMV ETHTRKLNK: SN=7, SBT=BASE_BOARD, TN=0, PN=1;
RMV ETHTRK: SN=7, SBT=BASE_BOARD, TN=0;
```

5.1.3.2.4 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.1.3.3 Activation Verification

Step 1 Run the **DSP ETHTRK** (in the old model)/**DSP ETHTRUNK** (in the new model) command to check the status of the configured Ethernet link aggregation group. If the value of **Ethernet Trunk Status** is **Up** and the value of **Number of Active Trunk Ports** is not 0, the Ethernet link aggregation function is normal and there are active ports in the group.

Step 2 Remove the optical fiber or Ethernet cable from an active port and run the **DSP ETHTRK** (in the old model)/**DSP ETHTRUNK** (in the new model) command to recheck the status of the Ethernet link aggregation group. In the command output, the number of this port is not displayed and the S1 interface is functional. Run the **DSP ETHTRKLNK** (in the old model)/**DSP ETHTRUNKLNK** (in the new model) command to check the status of the port from which the cable has been removed. The command output shows that the value of **Port Status** is **Down**.

Step 3 Reconnect the optical fiber or Ethernet cable to the port.

Step 4 Run the **DSP ETHTRK** (in the old model)/**DSP ETHTRUNK** (in the new model) and **DSP ETHTRKLNK** (in the old model)/**DSP ETHTRUNKLNK** (in the new model) commands. The command outputs show that the port to which the optical fiber or Ethernet cable is reconnected becomes active.

----End

5.1.3.4 Network Monitoring

After Ethernet link aggregation is deployed, you can use counters listed in **Table 5-2** to monitor the performance.

Table 5-2 Trunk related counters

Counter ID	Counter Name
1542456126	VS.TRUNK.TxPackets
1542456127	VS.TRUNK.RxPackets

Counter ID	Counter Name
1542456128	VS.TRUNK.RxErrPackets
1542460302	VS.TRUNK.TxTotalBW
1542460303	VS.TRUNK.RxTotalBW
1542460304	VS.TRUNK.RxMaxSpeed
1542460305	VS.TRUNK.RxMeanSpeed
1542460306	VS.TRUNK.TxMaxSpeed
1542460307	VS.TRUNK.TxMeanSpeed

If any of the alarms listed in **Table 5-3** is reported after Ethernet link aggregation is deployed, check whether trunk deployment is correct by referring to the alarm help.

Table 5-3 Alarms related to Ethernet link aggregation

Alarm ID	Alarm Name	Alarm Severity
ALM-25887	Ethernet Trunk Link Fault	Minor
ALM-25895	Ethernet Trunk Group Fault	Major

5.2 IP Route Backup

5.2.1 Network Analysis

5.2.1.1 Benefits

This function improves the reliability of an Ethernet link.

5.2.1.2 Impacts

The impacts between this function and other functions are described in **5.2.2.2.3 Function Impacts**.

This function has no impact on the network.

5.2.2 Requirements

5.2.2.1 Licenses

None

5.2.2.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.2.2.2.1 Prerequisite Functions

None

5.2.2.2.2 Mutually Exclusive Functions

None

5.2.2.2.3 Function Impacts

None

5.2.2.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards support this function. To learn which main control boards are NR-capable, see technical specifications of the BBU in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

This function does not depend on RF modules.

5.2.2.4 Networking

One physical port of the base station or two physical ports on the same board connect, through a Layer 2 network, to the router that supports active/standby gateways.

The base station supports source-based routing and destination-based routing. The routing policies vary depending on site conditions. For details, see *IPv4 Transmission* and *IPv6 Transmission*.

For IPv4 active and standby routes, BFD is enabled between the base station and gateway routers to detect the link status, and the BFD status is associated with the route status. In this way, route switchovers can be triggered when necessary.

IPv6 active and standby routes can be switched based on the status of the bound IPv6 BFD session or the physical port.

The active and standby IP routes can be configured only on the same main control board.

5.2.2.5 Others

A standby route to the base station is configured on both the active and standby gateways. Even if the active route is faulty, the gateway router can send packets to the base station through the standby route.

5.2.3 Operation and Maintenance

5.2.3.1 When to Use

This function is recommended when a base station is connected to Layer 3 routers that provide active/standby gateways.

5.2.3.2 Data Configuration (in the IPv4 New Model)

5.2.3.2.1 Data Preparation

The following table lists the key parameters that must be set in an **IPROUTE4** MO to configure IP route backup.

Parameter Name	Parameter ID	Setting Notes
Next Hop IP	**IPROUTE4.*NEXTHOP***	Set this parameter based on the network plan.
Preference	**IPROUTE4.*PREF***	Set this parameter based on the network plan. The active gateway takes precedence over the standby gateway. A smaller value of this parameter indicates a higher route priority.

The following table lists the parameters that must be set in a **BFD** MO to configure BFD.

Parameter Name	Parameter ID	Setting Notes
IP Version	**BFD.*IPVERSION***	Set this parameter to **IPV4**.

Parameter Name	Parameter ID	Setting Notes
Hop Type	**BFD**.***HT***	● The single-hop BFD session is used to perform point-to-point detection (generally in Layer 2 networking). ● The multi-hop BFD session is used to perform end-to-end connectivity check (generally in Layer 3 networking).
Source IP	**BFD**.***SRCIP***	This parameter specifies the source IP address of a BFD session. Set this parameter based on the network plan. ● A BFD session cannot be configured as a single-hop session if its source IP address is a logical IP address. ● Ensure that the **IPADDR4** MO corresponding to this IP address has been configured. ● This parameter must be set to a valid IP address and cannot be set to 0. It must be set to a device IP address (for example, the IP address of an Ethernet port), or a logical IP address (for example, the IP address of a loopback interface) of a specified board. It cannot be set to the OMCH IP address. When this parameter is set to a logical IP address, single-hop BFD sessions are not supported. This parameter must be set to a value different from that of the **BFD**.***DSTIP*** parameter.
Destination IP	**BFD**.***DSTIP***	This parameter specifies the destination IP address of the BFD session. Set this parameter based on the network plan. ● If single-hop BFD is used and the source IP address is configured on an Ethernet port, the source and destination IP addresses must be on the same network segment. ● The destination IP address of each BFD session must be unique.

Parameter Name	Parameter ID	Setting Notes
My Discriminator Generation Mode	**BFD**.***MYDSCRMGENMODE***	This parameter specifies the generation mode of my discriminator. Set this parameter based on the network plan. The following generation modes are supported: ● **STATIC_CONFIG**. In this mode, my discriminator is generated based on the configured value of **BFD**.***MYDISCREAMINATOR***. It is recommended that your discriminator not be set on the router. If your discriminator is set on the router, it needs to be the same as my discriminator of the base station. ● **DYNAMIC_GEN**. In this mode, my discriminator is generated randomly when negotiation of a session starts. ● **DYNAMIC_GEN_ENHANCE**. In this mode, my discriminator is generated randomly when negotiation of a session starts and, if my discriminator is the same as your discriminator, the base station discards this session and generates another my discriminator randomly to start negotiation. If your discriminator is set on the router, this parameter can only be set to **STATIC_CONFIG**.
My Discriminator	**BFD**.***MYDISCREAMINATOR***	This parameter specifies my discriminator of the BFD session. It is recommended that your discriminator not be set on the router. If your discriminator is set on the router, your discriminator needs to be the same as my discriminator of the base station. This parameter can be set only when the **BFD**.***MYDSCRMGENMODE*** parameter is set to **STATIC_CONFIG**.
Min TX Interval	**BFD**.***MINTI***	Set this parameter to **100**.
Min RX Interval	**BFD**.***MINRI***	Set this parameter to **100**.
Detection Multiplier	**BFD**.***DM***	Set this parameter to **3**.
Session Catalog	**BFD**.***CATLOG***	● If this parameter is set to **MAINTENANCE**, the BFD session is used only for performing a connectivity check. ● If this parameter is set to **RELIABILITY**, the BFD session is used to trigger route interlock. Route interlock enables the standby route to take over once the active route becomes faulty, and prevents service interruptions caused by route failures.

Parameter Name	Parameter ID	Setting Notes
DSCP	**BFD**.*DSCP*	Set this parameter based on the network plan.
BFD Authentication Switch	**BFD**.*BFDAUTHSW*	• Enable BFD authentication if high security is required for BFD control packets. • Otherwise, disable BFD authentication.
BFD Authentication Algorithm	**BFD**.*BFDAUTHTYPE*	Set this parameter based on the network plan. The following algorithms are supported: • MD5 • MeMD5 • SHA1 • MeSHA1 The base station and the peer device must negotiate to use the same algorithm.
BFD Authentication Key Chain ID	**BFD**.*KEYCHAINID*	Set this parameter to the same value as **BFDKEYCHAIN**.*KEYCHAINID*. Only one key chain is currently supported.

NOTE

MD5 and SHA1 are considered insufficiently secure in the industry. If these algorithms are used, transmitted data may be forged by attackers. It is recommended that the MeMD5 or MeSHA1 authentication algorithm be used to ensure the security of transmitted data.

5.2.3.2.2 Using MML Commands

Before using MML commands, refer to **5.2.2 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation:

Step 1 Run the **ADD IPROUTE4** command with **Route Type** set to **NEXTHOP**, **Next Hop IP** set to *the IP address of the active gateway*, and **Preference** set to **60**.

Step 2 Run the **ADD IPROUTE4** command with **Route Type** set to **NEXTHOP**, **Next Hop IP** set to *the IP address of the standby gateway*, and **Preference** set to **80**.

Step 3 Run the **ADD BFD** command with **My Discriminator Generation Mode** set to **STATIC_CONFIG**, **Hop Type** set to **SINGLE_HOP**, and **Session Catalog** set to **RELIABILITY** to start BFD for the active gateway router.

Step 4 Run the **ADD BFD** command with **My Discriminator Generation Mode** set to **STATIC_CONFIG**, **Hop Type** set to **SINGLE_HOP**, and **Session Catalog** set to **RELIABILITY** to start BFD for the standby gateway router.

Step 5 (Optional) Run the **SET GTRANSPARA** command with **Switch Back Time** set to **300**.

----End

NOTE

Set **Protocol Version** to the BFD session protocol version supported by the peer device, which can be either **DRAFT4** or **STANDARD**.

Deactivation:

Step 1 Run the **RMV BFD** command to remove the BFD session.

Step 2 Run the **RMV IPROUTE4** command to remove low-priority and high-priority IP routes. If the removal fails, perform operations as prompted.

----End

Reconfiguration:

Run the **MOD IPROUTE4** command to perform the reconfiguration.

Activation Command Examples

```
//Activating IP route backup for the base station
ADD IPROUTE4: RTIDX=0, DSTIP="10.10.10.10", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
NEXTHOP="10.10.12.2", PREF=60;
ADD IPROUTE4: RTIDX=1, DSTIP="10.10.10.10", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP,
NEXTHOP="10.10.13.2", PREF=80;
//Adding BFD sessions on the base station. For a single-hop BFD session, the source and destination IP
addresses must be on the same network segment.
ADD BFD: BFDSN=100, IPVERSION=IPV4, SRCIP="10.10.12.1", DSTIP="10.10.12.2",
MYDSCRMGENMODE=STATIC_CONFIG, MYDISCREAMINATOR=1, HT=SINGLE_HOP, CATLOG=RELIABILITY,
DSCP=48, VER=STANDARD, BFDAUTHSW=OFF;
ADD BFD: BFDSN=101, IPVERSION=IPV4, SRCIP="10.10.13.1", DSTIP="10.10.13.2",
MYDSCRMGENMODE=STATIC_CONFIG, MYDISCREAMINATOR=2, HT=SINGLE_HOP, CATLOG=RELIABILITY,
DSCP=48, VER=STANDARD, BFDAUTHSW=OFF;
SET GTRANSPARA: SBTIME=300;
```

Deactivation Command Examples

```
//Deactivating IP route backup for the base station
RMV BFD: BFDSN=100;
RMV BFD: BFDSN=101;
RMV IPROUTE4: RTIDX=0;
RMV IPROUTE4: RTIDX=1;
```

Reconfiguration Command Examples

```
//Modifying the IPv4 route
MOD IPROUTE4: RTIDX=0, PREF=70;
```

5.2.3.2.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.2.3.3 Data Configuration (in the IPv4 Old Model)

5.2.3.3.1 Data Preparation

The following table lists the key parameters that must be set in an **IPRT** MO to configure IP route backup.

Parameter Name	Parameter ID	Setting Notes
Next Hop IP	**IPRT**.***NEXTHOP***	Set this parameter to the IP address of the active/standby gateway.
Preference	**IPRT**.***PREF***	The active gateway takes precedence over the standby gateway. A smaller value of this parameter indicates a higher route priority.

5.2.3.3.2 Using MML Commands

Before using MML commands, refer to **5.2.2 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation:

Step 1 Run the **ADD IPRT** command with **Route Type** set to **NEXTHOP**, **Next Hop IP** set to *the IP address of the active gateway*, and **Preference** set to **60**.

Step 2 Run the **ADD IPRT** command with **Route Type** set to **NEXTHOP**, **Next Hop IP** set to *the IP address of the standby gateway*, and **Preference** set to **80**.

Step 3 Run the **ADD BFDSESSION** command with **My Discriminator Generation Mode** set to **STATIC_CONFIG**, **Hop Type** set to **SINGLE_HOP**, and **Session Catalog** set to **RELIABILITY** to start BFD for the active gateway router.

Step 4 Run the **ADD BFDSESSION** command with **My Discriminator Generation Mode** set to **STATIC_CONFIG**, **Hop Type** set to **SINGLE_HOP**, and **Session Catalog** set to **RELIABILITY** to start BFD for the standby gateway router.

Step 5 (Optional) Run the **SET GTRANSPARA** command with **Switch Back Time** set to **300**.

----End

NOTE

Set **Protocol Version** to the BFD session protocol version supported by the peer device, which can be either **DRAFT4** or **STANDARD**.

Deactivation:

Step 1 Run the **RMV BFDSESSION** command to disable BFD.

Step 2 Run the **RMV IPRT** command to remove low-priority and high-priority IP routes. If the removal fails, perform operations as prompted.

----End

Reconfiguration:

Run the **MOD IPRT** command to perform the reconfiguration.

Activation Command Examples

```
//Activating IP route backup for the base station
ADD IPRT: RTIDX=0, SN=7, SBT=BASE_BOARD, DSTIP="10.10.10.10", DSTMASK="255.255.255.255",
RTTYPE=NEXTHOP, NEXTHOP="10.10.12.2", PREF=60;
ADD IPRT: RTIDX=1, SN=7, SBT=BASE_BOARD, DSTIP="10.10.10.10", DSTMASK="255.255.255.255",
RTTYPE=NEXTHOP, NEXTHOP="10.10.13.2", PREF=80;
//Adding BFD sessions on the base station. For a single-hop BFD session, the source and destination IP
addresses must be on the same network segment.
ADD BFDSESSION: SN=7, BFDSN=0, SRCIP="10.10.12.1", DSTIP="10.10.12.2",
MYDSCRMGENMODE=STATIC_CONFIG, HT=SINGLE_HOP, CATLOG=RELIABILITY, DSCP=48, VER=STANDARD,
BFDAUTHSW=OFF;
ADD BFDSESSION: SN=7, BFDSN=1, SRCIP="10.10.13.1", DSTIP="10.10.13.2",
MYDSCRMGENMODE=STATIC_CONFIG, HT=SINGLE_HOP, CATLOG=RELIABILITY, DSCP=48, VER=STANDARD,
BFDAUTHSW=OFF;
SET GTRANSPARA: SBTIME=300;
```

Deactivation Command Examples

```
//Deactivating IP route backup for the base station
RMV BFDSESSION: SN=7, BFDSN=0;
RMV BFDSESSION: SN=7, BFDSN=1;
RMV IPRT: RTIDX=0;
RMV IPRT: RTIDX=1;
```

Reconfiguration Command Examples

```
//Modifying the IPv4 static route
MOD IPRT: RTIDX=0, PREF=70;
```

5.2.3.3.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.2.3.4 Data Configuration (IPv6)

5.2.3.4.1 Data Preparation

Configure destination-based IPv6 routes (when the active and standby IP routes are destination-based routes).

The following table describes the parameters that must be set in an **IPROUTE6** MO to configure an IPv6 route.

Parameter Name	Parameter ID	Setting Notes
Route Index	**IPROUTE6.*RTIDX***	Set this parameter based on the network plan.
Destination IPv6 Address	**IPROUTE6.*DSTIP***	The default route, of which the destination IPv6 address is 0::0 and the prefix length is 0, is not recommended. If the peer NE such as the core network equipment or peer eNodeB/gNodeB is located on the same network segment, an IP route to this network segment is recommended, with the destination IPv6 address being the network segment address and the prefix length being less than 128.
Prefix Length	**IPROUTE6.*PFXLEN***	Set this parameter based on the network plan.
Route Type	**IPROUTE6.*RTTYPE***	Set this parameter to **NEXTHOP**.
Next-Hop IPv6 Address	**IPROUTE6.*NEXTHOP***	Set this parameter to the IP address of the active/standby gateway.
Preference	**IPROUTE6.*PREF***	The active gateway takes precedence over the standby gateway. A smaller value of this parameter indicates a higher route priority.

Configure source-based IPv6 routes (when the active and standby IP routes are source-based routes).

The following table describes the parameters that must be set in an **SRCIPROUTE6** MO to configure a source-based IPv6 route.

Parameter Name	Parameter ID	Setting Notes
IPv6 Source Route Index	**SRCIPROUTE6.*SRCRTIDX***	Set this parameter based on the network plan.
Source IPv6 Address	**SRCIPROUTE6.*SRCIP***	Set this parameter based on the network plan.
Route Type	**SRCIPROUTE6.*RTTYPE***	Set this parameter to **NEXTHOP**.

Parameter Name	Parameter ID	Setting Notes
Next-Hop IPv6 Address	**SRCIPROUTE6**.***NEXTHOP***	Set this parameter to the IP address of the active/standby gateway connected to different physical ports.
Preference	**SRCIPROUTE6**.***PREF***	Set this parameter based on the network plan.

The following table lists the parameters that must be set in a **BFD** MO to configure BFD.

Parameter Name	Parameter ID	Setting Notes
IP Version	**BFD**.***IPVERSION***	Set this parameter to **IPV6**.
Hop Type	**BFD**.***HT***	● The single-hop BFD session is used to perform point-to-point detection (generally in Layer 2 networking). ● The multi-hop BFD session is used to perform end-to-end connectivity check (generally in Layer 3 networking).
Source IPv6 Address	**BFD**.***SRCIPV6***	This parameter specifies the source IPv6 address of a BFD session. Set this parameter based on the network plan. ● A BFD session cannot be configured as a single-hop session if its source IPv6 address is a logical IPv6 address. ● The source IPv6 address must have been configured in the **IPADDR6** MO. ● This parameter must be set to a valid IP address and cannot be set to 0. It must be set to a device IP address (for example, the IP address of an Ethernet port), or a logical IP address (for example, the IP address of a loopback interface) of a specified board. It cannot be set to the OMCH IP address. When this parameter is set to a logical IP address, single-hop BFD sessions are not supported. This parameter must be set to a value different from that of the **BFD**.***DSTIPV6*** parameter.
Destination IPv6 Address	**BFD**.***DSTIPV6***	This parameter specifies the destination IPv6 address of the BFD session. Set this parameter based on the network plan. If **Hop Type** is **SINGLE_HOP** and the source IPv6 address is the IPv6 address of an Ethernet port, the source and destination IPv6 addresses must be on the same network segment.

Parameter Name	Parameter ID	Setting Notes
My Discriminator Generation Mode	**BFD**.***MYDSCRMGENMODE***	This parameter specifies the generation mode of my discriminator. Set this parameter based on the network plan. The following generation modes are supported: ● **STATIC_CONFIG**. In this mode, my discriminator is generated based on the configured value of **BFD**.***MYDISCREAMINATOR***. It is recommended that your discriminator not be set on the router. If your discriminator is set on the router, your discriminator needs to be the same as my discriminator of the base station. ● **DYNAMIC_GEN**. In this mode, my discriminator is generated randomly when negotiation of a session starts. ● **DYNAMIC_GEN_ENHANCE**. In this mode, my discriminator is generated randomly when negotiation of a session starts and, if my discriminator is the same as your discriminator, the base station discards this session and generates another my discriminator randomly to start negotiation. If your discriminator is set on the router, this parameter can only be set to **STATIC_CONFIG**.
My Discriminator	**BFD**.***MYDISCREAMINATOR***	This parameter specifies my discriminator of the BFD session. It is recommended that your discriminator not be set on the router. If your discriminator is set on the router, your discriminator needs to be the same as my discriminator of the base station. This parameter can be set only when the **BFD**.***MYDSCRMGENMODE*** parameter is set to **STATIC_CONFIG**.
Min TX Interval	**BFD**.***MINTI***	Set this parameter to **100**.
Min RX Interval	**BFD**.***MINRI***	Set this parameter to **100**.
Detection Multiplier	**BFD**.***DM***	Set this parameter to **3**.

Parameter Name	Parameter ID	Setting Notes
Session Catalog	**BFD**.*CATLOG*	• If this parameter is set to **MAINTENANCE**, the BFD session is used only for performing a connectivity check. • If this parameter is set to **RELIABILITY**, the BFD session is used to trigger route interlock. Route interlock enables the standby route to take over once the active route becomes faulty, and prevents service interruptions caused by route failures.
DSCP	**BFD**.*DSCP*	Set this parameter based on the network plan.
BFD Authentication Switch	**BFD**.*BFDAUTHSW*	• Enable BFD authentication if high security is required for BFD control packets. • Otherwise, disable BFD authentication.
BFD Authentication Algorithm	**BFD**.*BFDAUTHTYPE*	Set this parameter based on the network plan. The following algorithms are supported: • SHA1 • MeSHA1 The base station and the peer device must negotiate to use the same algorithm.
BFD Authentication Key Chain ID	**BFD**.*KEYCHAINID*	Set this parameter to the same value as **BFDKEYCHAIN**.*KEYCHAINID*. Only one key chain is currently supported.

5.2.3.4.2 Using MML Commands

Before using MML commands, refer to **5.2.2 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

When IP route backup uses destination-based routes:

Activation:

Step 1 Run the **ADD IPROUTE6** command with **Route Type** set to **NEXTHOP**, **Next-Hop IPv6 Address** set to *the IP address of the active gateway*, and **Preference** set to **60**.

Step 2 Run the **ADD IPROUTE6** command with **Route Type** set to **NEXTHOP**, **Next-Hop IPv6 Address** set to *the IP address of the standby gateway*, and **Preference** set to **80**.

Step 3 Run the **ADD BFD** command with **My Discriminator Generation Mode** set to **STATIC_CONFIG**, **Hop Type** set to **SINGLE_HOP**, and **Session Catalog** set to **RELIABILITY** to start BFD for the active gateway router.

Step 4 Run the **ADD BFD** command with **My Discriminator Generation Mode** set to **STATIC_CONFIG**, **Hop Type** set to **SINGLE_HOP**, and **Session Catalog** set to **RELIABILITY** to start BFD for the standby gateway router.

Step 5 (Optional) Run the **SET GTRANSPARA** command with **Switch Back Time** set to **300**.

----End

Deactivation:

Step 1 Run the **RMV BFD** command to remove the BFD session.

Step 2 Run the **RMV IPROUTE6** command to remove low-priority and high-priority IP routes. If the removal fails, perform operations as prompted.

----End

Reconfiguration:

Run the **MOD IPROUTE6** command to perform the reconfiguration.

When IP route backup uses source-based routes:

Activation:

Step 1 Run the **ADD IPROUTE6** command with **Route Type** set to **NEXTHOP**, **Next-Hop IPv6 Address** set to *the IP address of the active gateway*, and **Preference** set to **60**.

Step 2 Run the **ADD IPROUTE6** command with **Route Type** set to **NEXTHOP**, **Next-Hop IPv6 Address** set to *the IP address of the standby gateway*, and **Preference** set to **80**.

Step 3 Run the **ADD SRCIPROUTE6** command with **Route Type** set to **NEXTHOP**, **Next-Hop IPv6 Address** set to *the IP address of the active gateway*, and **Preference** set to **60**.

Step 4 Run the **ADD SRCIPROUTE6** command with **Route Type** set to **NEXTHOP**, **Next-Hop IPv6 Address** set to *the IP address of the standby gateway*, and **Preference** set to **80**.

----End

Deactivation:

Step 1 Run the **RMV BFD** command to remove the BFD session.

Step 2 Run the **RMV SRCIPROUTE6** command to remove low-priority and high-priority IP routes. If the removal fails, perform operations as prompted.

----End

Reconfiguration:

Run the **MOD SRCIPROUTE6** command to perform the reconfiguration.

Activation Command Examples

When IP route backup uses destination-based routes:

```
//Activating IP route backup for the base station
ADD IPROUTE6: RTIDX=0, DSTIP="2001:db8:100:ad1:200:100:3001:0", PFXLEN=112, RTTYPE=NEXTHOP,
NEXTHOP="2001:db8:100:ad1:200:100:100:1", PREF=60;
ADD IPROUTE6: RTIDX=1, DSTIP="2001:db8:100:ad1:200:100:3001:0", PFXLEN=112, RTTYPE=NEXTHOP,
NEXTHOP="2001:db8:100:ad1:200:100:200:1", PREF=80;
//Adding BFD sessions on the base station. For a single-hop BFD session, the source and destination IP
addresses must be on the same network segment.
ADD BFD: BFDSN=100, IPVERSION=IPV6, SRCIPV6="2001:db8:100:ad1:200:100:100:12",
DSTIPV6="2001:db8:100:ad1:200:100:100:1", MYDSCRMGENMODE=STATIC_CONFIG,
MYDISCREAMINATOR=1, HT=SINGLE_HOP, CATLOG=RELIABILITY, DSCP=48, VER=STANDARD,
BFDAUTHSW=OFF;
ADD BFD: BFDSN=101, IPVERSION=IPV6, SRCIPV6="2001:db8:100:ad1:200:100:200:12",
DSTIPV6="2001:db8:100:ad1:200:100:200:1", MYDSCRMGENMODE=STATIC_CONFIG,
MYDISCREAMINATOR=2, HT=SINGLE_HOP, CATLOG=RELIABILITY, DSCP=48, VER=STANDARD,
BFDAUTHSW=OFF;
SET GTRANSPARA: SBTIME=300;
```

When IP route backup uses source-based routes:

```
//Activating IP route backup for the base station
ADD SRCIPROUTE6: SRCRTIDX=0, SRCIP="2001:db8:100:ad1:200:100:200:100", RTTYPE=NEXTHOP,
NEXTHOP="2001:db8:100:ad1:200:100:100:1", PREF=60;
ADD SRCIPROUTE6: SRCRTIDX=1, SRCIP="2001:db8:100:ad1:200:100:200:100", RTTYPE=NEXTHOP,
NEXTHOP="2001:db8:100:ad1:200:100:200:1", PREF=80;
```

Deactivation Command Examples

When IP route backup uses destination-based routes:

```
//Deactivating IP route backup and removing the BFD session for the base station
RMV BFD: BFDSN=100;
RMV BFD: BFDSN=101;
RMV IPROUTE6: RTIDX=0;
RMV IPROUTE6: RTIDX=1;
```

When IP route backup uses source-based routes:

```
//Deactivating IP route backup and removing the BFD session for the base station
RMV SRCIPROUTE6: SRCRTIDX=0;
RMV SRCIPROUTE6: SRCRTIDX=1;
```

Reconfiguration Command Examples

When IP route backup uses destination-based routes:

```
MOD IPROUTE6: RTIDX=0, PREF=70;
```

When IP route backup uses source-based routes:

```
MOD SRCIPROUTE6: SRCRTIDX=0, PREF=70;
```

5.2.3.4.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.2.3.5 Activation Verification

5.2.3.5.1 Activation Verification for IPv4

Ensure that the active and standby IP routes are normal before verifying the IP route backup function. Run the **DSP IPRT** (in the old model)/**DSP IPROUTE4** (in the new model) command to check whether the active and standby IP routes are in the routing table. If the active and standby IP routes are in the routing table, both IP routes are functional, and the verification can continue. The verification procedure is as follows:

Step 1 Check the active IP route. Run the **TRACERT** command to check the active route. In the command output, the route with the first-hop IP address being the next-hop IP address is the active route.

Step 2 Cause a fault in the active IP route. Run the **DSP IPRT** (in the old model)/**DSP IPROUTE4** (in the new model) command to check that the active route is invalid.

Step 3 Verify the switchover. Run the **TRACERT** command. The switchover is successful if the first hop IP address is the next hop IP address of the standby IP route.

Step 4 Restore the transmission link of the IP route with the higher priority.

----End

5.2.3.5.2 Activation Verification for IPv6

To observe the active and standby routes of the base station, trigger a route fault and check whether services can be switched over to the other route. If the two IP routes are both functional, data is transmitted on the IP route with the higher priority.

When IP route backup uses destination-based routes:

Ensure that the active and standby IP routes are normal before verifying the IP route backup function. Run the **DSP IPROUTE6** command to check whether the active and standby routes are both in the route forwarding table. If the active and standby IP routes are in the routing table, both IP routes are functional, and the verification can continue. The verification procedure is as follows:

Step 1 Check the active IP route. Run the **TRACERT6** command to check the active route. In the command output, the route with the first-hop IP address being the next-hop IP address is the active route.

Step 2 Cause a fault in the active IP route. Run the **DSP IPROUTE6** command to check that the active route is invalid.

Step 3 Verify the switchover. Run the **TRACERT6** command. The switchover is successful if the first hop IP address is the next hop IP address of the standby IP route.

Step 4 Restore the transmission link of the IPv6 route with the higher priority.

----End

When IP route backup uses source-based routes:

Ensure that the active and standby IP routes are normal before verifying the IP route backup function. Run the **DSP SRCIPROUTE6** command to check whether the active and standby routes are both in the route forwarding table. If the active and standby IP routes are in the routing table, both IP routes are functional, and the verification can continue. The verification procedure is as follows:

Step 1 Check the active IP route. Run the **TRACERT6** command to check the active route. In the command output, the route with the first-hop IP address being the next-hop IP address is the active route.

Step 2 Cause a fault in the active IP route. Run the **DSP SRCIPROUTE6** command to check that the active route is invalid.

Step 3 Verify the switchover. Run the **TRACERT6** command. The switchover is successful if the first hop IP address is the next hop IP address of the standby IP route.

Step 4 Restore the transmission link of the source-based IPv6 route with the higher priority.

----End

5.2.3.6 Network Monitoring

None

5.3 OM Channel Backup

5.3.1 Network Analysis

5.3.1.1 Benefits

This function ensures the reliability of an OM channel.

5.3.1.2 Impacts

The impacts between this function and other functions are described in **5.3.2.2.3 Function Impacts**.

This function has no impact on the network.

5.3.2 Requirements

5.3.2.1 Licenses

Feature ID	Feature Name	License Control Item	Sales Unit
FOFD-010060	Transmission Network Detection and Reliability Improvement	NR0S0TNDER00	per gNodeB

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.3.2.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.2.1 Prerequisite Functions

None

5.3.2.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Source-based routing	None	*IP NR Engineering Guide*	An OMCH cannot be bound to a source-based IP route.

5.3.2.2.3 Function Impacts

None

5.3.2.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards support this function. To learn which main control boards are NR-capable, see technical specifications of the BBU in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

This function does not depend on RF modules.

5.3.2.4 Networking

Two OM channel IP addresses must be planned on the gNodeB side for different transmission links.

OM channels cannot be bound to the active and standby IPv6 routes in the 6in4 IPsec scenario. 6in4 IPsec indicates that the inner IP header is IPv6 and the outer IP header is IPv4. For details about the 6in4 IPsec, see *IPsec*.

5.3.2.5 Others

The MAE must be configured with the IP addresses of the active and standby OM channels and the bound routes.

5.3.3 Operation and Maintenance

5.3.3.1 Data Configuration (IPv4)

5.3.3.1.1 Data Preparation

The following table lists the data to be prepared when the OM channel switchback function is not required.

Table 5-4 Data to be prepared for OM channel backup

Parameter Name	Parameter ID	Setting Notes
Standby Status	**OMCH.*FLAG***	Set this parameter to **MASTER** for the active OM channel and to **SLAVE** for the standby OM channel.
Local IP	**OMCH.*IP***	Set this parameter to the local IP address of the active OM channel for the active OM channel and to the local IP address of the standby OM channel for the standby OM channel.

Parameter Name	Parameter ID	Setting Notes
Local Mask	**OMCH**.***MASK***	Set this parameter based on the network plan.
Peer IP	**OMCH**.***PEERIP***	Set this parameter to the MAE address.
Peer Mask	**OMCH**.***PEERMASK***	Set this parameter based on the network plan.
Binding Route	**OMCH**.***BRT***	● Set this parameter to **YES** when the OM channels use destination-based routes. ● Set this parameter to **NO** when the OM channels use source-based routes.
Route Index	**OMCH**.***RTIDX***	Set this parameter to the index of the route destined for the active MAE.
Binding Secondary Route	**OMCH**.***BINDSECONDARYRT***	Set this parameter to **YES** if the MAE uses the remote HA system and the IP address of the standby MAE is beyond the network segment range of the route bound to the OM channel. Otherwise, set this parameter to **NO**.
Secondary Route Index	**OMCH**.***SECONDARYRTIDX***	Set this parameter to the index of the route to the standby MAE. This parameter is valid only when **Binding Secondary Route** is set to **YES**.

When the OM channel switchback function is required:

The following table describes the key parameters that must be set in the **GTRANSPARA** MO to set the global transmission parameters of an NE.

Table 5-5 Data to be prepared for OM channel switchback

Parameter Name	Parameter ID	Setting Notes
OMCH Switch Back	**GTRANSPARA**.***OMCHSWITCHBACK***	If OM channel switchback is required, set this parameter to **ENABLE**.
OMCH Switch Back Wait Time	**GTRANSPARA**.***OMCHSBWAITTIME***	Set this parameter to its default value, 60, in seconds.

The following table describes the key parameters that must be set in the **OMCH** MO to set an OM channel.

Table 5-6 Data to be prepared for OM channel backup

Parameter Name	Parameter ID	Setting Notes
Standby Status	**OMCH**.***FLAG***	● Set this parameter to **MASTER** for the active OM channel. ● Set this parameter to **SLAVE** for the standby OM channel.
Local IP	**OMCH**.***IP***	● Set this parameter to the IP address of the active OM channel when the active OM channel is used. ● Set this parameter to the local IP address of the standby OM channel when the standby OM channel is used.
Local Mask	**OMCH**.***MASK***	Set this parameter based on the network plan.
Peer IP	**OMCH**.***PEERIP***	Set this parameter to the MAE address.
Peer Mask	**OMCH**.***PEERMASK***	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Check Type	**OMCH**.***CHECKTYPE***	Set this parameter to **AUTO_UDPSESSION** for only the active OM channel. If this parameter is set to **AUTO_UDPSESSION** for the active OM channel, this parameter must be set to **NONE** for the standby OM channel.

The following table lists the data to be prepared when **GTRANSPARA**.***TRANSCFGMODE*** is set to **OLD**.

The following table describes the key parameters that must be set in an **SRCIPRT** MO to configure a source-based IP route.

Table 5-7 Data to be prepared for the source IP routes of active and standby OM channels

Parameter Name	Parameter ID	Setting Notes
Source Route Index	**SRCIPRT**.***SRCRTIDX***	Set this parameter based on the network plan.
Source IP Address	**SRCIPRT**.***SRCIP***	When OM channel backup is used, set this parameter to the local IP address of the OM channel.
Subboard Type	**SRCIPRT**.***SBT***	Set this parameter to **BASE_BOARD(Base Board)**.
Route Type	**SRCIPRT**.***RTTYPE***	Set this parameter to **NEXTHOP(Next Hop)** in Ethernet scenarios.

Parameter Name	Parameter ID	Setting Notes
Next Hop IP	**SRCIPRT**.***NEXTHOP***	This parameter is valid only when the **SRCIPRT**.***RTTYPE*** parameter is set to **NEXTHOP(Next Hop)**. Set this parameter based on the network plan. Generally, this parameter is set to the IP address of the gateway on the transmission network connecting to the base station.

The following table lists the data to be prepared when **GTRANSPARA**.***TRANSCFGMODE*** is set to **NEW**.

The following table describes the key parameters that must be set in the **SRCIPROUTE4** MO to configure source IPv4 routes.

Table 5-8 Data to be prepared for the source IPv4 routes of active and standby OM channels

Parameter Name	Parameter ID	Setting Notes
Source Route Index	**SRCIPROUTE4**.***SRCRTIDX***	Set this parameter based on the network plan.
Source IP Address	**SRCIPROUTE4**.***SRCIP***	When OM channel backup is used, set this parameter to the local IP address of the OM channel.
Route Type	**SRCIPROUTE4**.***RTTYPE***	Set this parameter to **NEXTHOP(Next Hop)** in Ethernet scenarios.

Parameter Name	Parameter ID	Setting Notes
Next Hop IP	**SRCIPROUTE4**.***NEXTHOP***	This parameter is valid only when the **SRCIPROUTE4**.***RTTYPE*** parameter is set to **NEXTHOP(Next Hop)** for the source IPv4 route. Set this parameter based on the network plan. Generally, this parameter is set to the IP address of the gateway on the transmission network connecting to the base station.

5.3.3.1.2 Using MML Commands

Before using MML commands, refer to **5.3.2 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

When the OM channel switchback function is not required:

Activation:

Step 1 Run the **ADD OMCH** command with **Standby Status** set to **MASTER**, **Local IP** set to *the IP address of the active OM channel*, **Binding Route** set to **YES**, and **Route Index** set to *the index of the route bound to the active OM channel*.

If the MAE uses the remote HA system and the IP address of the standby MAE is beyond the network segment range of the route bound to the active OMCH, set **Binding Secondary Route** to **YES** and **Secondary Route Index** to *the index of the route from the active OM channel to the standby MAE*.

Step 2 Run the **ADD OMCH** command with **Standby Status** set to **SLAVE**, **Local IP** to *the IP address of the standby OM channel*, **Binding Route** to **YES**, and **Route Index** to *the index of the route bound to the standby OM channel*.

If the MAE uses the remote HA system and the IP address of the standby MAE is beyond the network segment range of the route bound to the standby OMCH, set **Binding Secondary Route** to **YES** and **Secondary Route Index** to *the index of the route from the standby OM channel to the standby MAE*.

----End

Deactivation:

Run the **RMV OMCH: FLAG=SLAVE** command to remove the standby OM channel.

Reconfiguration:

If the OM channel route needs to be reconfigured, run the **MOD OMCH** command on the base station side.

When the OM channel switchback function is required:

Activation:

Step 1 Run the **SET GTRANSPARA** command with **OMCH Switch Back** set to **ENABLE** and **OMCH Switch Back Wait Time** set to **60**.

Step 2 Configure the source-based IP route used by the OM channel.

Configuration when the MAE uses the remote HA system:

- When **GTRANSPARA**.***TRANSCFGMODE*** is set to **OLD**: Run the **ADD SRCIPRT** command with **SRCIP** set to the value of **OMCH**.***IP***.
- When **GTRANSPARA**.***TRANSCFGMODE*** is set to **NEW**: Run the **ADD SRCIPROUTE4** command (new model) with **SRCIP** set to the value of **OMCH**.***IP***.

Step 3 Run the **ADD OMCH** command with **Standby Status** set to **MASTER**, **Local IP** set to *the IP address of the active OM channel*, and **Check Type** set to **AUTO_UDPSESSION**.

Step 4 Run the **ADD OMCH** command with **Standby Status** set to **SLAVE**, **Local IP** set to *the IP address of the standby OM channel*, and **Check Type** set to **NONE**.

----End

Deactivation:

Run the **RMV OMCH: FLAG=SLAVE** command to remove the standby OM channel.

When the OM channel switchback function is configured: Run the **SET GTRANSPARA: OMCHSWITCHBACK=DISABLE** command to disable the OM channel switchback function.

Reconfiguration:

If the OM channel route needs to be reconfigured, run the **MOD OMCH** command on the base station side.

To change the OM channel binding mode to the OM channel switchback, perform the following steps:

Step 1 This reconfiguration may interrupt the OM channel. You are advised to run the **CFM CB** command to enable the automatic configuration data rollback function before data reconstruction.

Step 2 Run the **SET GTRANSPARA** command to turn on the OM channel switchback switch.

Step 3 Run the **ADD SRCIPRT** (old model)/**ADD SRCIPROUTE4** (new model) command to add the source-based IP routes used by the active and standby OM channels.

Step 4 Run the **MOD OMCH** command with **Check Type** of the active OM channel set to **AUTO_UDPSESSION**, **Check Type** of the standby OM channel set to **NONE**, and **Binding Route** set to **NO**.

----End

Activation Command Examples

When the old transmission configuration model is used:

When the OM channel switchback function is not required:

Activating OM channel backup when the MAE does not use the remote HA system

```
ADD IPRT: RTIDX=0, SN=7, SBT=BASE_BOARD, DSTIP="100.100.100.1", DSTMASK="255.255.255.0",
RTTYPE=NEXTHOP, NEXTHOP="10.10.12.2";
ADD IPRT: RTIDX=1, SN=7, SBT=BASE_BOARD, DSTIP="100.100.100.1", DSTMASK="255.255.255.0",
RTTYPE=NEXTHOP, NEXTHOP="10.10.13.2";
ADD OMCH: FLAG=MASTER, IP="10.10.10.1", MASK="255.255.255.255", PEERIP="100.100.100.1",
PEERMASK="255.255.255.0", BEAR=IPV4, BRT=YES, RTIDX=0, BINDSECONDARYRT=NO, CHECKTYPE=NONE;
ADD OMCH: FLAG=SLAVE, IP="10.10.11.1", MASK="255.255.255.255", PEERIP="100.100.100.1",
PEERMASK="255.255.255.0", BEAR=IPV4, BRT=YES, RTIDX=1, BINDSECONDARYRT=NO, CHECKTYPE=NONE;
```

Activating OM channel backup when the MAE uses the remote HA system

```
ADD IPRT: RTIDX=0, SN=7, SBT=BASE_BOARD, DSTIP="100.100.100.1", DSTMASK="255.255.255.0",
RTTYPE=NEXTHOP, NEXTHOP="10.10.12.2";
ADD IPRT: RTIDX=1, SN=7, SBT=BASE_BOARD, DSTIP="100.100.100.1", DSTMASK="255.255.255.0",
RTTYPE=NEXTHOP, NEXTHOP="10.10.13.2";
ADD IPRT: RTIDX=2, SN=7, SBT=BASE_BOARD, DSTIP="100.100.200.1", DSTMASK="255.255.255.0",
RTTYPE=NEXTHOP, NEXTHOP="10.10.12.2";
ADD IPRT: RTIDX=3, SN=7, SBT=BASE_BOARD, DSTIP="100.100.200.1", DSTMASK="255.255.255.0",
RTTYPE=NEXTHOP, NEXTHOP="10.10.13.2";
ADD OMCH: FLAG=MASTER, IP="10.10.10.1", MASK="255.255.255.255", PEERIP="100.100.100.1",
PEERMASK="255.255.255.0", BEAR=IPV4, BRT=YES, RTIDX=0, BINDSECONDARYRT=YES,
SECONDARYRTIDX=2, CHECKTYPE=NONE;
ADD OMCH: FLAG=SLAVE, IP="10.10.11.1", MASK="255.255.255.255", PEERIP="100.200.100.1",
PEERMASK="255.255.255.0", BEAR=IPV4, BRT=YES, RTIDX=1, BINDSECONDARYRT=YES,
SECONDARYRTIDX=3, CHECKTYPE=NONE;
```

When the OM channel switchback function is required:

```
//Activating OM channel backup when the MAE uses the remote HA system or not
SET GTRANSPARA: OMCHSWITCHBACK=ENABLE, OMCHSBWAITTIME=60;
ADD SRCIPRT: SRCRTIDX=0, SN=7, SBT=BASE_BOARD, SRCIP="10.10.10.1", RTTYPE=NEXTHOP,
NEXTHOP="10.10.12.2";
ADD SRCIPRT: SRCRTIDX=1, SN=7, SBT=BASE_BOARD, SRCIP="10.10.11.1", RTTYPE=NEXTHOP,
NEXTHOP="10.10.13.2";
ADD OMCH: FLAG=MASTER, IP="10.10.10.1", MASK="255.255.255.255", PEERIP="100.100.100.1",
PEERMASK="255.255.255.0", BEAR=IPV4, CHECKTYPE= AUTO_UDPSESSION;
ADD OMCH: FLAG=SLAVE, IP="10.10.11.1", MASK="255.255.255.255", PEERIP="100.100.100.1",
PEERMASK="255.255.255.0", BEAR=IPV4, CHECKTYPE=NONE;
```

When the new transmission configuration model is used:

When the OM channel switchback function is not required:

Activating OM channel backup when the MAE does not use the remote HA system

```
ADD IPROUTE4: RTIDX=0, DSTIP="100.100.100.1", DSTMASK="255.255.255.0", RTTYPE=NEXTHOP,
NEXTHOP="10.10.12.2";
ADD IPROUTE4: RTIDX=1, DSTIP="100.100.100.1", DSTMASK="255.255.255.0", RTTYPE=NEXTHOP,
NEXTHOP="10.10.13.2";
ADD OMCH: FLAG=MASTER, IP="10.10.10.1", MASK="255.255.255.255", PEERIP="100.100.100.1",
PEERMASK="255.255.255.0", BEAR=IPV4, BRT=YES, RTIDX=0, BINDSECONDARYRT=NO, CHECKTYPE=NONE;
ADD OMCH: FLAG=SLAVE, IP="10.10.11.1", MASK="255.255.255.255", PEERIP="100.100.100.1",
PEERMASK="255.255.255.0", BEAR=IPV4, BRT=YES, RTIDX=1, BINDSECONDARYRT=NO, CHECKTYPE=NONE;
```


Activating OM channel backup when the MAE uses the remote HA system

```
ADD IPROUTE4: RTIDX=0, DSTIP="100.100.100.1", DSTMASK="255.255.255.0", RTTYPE=NEXTHOP,
NEXTHOP="10.10.12.2";
ADD IPROUTE4: RTIDX=1, DSTIP="100.100.100.1", DSTMASK="255.255.255.0", RTTYPE=NEXTHOP,
NEXTHOP="10.10.13.2";
ADD IPROUTE4: RTIDX=2, DSTIP="100.100.200.1", DSTMASK="255.255.255.0", RTTYPE=NEXTHOP,
NEXTHOP="10.10.12.2";
ADD IPROUTE4: RTIDX=3, DSTIP="100.100.200.1", DSTMASK="255.255.255.0", RTTYPE=NEXTHOP,
NEXTHOP="10.10.13.2";
ADD OMCH: FLAG=MASTER, IP="10.10.10.1", MASK="255.255.255.255", PEERIP="100.100.100.1",
PEERMASK="255.255.255.0", BEAR=IPV4, BRT=YES, RTIDX=0, BINDSECONDARYRT=YES,
SECONDARYRTIDX=2, CHECKTYPE=NONE;
ADD OMCH: FLAG=SLAVE, IP="10.10.11.1", MASK="255.255.255.255", PEERIP="100.200.100.1",
PEERMASK="255.255.255.0", BEAR=IPV4, BRT=YES, RTIDX=1, BINDSECONDARYRT=YES,
SECONDARYRTIDX=3, CHECKTYPE=NONE;
```

When the OM channel switchback function is required:

```
//Activating OM channel backup when the MAE uses the remote HA system or not
SET GTRANSPARA: OMCHSWITCHBACK=ENABLE, OMCHSBWAITTIME=60;
ADD SRCIPROUTE4: SRCRTIDX=0, SRCIP="10.10.10.1", RTTYPE=NEXTHOP, NEXTHOP="10.10.12.2";
ADD SRCIPROUTE4: SRCRTIDX=0, SRCIP="10.10.11.1", RTTYPE=NEXTHOP, NEXTHOP="10.10.13.2";
ADD OMCH: FLAG=MASTER, IP="10.10.10.1", MASK="255.255.255.255", PEERIP="100.100.100.1",
PEERMASK="255.255.255.0", BEAR=IPV4, CHECKTYPE= AUTO_UDPSESSION;
ADD OMCH: FLAG=SLAVE, IP="10.10.11.1", MASK="255.255.255.255", PEERIP="100.100.100.1",
PEERMASK="255.255.255.0", BEAR=IPV4, CHECKTYPE=NONE;
```

Deactivation Command Examples

```
//Removing the standby OM channel
RMV OMCH: FLAG=SLAVE;
//Disabling the OM channel switchback function when it is configured
SET GTRANSPARA: OMCHSWITCHBACK=DISABLE;
```

Reconfiguration Command Examples

```
//Modifying the OM channel
MOD OMCH: FLAG=MASTER, BEAR=IPV4, BRT=NO;
```

To change the OM channel binding mode to the OM channel switchback, perform the following steps:

Old model

```
//Enabling the automatic data configuration rollback function
CFM CB: MODE=UNFORCED, NAME="CB1", COMMENT="omch back ", AUTORBKSW=ENABLE,
RBKTIME=60, RBKCONDITION=OMCH_FAULT;
//Turning on the OM channel switchback switch
SET GTRANSPARA: OMCHSWITCHBACK=ENABLE, OMCHSBWAITTIME=60;
//Adding source-based IP routes used by the active and standby OM channels
ADD SRCIPRT: SRCRTIDX=0, SN=7, SBT=BASE_BOARD, SRCIP="10.10.10.1", RTTYPE=NEXTHOP,
NEXTHOP="10.10.12.2";
ADD SRCIPRT: SRCRTIDX=1, SN=7, SBT=BASE_BOARD, SRCIP="10.10.11.1", RTTYPE=NEXTHOP,
NEXTHOP="10.10.13.2";
//Setting Check Type of the active OM channel to AUTO_UDPSESSION, Check Type of the standby OM
channel to NONE, and Binding Route to NO
MOD OMCH: FLAG=MASTER, IP="10.10.10.1", MASK="255.255.255.255", PEERIP="100.100.100.1",
PEERMASK="255.255.255.0", BEAR=IPV4, BRT=NO, CHECKTYPE=AUTO_UDPSESSION;
MOD OMCH: FLAG=SLAVE, IP="10.10.11.1", MASK="255.255.255.255", PEERIP="100.100.100.1",
PEERMASK="255.255.255.0", BEAR=IPV4, BRT=NO, CHECKTYPE=NONE;
//Disabling the automatic data configuration rollback function
MOD CB: NAME="CB1", AUTORBKSW=DISABLE;
```

New model

```
//Enabling the automatic data configuration rollback function
CFM CB: MODE=UNFORCED, NAME="CB1", COMMENT="omch back ", AUTORBKSW=ENABLE,
```

```
RBKTIME=60, RBKCONDITION=OMCH_FAULT;
//Turning on the OM channel switchback switch
SET GTRANSPARA: OMCHSWITCHBACK=ENABLE, OMCHSBWAITTIME=60;
//Adding source-based IP routes used by the active and standby OM channels
ADD SRCIPROUTE4: SRCRTIDX=0, SRCIP="10.10.10.1", RTTYPE=NEXTHOP, NEXTHOP="10.10.12.2";
ADD SRCIPROUTE4: SRCRTIDX=0, SRCIP="10.10.11.1", RTTYPE=NEXTHOP, NEXTHOP="10.10.13.2";
//Setting Check Type of the active OM channel to AUTO_UDPSESSION, Check Type of the standby OM
channel to NONE, and Binding Route to NO
MOD OMCH: FLAG=MASTER, IP="10.10.10.1", MASK="255.255.255.255", PEERIP="100.100.100.1",
PEERMASK="255.255.255.0", BEAR=IPV4, BRT=NO, CHECKTYPE=AUTO_UDPSESSION;
MOD OMCH: FLAG=SLAVE, IP="10.10.11.1", MASK="255.255.255.255", PEERIP="100.100.100.1",
PEERMASK="255.255.255.0", BEAR=IPV4, BRT=NO, CHECKTYPE=NONE;
//Disabling the automatic data configuration rollback function
MOD CB: NAME="CB1", AUTORBKSW=DISABLE;
```

5.3.3.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.3.3.2 Data Configuration (IPv6)

5.3.3.2.1 Data Preparation

When the OM channel switchback function is not required:

Table 5-9 Data to be prepared for OM channel backup

Parameter Name	Parameter ID	Setting Notes
Standby Status	**OMCH**.***FLAG***	Set this parameter to **MASTER** for the active OM channel and to **SLAVE** for the standby OM channel.
Bearer Type	**OMCH**.***BEAR***	• If two IPv6 OM channels work in active/standby mode, set this parameter to **IPV6** for both OM channels. • If a single IPv4 OM channel and a single IPv6 OM channel work in active/standby mode, set this parameter to **IPV6** for the active OM channel and to **IPV4** for the standby OM channel.

Parameter Name	Parameter ID	Setting Notes
Local IPv6 Address	**OMCH**.***IP6***	This parameter is mandatory when **OMCH**.***BEAR*** is set to **IPV6**. Set this parameter to the local IP address of the active OM channel for the active OM channel and to the local IP address of the standby OM channel for the standby OM channel. The base station OM IPv6 address configured on the MAE topology view must be the same as this local IPv6 address.
Peer IPv6 Address	**OMCH**.***PEERIP6***	This parameter is mandatory when **OMCH**.***BEAR*** is set to **IPV6**. This address can be an MAE address or an MAE network segment address.
Peer IPv6 Address Prefix Length	**OMCH**.***PEERIP6PFXLEN***	This parameter is mandatory when **OMCH**.***BEAR*** is set to **IPV6**. Set this parameter based on the network plan.
Local IP	**OMCH**.***IP***	This parameter is mandatory when **OMCH**.***BEAR*** is set to **IPV4**. Set this parameter to the local IP address of the active OM channel for the active OM channel and to the local IP address of the standby OM channel for the standby OM channel.
Local Mask	**OMCH**.***MASK***	This parameter is mandatory when **OMCH**.***BEAR*** is set to **IPV4**. Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Peer IP	**OMCH**.***PEERIP***	This parameter is mandatory when **OMCH**.***BEAR*** is set to **IPV4**. Set this parameter to the MAE address.
Peer Mask	**OMCH**.***PEERMASK***	This parameter is mandatory when **OMCH**.***BEAR*** is set to **IPV4**. Set this parameter based on the network plan.
Binding Route	**OMCH**.***BRT***	● Set this parameter to **YES** when the OM channels use destination-based routes. ● Set this parameter to **NO** when the OM channels use source-based routes.
Route Index	**OMCH**.***RTIDX***	Set this parameter to the index of the route destined for the active MAE.
Binding Secondary Route	**OMCH**.***BINDSECONDARYRT***	When the OM channels use destination-based routes, set this parameter to **YES** if the MAE uses the remote HA system and the IP address of the standby MAE is beyond the network segment range of the route bound to the OM channel. Otherwise, set this parameter to **NO**.
Secondary Route Index	**OMCH**.***SECONDARYRTIDX***	Set this parameter to the index of the route to the standby MAE. This parameter is valid only when **Binding Secondary Route** is set to **YES**.

When the OM channel switchback function is required:

The following table describes the key parameters that must be set in the **GTRANSPARA** MO to set the global transmission parameters of an NE.

Table 5-10 Data to be prepared for OM channel switchback

Parameter Name	Parameter ID	Setting Notes
OMCH Switch Back	**GTRANSPARA**.***OMCHSWITCHBACK***	If OM channel switchback is required, set this parameter to **ENABLE**.
OMCH Switch Back Wait Time	**GTRANSPARA**.***OMCHSBWAITTIME***	Use the default value **60** (in the unit of second).

The following table describes the key parameters that must be set in the **OMCH** MO to set an OM channel.

Table 5-11 Data to be prepared for OM channel backup

Parameter Name	Parameter ID	Setting Notes
Standby Status	**OMCH**.***FLAG***	• Set this parameter to **MASTER** for the active OM channel. • Set this parameter to **SLAVE** for the standby OM channel.
Bearer Type	**OMCH**.***BEAR***	• If two IPv6 OM channels work in active/standby mode, set this parameter to **IPV6** for both OM channels. • If a single IPv4 OM channel and a single IPv6 OM channel work in active/standby mode, set this parameter to **IPV6** for the active OM channel and to **IPV4** for the standby OM channel.

Parameter Name	Parameter ID	Setting Notes
Local IPv6 Address	**OMCH**.***IP6***	• Set this parameter to the local IPv6 address of the active OM channel when the active OM channel is used. • Set this parameter to the local IPv6 address of the standby OM channel when the standby OM channel is used. The base station OM IPv6 address configured on the MAE-Access topology view must be the same as this local IPv6 address.
Peer IPv6 Address	**OMCH**.***PEERIP6***	This address can be an MAE-Access address or an MAE-Access network segment address.
Peer IPv6 Address Prefix Length	**OMCH**.***PEERIP6PFXLEN***	Set this parameter based on the network plan.
Local IP	**OMCH**.***IP***	• Set this parameter to the local IP address of the active OM channel when the active OM channel is used. • Set this parameter to the local IP address of the standby OM channel when the standby OM channel is used.
Local Mask	**OMCH**.***MASK***	Set this parameter based on the network plan.
Peer IP	**OMCH**.***PEERIP***	Set this parameter to the MAE-Access address.
Peer Mask	**OMCH**.***PEERMASK***	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Check Type	**OMCH**.***CHECKTYPE***	The **OMCH**.***CHECKTYPE*** parameter can be set to **AUTO_UDPSESSION** only for the active OM channel. If the **OMCH**.***CHECKTYPE*** parameter is set to **AUTO_UDPSESSION** for the active OM channel, the **OMCH**.***CHECKTYPE*** parameter must be set to **NONE** for the standby OM channel.

5.3.3.2.2 Using MML Commands

Before using MML commands, refer to **5.3.2 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

When the OM channel switchback function is not required:

Activation:

Step 1 Run the **ADD OMCH** command with **Standby Status** set to **MASTER(Master)**.

If two IPv6 OM channels work in active/standby mode, set **Bearer Type** to **IPV6** and **Local IPv6 Address** to the IP address of the active OM channel.

If a single IPv4 OM channel and a single IPv6 OM channel work in active/standby mode, set **Bearer Type** to **IPV6** and **Local IPv6 Address** to the IP address of the active OM channel. In this example, the active OM channel uses IPv6, and the standby OM channel uses IPv4.

When the OM channels use destination-based routes, it is recommended that **Binding Route** be set to **YES** and **Route Index** be set to the route index of the active OM channel.

When the OM channels use source-based IP route, set **Binding Route** to **NO**.

Set **BINDSECONDARYRT** to **YES** and **SECONDARYRTIDX** to the index of the route of the active OMCH to the standby MAE only when the MAE uses the remote HA system and the IP address of the standby MAE is beyond the network segment range of the route bound to the active OMCH.

Step 2 Run the **ADD OMCH** command with **Standby Status** set to **SLAVE(Slave)**.

If two IPv6 OM channels work in active/standby mode, set **Bearer Type** to **IPV6** and **Local IPv6 Address** to the IP address of the standby OM channel.

If a single IPv4 OM channel and a single IPv6 OM channel work in active/standby mode, set **Bearer Type** to **IPV4** and **Local IP** to the IP address of the standby OM channel. In this example, the active OM channel uses IPv6, and the standby OM channel uses IPv4.

When the OM channels use destination-based routes, it is recommended that **Binding Route** be set to **YES** and **Route Index** be set to the route index of the active OM channel.

When the OM channels use source-based IP route, set **Binding Route** to **NO**.

Set **BINDSECONDARYRT** to **YES** and **SECONDARYRTIDX** to the index of the route of the standby OMCH to the standby MAE only when the MAE uses the remote HA system and the IP address of the standby MAE is beyond the network segment range of the route bound to the standby OMCH.

----End

Deactivation:

Run the **RMV OMCH: FLAG=SLAVE** command to remove the standby OM channel.

Reconfiguration:

To adjust the routes of the OM channels, run the **MOD OMCH** command on the base station side.

When the OM channel switchback function is required:

Activation:

Step 1 Run the **SET GTRANSPARA** command with **OMCHSWITCHBACK** set to **ENABLE(Enable)** and **OMCHSBWAITTIME** set to 60 seconds.

Step 2 Run the **ADD OMCH** command with **FLAG** set to **MASTER(Master)**.

When two IPv6 OM channels work in active/standby mode, set **BEAR** to **IPV6**, **IP6** to the IP address of the active OM channel, and **CHECKTYPE** to **AUTO_UDPSESSION**.

If a single IPv4 OM channel and a single IPv6 OM channel work in active/standby mode, set **BEAR** to **IPV6**, **IP6** to the IP address of the active OM channel, and **CHECKTYPE** to **AUTO_UDPSESSION**. In this example, the active OM channel uses IPv6, and the standby OM channel uses IPv4.

When the OM channels use destination-based routes, it is recommended that **BRT** be set to **YES** and **RTIDX** be set to the route index of the active OM channel.

When the OM channels use source-based IP routes, set **BRT** to **NO**.

Set **BINDSECONDARYRT** to **YES** and **SECONDARYRTIDX** to the index of the route of the active OMCH to the standby MAE only when the MAE uses the remote HA system and the IP address of the standby MAE is beyond the network segment range of the route bound to the active OMCH.

Step 3 Run the **ADD OMCH** command with **Standby Status** set to **SLAVE(Slave)**.

If two IPv6 OM channels work in active/standby mode, set **Bearer Type** to **IPV6** and **Local IPv6 Address** to the IP address of the standby OM channel.

If a single IPv4 OM channel and a single IPv6 OM channel work in active/standby mode, set **Bearer Type** to **IPV4** and **Local IP** to the IP address of the standby OM channel. In this example, the active OM channel uses IPv6, and the standby OM channel uses IPv4.

When the OM channels use destination-based routes, it is recommended that **BRT** be set to **YES** and **RTIDX** be set to the route index of the active OM channel.

When the OM channels use source-based IP route, set **Binding Route** to **NO**.

Set **BINDSECONDARYRT** to **YES** and **SECONDARYRTIDX** to the index of the route of the standby OMCH to the standby MAE only when the MAE uses the remote HA system and the IP address of the standby MAE is beyond the network segment range of the route bound to the standby OMCH.

----End

Deactivation:

Run the **RMV OMCH: FLAG=SLAVE;** command to remove the standby OM channel.

When the OM channel switchback function is configured: Run the **SET GTRANSPARA: OMCHSWITCHBACK=DISABLE;** command to disable the OM channel switchback function.

Reconfiguration:

To adjust the routes of the OM channels, run the **MOD OMCH** command on the base station side.

Activation Command Examples

Activating OM channel backup when the MAE does not use the remote HA system

Configuring two IPv6 OM channels working in active/standby mode (The OM channels use destination-based routes.)

When OM channel switchback is not required:

```
ADD IPROUTE6: RTIDX=0, DSTIP="2001:db8:100:ad1:200:100:211:10", PFXLEN=124, RTTYPE=NEXTHOP,
NEXTHOP="2001:db8:100:ad1:200:100:100:1", PREF=60;
ADD IPROUTE6: RTIDX=1, DSTIP="2001:db8:100:ad1:200:100:211:10", PFXLEN=124, RTTYPE=NEXTHOP,
NEXTHOP="2001:db8:100:ad1:200:100:200:1", PREF=80;
ADD OMCH: FLAG=MASTER, BEAR=IPV6, IP6="2001:db8:100:ad1:200:100:100:10",
PEERIP6="2001:db8:100:ad1:200:100:211:10", PEERIP6PFXLEN=124, BRT=YES, RTIDX=0,
BINDSECONDARYRT=NO;
```

Activating OM channel backup when the MAE does not use the remote HA system

Configuring two IPv6 OM channels working in active/standby mode (The OM channels use destination-based routes.)

When OM channel switchback is required:

```
ADD IPROUTE6: RTIDX=0, DSTIP="2001:db8:100:ad1:200:100:211:10", PFXLEN=124, RTTYPE=NEXTHOP,
NEXTHOP="2001:db8:100:ad1:200:100:100:1", PREF=60;
```

```
ADD IPROUTE6: RTIDX=1, DSTIP="2001:db8:100:ad1:200:100:211:10", PFXLEN=124, RTTYPE=NEXTHOP,
NEXTHOP="2001:db8:100:ad1:200:100:200:1", PREF=80;
SET GTRANSPARA: OMCHSWITCHBACK=ENABLE, OMCHSBWAITTIME=60;
ADD OMCH: FLAG=MASTER, BEAR=IPV6,IP6="2001:db8:100:ad1:200:100:100:10",
PEERIP6="2001:db8:100:ad1:200:100:211:10", PEERIP6PFXLEN=124, BRT=YES, RTIDX=0,
BINDSECONDARYRT=NO, CHECKTYPE=AUTO_UDPSESSION;
ADD OMCH: FLAG=SLAVE, BEAR=IPV6, IP6="2001:db8:100:ad1:200:100:200:10",
PEERIP6="2001:db8:100:ad1:200:100:211:10", PEERIP6PFXLEN=124, BRT=YES, RTIDX=1,
BINDSECONDARYRT=NO;
```

Activating OM channel backup when the MAE does not use the remote HA system

Configuring two IPv6 OM channels working in active/standby mode (The OM channels use source-based routes.)

When OM channel switchback is not required:

```
ADD SRCIPROUTE6: SRCRTIDX=0, SRCIP="2001:db8:100:ad1:200:100:100:10", RTTYPE=NEXTHOP,
NEXTHOP="2001:db8:100:ad1:200:100:100:1";
ADD SRCIPROUTE6: SRCRTIDX=1, SRCIP="2001:db8:100:ad1:200:100:200:10", RTTYPE=NEXTHOP,
NEXTHOP="2001:db8:100:ad1:200:100:200:1";
ADD OMCH: FLAG=MASTER, BEAR=IPV6, IP6="2001:db8:100:ad1:200:100:100:10",
PEERIP6="2001:db8:100:ad1:200:100:211:10", PEERIP6PFXLEN=124, BRT=NO;
```

Activating OM channel backup when the MAE does not use the remote HA system

Configuring two IPv6 OM channels working in active/standby mode (The OM channels use source-based routes.)

When OM channel switchback is required:

```
ADD SRCIPROUTE6: SRCRTIDX=0, SRCIP="2001:db8:100:ad1:200:100:100:10", RTTYPE=NEXTHOP,
NEXTHOP="2001:db8:100:ad1:200:100:100:1";
ADD SRCIPROUTE6: SRCRTIDX=1, SRCIP="2001:db8:100:ad1:200:100:200:10", RTTYPE=NEXTHOP,
NEXTHOP="2001:db8:100:ad1:200:100:200:1";
SET GTRANSPARA: OMCHSWITCHBACK=ENABLE, OMCHSBWAITTIME=60;
ADD OMCH: FLAG=MASTER, BEAR=IPV6, IP6="2001:db8:100:ad1:200:100:100:10",
PEERIP6="2001:db8:100:ad1:200:100:211:10", PEERIP6PFXLEN=124, BRT=NO,
CHECKTYPE=AUTO_UDPSESSION;
ADD OMCH: FLAG=SLAVE, BEAR=IPV6,IP6="2001:db8:100:ad1:200:100:200:10",
PEERIP6="2001:db8:100:ad1:200:100:211:10", PEERIP6PFXLEN=124, BRT=NO;
```

Activating OM channel backup when the MAE does not use the remote HA system

Configuring a single IPv4 OM channel and a single IPv6 OM channel to work in active/standby mode (The OM channels use destination-based routes.)

When OM channel switchback is not required:

```
ADD IPROUTE6: RTIDX=0, DSTIP="2001:db8:100:ad1:200:100:211:10", PFXLEN=124, RTTYPE=NEXTHOP,
NEXTHOP="2001:db8:100:ad1:200:100:100:1";
ADD IPROUTE4: RTIDX=1, DSTIP="100.100.100.1", DSTMASK="255.255.255.0", RTTYPE=NEXTHOP,
NEXTHOP="10.10.13.2";
ADD OMCH: FLAG=MASTER, BEAR=IPV6, IP6="2001:db8:100:ad1:200:100:100:10",
PEERIP6="2001:db8:100:ad1:200:100:211:10", PEERIP6PFXLEN=124, BRT=YES, RTIDX=0,
BINDSECONDARYRT=NO;
```

Activating OM channel backup when the MAE does not use the remote HA system

Configuring a single IPv4 OM channel and a single IPv6 OM channel to work in active/standby mode (The OM channels use destination-based routes.)

When OM channel switchback is required:

```
ADD IPROUTE6: RTIDX=0, DSTIP="2001:db8:100:ad1:200:100:211:10", PFXLEN=124, RTTYPE=NEXTHOP,
NEXTHOP="2001:db8:100:ad1:200:100:100:1";
ADD IPROUTE4: RTIDX=1, DSTIP="100.100.100.1", DSTMASK="255.255.255.0", RTTYPE=NEXTHOP,
NEXTHOP="10.10.13.2";
```

```
SET GTRANSPARA: OMCHSWITCHBACK=ENABLE, OMCHSBWAITTIME=60;
ADD OMCH: FLAG=MASTER, BEAR=IPV6,IP6="2001:db8:100:ad1:200:100:100:10",
PEERIP6="2001:db8:100:ad1:200:100:211:10", PEERIP6PFXLEN=124, BRT=YES, RTIDX=0,
BINDSECONDARYRT=NO, CHECKTYPE=AUTO_UDPSESSION;
ADD OMCH: FLAG=SLAVE, IP="10.10.11.1", MASK="255.255.255.255", PEERIP="100.100.100.1",
PEERMASK="255.255.255.0", BEAR=IPV4, BRT=YES, RTIDX=1, BINDSECONDARYRT=NO;
```

Activating OM channel backup when the MAE does not use the remote HA system

Configuring a single IPv4 OM channel and a single IPv6 OM channel to work in active/standby mode (The OM channels use source-based routes.)

When OM channel switchback is not required:

```
ADD SRCIPROUTE6: SRCRTIDX=0, SRCIP="2001:db8:100:ad1:200:100:100:10", RTTYPE=NEXTHOP,
NEXTHOP="2001:db8:100:ad1:200:100:100:1";
ADD SRCIPROUTE4: SRCRTIDX=2, SRCIP="10.10.11.1", RTTYPE=NEXTHOP, NEXTHOP="10.10.13.2";
ADD OMCH: FLAG=MASTER, BEAR=IPV6, IP6="2001:db8:100:ad1:200:100:100:10",
PEERIP6="2001:db8:100:ad1:200:100:211:10", PEERIP6PFXLEN=124, BRT=NO;
```

Activating OM channel backup when the MAE does not use the remote HA system

Configuring a single IPv4 OM channel and a single IPv6 OM channel to work in active/standby mode (The OM channels use source-based routes.)

When OM channel switchback is required:

```
ADD SRCIPROUTE6: SRCRTIDX=0, SRCIP="2001:db8:100:ad1:200:100:100:10", RTTYPE=NEXTHOP,
NEXTHOP="2001:db8:100:ad1:200:100:100:1";
ADD SRCIPROUTE4: SRCRTIDX=2, SRCIP="10.10.11.1", RTTYPE=NEXTHOP, NEXTHOP="10.10.13.2";
SET GTRANSPARA: OMCHSWITCHBACK=ENABLE, OMCHSBWAITTIME=60;
ADD OMCH: FLAG=MASTER, BEAR=IPV6,IP6="2001:db8:100:ad1:200:100:100:10",
PEERIP6="2001:db8:100:ad1:200:100:211:10", PEERIP6PFXLEN=124, BRT=NO,
CHECKTYPE=AUTO_UDPSESSION;
ADD OMCH: FLAG=SLAVE, IP="10.10.11.1", MASK="255.255.255.255", PEERIP="100.100.100.1",
PEERMASK="255.255.255.0", BEAR=IPV4, BRT=YES, RTIDX=1, BINDSECONDARYRT=NO;
```

Activating OM channel backup when the MAE uses the remote HA system

Configuring two IPv6 OM channels working in active/standby mode (The OM channels use destination-based routes.)

```
//Configuring a route from the active OM channel to the active MAE
ADD IPROUTE6: RTIDX=0, DSTIP="2001:db8:100:ad1:200:100:211:10", PFXLEN=124, RTTYPE=NEXTHOP,
NEXTHOP="2001:db8:100:ad1:200:100:100:1", PREF=60;
//Configuring a route from the standby OM channel to the active MAE
ADD IPROUTE6: RTIDX=1, DSTIP="2001:db8:100:ad1:200:100:211:10", PFXLEN=124, RTTYPE=NEXTHOP,
NEXTHOP="2001:db8:100:ad1:200:100:200:1", PREF=80;
//Configuring a route from the active OM channel to the standby MAE
ADD IPROUTE6: RTIDX=2, DSTIP="2001:db8:100:ad1:200:100:222:10", PFXLEN=124, RTTYPE=NEXTHOP,
NEXTHOP="2001:db8:100:ad1:200:100:100:1", PREF=60;
//Configuring a route from the standby OM channel to the standby MAE
ADD IPROUTE6: RTIDX=3, DSTIP="2001:db8:100:ad1:200:100:222:10", PFXLEN=124, RTTYPE=NEXTHOP,
NEXTHOP="2001:db8:100:ad1:200:100:200:1", PREF=80;
//Configuring active and standby OM channels
ADD OMCH: FLAG=MASTER, BEAR=IPV6, IP6="2001:db8:100:ad1:200:100:100:10",
PEERIP6="2001:db8:100:ad1:200:100:211:10", PEERIP6PFXLEN=124, BRT=YES, RTIDX=0,
BINDSECONDARYRT=YES, SECONDARYRTIDX=2;
ADD OMCH: FLAG=SLAVE, BEAR=IPV6, IP6="2001:db8:100:ad1:200:100:200:10",
PEERIP6="2001:db8:100:ad1:200:100:222:10", PEERIP6PFXLEN=124, BRT=YES, RTIDX=1,
BINDSECONDARYRT=YES, SECONDARYRTIDX=3;
```

Activating OM channel backup when the MAE uses the remote HA system

Configuring two IPv6 OM channels working in active/standby mode (The OM channels use source-based routes.)

```
ADD SRCIPROUTE6: SRCRTIDX=0, SRCIP="2001:db8:100:ad1:200:100:100:10", RTTYPE=NEXTHOP,
NEXTHOP="2001:db8:100:ad1:200:100:100:1";
```

```
ADD SRCIPROUTE6: SRCRTIDX=1, SRCIP="2001:db8:100:ad1:200:100:200:10", RTTYPE=NEXTHOP,
NEXTHOP="2001:db8:100:ad1:200:100:200:1";
ADD OMCH: FLAG=MASTER, BEAR=IPV6, IP6="2001:db8:100:ad1:200:100:100:10",
PEERIP6="2001:db8:100:ad1:200:100:211:10", PEERIP6PFXLEN=124, BRT=NO;
ADD OMCH: FLAG=SLAVE, BEAR=IPV6, IP6="2001:db8:100:ad1:200:100:200:10",
PEERIP6="2001:db8:100:ad1:200:100:222:10", PEERIP6PFXLEN=124, BRT=NO;
```

Activating OM channel backup when the MAE uses the remote HA system

Configuring a single IPv4 OM channel and a single IPv6 OM channel to work in active/standby mode (The OM channels use destination-based routes.)

```
//Configuring a route from the active OM channel to the active MAE
ADD IPROUTE6: RTIDX=0, DSTIP="2001:db8:100:ad1:200:100:211:10", PFXLEN=124, RTTYPE=NEXTHOP,
NEXTHOP="2001:db8:100:ad1:200:100:100:1";
//Configuring a route from the standby OM channel to the active MAE
ADD IPROUTE4: RTIDX=1, DSTIP="100.100.100.1", DSTMASK="255.255.255.0", RTTYPE=NEXTHOP,
NEXTHOP="10.10.13.2";
//Configuring a route from the active OM channel to the standby MAE
ADD IPROUTE6: RTIDX=2, DSTIP="2001:db8:100:ad1:200:100:222:10", PFXLEN=124, RTTYPE=NEXTHOP,
NEXTHOP="2001:db8:100:ad1:200:100:100:1";
//Configuring a route from the standby OM channel to the standby MAE
ADD IPROUTE4: RTIDX=3, DSTIP="100.100.200.1", DSTMASK="255.255.255.0", RTTYPE=NEXTHOP,
NEXTHOP="10.10.13.2";
//Configuring active and standby OM channels
ADD OMCH: FLAG=MASTER, BEAR=IPV6, IP6="2001:db8:100:ad1:200:100:100:10",
PEERIP6="2001:db8:100:ad1:200:100:211:10", PEERIP6PFXLEN=124, BRT=YES, RTIDX=0,
BINDSECONDARYRT=YES, SECONDARYRTIDX=2;
ADD OMCH: FLAG=SLAVE, IP="10.10.11.1", MASK="255.255.255.255", PEERIP="100.100.100.1",
PEERMASK="255.255.255.0", BEAR=IPV4, BRT=YES, RTIDX=1, BINDSECONDARYRT=YES,
SECONDARYRTIDX=3;
```

Activating OM channel backup when the MAE uses the remote HA system

Configuring a single IPv4 OM channel and a single IPv6 OM channel to work in active/standby mode (The OM channels use source-based routes.)

```
ADD SRCIPROUTE6: SRCRTIDX=0, SRCIP="2001:db8:100:ad1:200:100:100:10", RTTYPE=NEXTHOP,
NEXTHOP="2001:db8:100:ad1:200:100:100:1";
ADD SRCIPROUTE4: SRCRTIDX=2, SRCIP="10.10.11.1", RTTYPE=NEXTHOP, NEXTHOP="10.10.13.2";
ADD OMCH: FLAG=MASTER, BEAR=IPV6, IP6="2001:db8:100:ad1:200:100:100:10",
PEERIP6="2001:db8:100:ad1:200:100:211:10", PEERIP6PFXLEN=124, BRT=NO;
ADD OMCH: FLAG=SLAVE, IP="10.10.11.1", MASK="255.255.255.255", PEERIP="100.100.100.1",
PEERMASK="255.255.255.0", BEAR=IPV4, BRT=YES, RTIDX=1;
```

Deactivation Command Examples

```
//Removing the standby OM channel
RMV OMCH: FLAG=SLAVE;
```

Optimization Command Examples

```
//Modifying the OM channel
MOD OMCH: FLAG=MASTER, BEAR=IPV6, BRT=NO;
```

5.3.3.2.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.3.3.3 Activation Verification

The verification method is based on the following assumptions:

- The local IP addresses of the active and standby OM channels are on different network segments.
- When IPv4 is used, the next-hop IP addresses of the IP routes are different. These IP addresses are defined by **IPRT** (in the old model)/**IPROUTE4** (in the new model) and bound to the active and standby OM channels.
- When IPv6 is used, the next-hop IP addresses of the IPv6 routes or source-based IPv6 routes are different. These IP addresses are defined by **IPROUTE6** or **SRCIPROUTE6** and bound to the active and standby OM channels.

If the actual networking is different from the assumed networking, perform the verification based on the actual networking. That is, an OM channel switchover can be triggered when the OM channel in use is faulty and the transmission link of the other OM channel is normal.

To check whether OM channel backup has taken effect, perform the following steps:

Step 1 Check which OM channel is in use. Run the **DSP OMCH** command to query the status of the active and standby OM channels. An OM channel is considered in use if its values of **OM Channel Status** and **Used State** are **Normal** and **In Use**, respectively.

Step 2 Trigger an OM channel switchover.

Generate a transmission link fault on the OM channel in use and verify that the normal OM channel can take over the faulty OM channel.

- If the OM channel in use is the active channel, generate a route fault on the active OM channel. Wait approximately 10 minutes and run the **DSP OMCH** command to check the status of the standby OM channel. The switchover is successful if the value of **OM Channel Status** is **Normal** and the value of **Used State** is **In Use**.
- If the OM channel in use is the standby channel, generate a route fault on the standby OM channel. Wait approximately 10 minutes and run the **DSP OMCH** command to check the status of the active OM channel. The switchover is successful if the value of **OM Channel Status** is **Normal** and the value of **Used State** is **In Use**.

Step 3 Restore the faulty link in **Step 2**.

Step 4 When the OM channel switchback function is configured, trigger an OMCH switchback.

This example aims to verify that the OM channel can be switched back to the active channel after the fault in the active channel is rectified in **Step 3**.

If **GTRANSPARA**.***OMCHSBWAITTIME*** is set to **60** (in seconds), wait about 5 minutes and run the **DSP OMCH** command to check the status of the active OM channel. The switchback is successful if the value of **OM Channel Status** is **Normal** and the value of **Used State** is **In Use**. Otherwise, check the value of

Automatic UDP Session Detection Status. If the value is **DETECTING** or **SWITCHING**, the detection or switchback is ongoing for the active OM channel.

----End

5.3.3.4 Network Monitoring

After OM channel backup is deployed, you can use counters to monitor the performance. For details, see **4.3.3.4 Network Monitoring**.

5.3.3.5 Possible Issues

During a switchover or switchback of the active and standby OM channels, the MAE reports ALM-301 NE Is Disconnected. After the switchover or switchback is successfully completed, the alarm is cleared. The base station reports EVT-25893 Remote Maintenance Link Switchover to help determine whether the OM channel switchover is successful.

6 Transmission Maintenance and Detection

6.1 BFD

6.1.1 Network Analysis

6.1.1.1 Benefits

BFD provides a fast connectivity check mechanism for redundant links (such as active and standby IP routes) on networks.

6.1.1.2 Impacts

The impacts between this function and other functions are described in **6.1.2.2.3 Function Impacts**.

To check network connectivity, BFD needs to construct and send IP packets, which occupies some transmission bandwidths.

6.1.2 Requirements

6.1.2.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-010060	Transmission Network Detection and Reliability Improvement	NR0S0TNDER00	per gNodeB

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

6.1.2.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.1.2.2.1 Prerequisite Functions

None

6.1.2.2.2 Mutually Exclusive Functions

None

6.1.2.2.3 Function Impacts

None

6.1.2.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards support this function. To learn which main control boards are NR-capable, see technical specifications of the BBU in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

This function does not depend on RF modules.

6.1.2.4 Networking

None

6.1.2.5 Others

None

6.1.3 Operation and Maintenance

6.1.3.1 When to Use

It is recommended that this feature be enabled only when BFD is required for quick fault location. This is because BFD consumes CPU resources and transmission bandwidth in IP transmission scenarios.

You are advised to enable the SBFD authentication function for regions and operators with high security requirements. The peer gateway must support BFD authentication.

6.1.3.2 Data Configuration (in the New Model)

6.1.3.2.1 Data Preparation

Parameter Name	Parameter ID	Setting Notes
IP Version	**BFD**.***IPVERSION***	This parameter specifies the IP protocol version of a BFD session. Set this parameter based on the network plan.
Hop Type	**BFD**.***HT***	● The single-hop BFD session is used to perform point-to-point detection (generally in Layer 2 networking). ● The multi-hop BFD session is used to perform end-to-end connectivity check (generally in Layer 3 networking).

Parameter Name	Parameter ID	Setting Notes
Source IP	**BFD**.***SRCIP***	This parameter specifies the source IP address of a BFD session. Set this parameter based on the network plan. ● A BFD session cannot be configured as a single-hop session if its source IP address is a logical IP address. ● Ensure that the **IPADDR4** MO corresponding to this IP address has been configured. ● This parameter value must be a valid IP address and cannot be 0.0.0.0. It must be a device port IP address (for example, the IP address of an Ethernet port), or a logical IP address (for example, the IP address of a loopback interface) of a specified board. But it cannot be set to the IP address of the OM channel. When this parameter is set to a logical IP address, single-hop BFD sessions are not supported. This parameter must be set to a value different from that of the **BFD**.***DSTIP*** parameter.
Destination IP	**BFD**.***DSTIP***	This parameter specifies the destination IP address of the BFD session. Set this parameter based on the network plan. ● If single-hop BFD is used and the source IP address is configured on an Ethernet port, the source and destination IP addresses must be on the same network segment. ● The destination IP address of each BFD session must be unique.
Source IPv6 Address	**BFD**.***SRCIPV6***	This parameter specifies the source IPv6 address of a BFD session. The source IPv6 address of a BFD session must be the device IPv6 address or logical IPv6 address of a specified board. A BFD session cannot be configured as a single-hop session if its source IPv6 address is a logical IPv6 address.
Destination IPv6 Address	**BFD**.***DSTIPV6***	This parameter specifies the destination IPv6 address of a BFD session. The destination IPv6 address of a BFD session must be a valid IPv6 address and cannot be all 0s or an existing IPv6 address in the system.

Parameter Name	Parameter ID	Setting Notes
My Discriminator Generation Mode	**BFD**.***MYDSCRMGENMODE***	This parameter specifies the generation mode of my discriminator. Set this parameter based on the network plan. The following generation modes are supported: ● **STATIC_CONFIG**. In this mode, my discriminator is generated based on the configured value of **BFD**.***MYDISCREAMINATOR***. It is recommended that your discriminator not be set on the router. If your discriminator is set on the router, your discriminator needs to be the same as my discriminator of the base station. ● **DYNAMIC_GEN**. In this mode, my discriminator is generated randomly when negotiation of a session starts. ● **DYNAMIC_GEN_ENHANCE**. In this mode, my discriminator is generated randomly when negotiation of a session starts and, if my discriminator is the same as your discriminator, the base station discards this session and generates another my discriminator randomly to start negotiation. If your discriminator is set on the router, this parameter can only be set to **STATIC_CONFIG**.
My Discriminator	**BFD**.***MYDISCREAMINATOR***	This parameter specifies my discriminator of the BFD session. It is recommended that your discriminator not be set on the router. If your discriminator is set on the router, your discriminator needs to be the same as my discriminator of the base station. This parameter can be set only when the **BFD**.***MYDSCRMGENMODE*** parameter is set to **STATIC_CONFIG**.
Min TX Interval	**BFD**.***MINTI***	Set this parameter to **100**.
Min RX Interval	**BFD**.***MINRI***	Set this parameter to **100**.
Detection Multiplier	**BFD**.***DM***	Set this parameter to **3**.

Parameter Name	Parameter ID	Setting Notes
Session Catalog	**BFD**.***CATLOG***	• If this parameter is set to **MAINTENANCE**, the BFD session is used only for performing a connectivity check. • If this parameter is set to **RELIABILITY**, the BFD session is used to trigger route interlock. Route interlock enables the standby route to take over once the active route becomes faulty, and prevents service interruptions caused by route failures. • The reliability type is not supported in IPv6 multi-hop scenarios.
DSCP	**BFD**.***DSCP***	Set this parameter based on the network plan.
BFD Authentication Switch	**BFD**.***BFDAUTHSW***	• Enable BFD authentication if high security is required for BFD control packets. • Otherwise, disable BFD authentication.
BFD Authentication Algorithm	**BFD**.***BFDAUTHTYPE***	Set this parameter based on the network plan. The following algorithms are supported in IPv4 scenarios: • MD5 • MeMD5 • SHA1 • MeSHA1 The following algorithms are supported in IPv6 scenarios: • SHA1 • MeSHA1 The algorithms must be consistent between the base station and the peer device.
BFD Authentication Key Chain ID	**BFD**.***KEYCHAINID***	Set this parameter to the same value as **BFDKEYCHAIN**.***KEYCHAINID***. Only one key chain is currently supported.
BFD Authentication Key Chain ID	**BFDKEYCHAIN**.***KEYCHAINID***	Only one key chain is currently supported.
Key Chain Description	**BFDKEYCHAIN**.***KEYCHAINDESC***	Only one key chain is currently supported.
KEY Identity	**BFDKEY**.***KEYID***	The parameter settings must be consistent between the base station and the peer device.
KEY String	**BFDKEY**.***KEY***	The parameter settings must be consistent between the base station and the peer device.

6.1.3.2.2 Using MML Commands

Before using MML commands, refer to **6.1.2 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

- Activation

  Run the **ADD BFD** command with **My Discriminator Generation Mode** set to **STATIC_CONFIG**, **Hop Type** set to **SINGLE_HOP**, and **Session Catalog** set to **RELIABILITY** to add a BFD session.

  NOTE

  Set **Protocol Version** to the BFD session protocol version supported by the peer device, which can be either **DRAFT4** or **STANDARD**.

- Deactivation

  Run the **RMV BFD** command to remove the BFD session.

- Reconfiguration

  Run the **MOD BFD** command to reconfigure the **MINTI**, **MINRI**, and **DM** parameters, if required.

  – If BFD packets are sent at a too high frequency, excessive bandwidth is required, which will affect the performance of the peer device. In this case, increase the value of the **MINTI** parameter (negotiation with the operator is required).
  – If BFD packets are sent at a too low frequency, detection results are less accurate. In this case, decrease the value of the **MINTI** parameter (negotiation with the operator is required).
  – If fault detection sensitivity is too high at the peer end, this will result in a ping-pong effect. In this case, increase the value of the **MINRI** and **DM** parameters (negotiation with the operator is required).
  – If fault detection sensitivity is too low at the peer end, fault detection is difficult and will require too much time. In this case, decrease the value of the **MINRI** and **DM** parameters (negotiation with the operator is required).

Activation Command Examples

```
//Adding a BFD authentication key chain on the base station (BFD Authentication Key Chain ID = 0; Key Chain Description = bfdkeychain)
ADD BFDKEYCHAIN: KEYCHAINID=0, KEYCHAINDESC="bfdkeychain";
//Adding a key to the BFD key chain on the base station (BFD Authentication Key Chain ID = 0; KEY Identity = 0; KEY String = 123)
ADD BFDKEY: KEYCHAINID=0, KEYID=0, KEY="*****";
//Adding a BFD session on the base station (Session ID = 100; Source IP = 192.168.5.5; Destination IP = 192.168.5.6; My Discriminator Generation Mode = STATIC_CONFIG; My Discriminator = 1; Hop Type = SINGLE_HOP; Session Catalog = RELIABILITY; DSCP = 0; Protocol Version = STANDARD; BFD Authentication Switch = ON; BFD Authentication Algorithm = MeSHA1; BFD Authentication Key Chain ID = 0)
ADD BFD: BFDSN=100, IPVERSION=IPV4, SRCIP="192.168.5.5", DSTIP="192.168.5.6",
```



```
MYDSCRMGENMODE=STATIC_CONFIG, MYDISCREAMINATOR=1, HT=SINGLE_HOP, CATLOG=RELIABILITY,
DSCP=0, VER=STANDARD, BFDAUTHSW=ON, BFDAUTHTYPE=MeSHA1, KEYCHAINID=0;
//Configuring IPv6 BFD
ADD BFD: BFDSN=100, IPVERSION=IPV6, SRCIPV6="2001:db8:100:ad1:200:100:200:12",
DSTIPV6="2001:db8:100:ad1:200:100:200:1", MYDSCRMGENMODE=STATIC_CONFIG,
MYDISCREAMINATOR=1, HT=SINGLE_HOP, CATLOG=RELIABILITY, DSCP=48, VER=STANDARD,
BFDAUTHSW=ON, BFDAUTHTYPE=MeSHA1, KEYCHAINID=0;
SET GTRANSPARA: SBTIME=300;
```

Deactivation Command Examples

```
//Removing the BFD session from the base station (Session ID = 100)
RMV BFD: BFDSN=100;
```

Reconfiguration Command Examples

```
//Modifying the BFD
MOD BFD: BFDSN=100, MINTI=200, MINRI=200, DM=5;
```

6.1.3.2.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.1.3.3 Data Configuration (in the Old Model)

6.1.3.3.1 Data Preparation

Parameter Name	Parameter ID	Setting Notes
Hop Type	**BFDSESSION**.***HT***	• The SBFD session is used to perform point-to-point detection (generally in Layer 2 networking). • The MBFD session is used to perform end-to-end connectivity check (generally in Layer 3 networking).

Parameter Name	Parameter ID	Setting Notes
Source IP	**BFDSESSION**.***SRCIP***	This parameter specifies the source IP address of the BFD session. Set this parameter based on the network plan. ● A BFD session cannot be configured as a single-hop session if its source IP address is a logical IP address. ● Ensure that the **DEVIP** MO corresponding to this IP address has been configured. ● This parameter value must be a valid IP address and cannot be 0.0.0.0. It must be a device IP address (for example, the IP address of an Ethernet port), or a logical IP address (for example, the IP address of a loopback interface) of a specified board. But it cannot be set to the IP address of the OM channel. When this parameter is set to a logical IP address, SBFD sessions are not supported. This parameter must be set to a value different from that of the **BFDSESSION**.***DSTIP*** parameter.
Destination IP	**BFDSESSION**.***DSTIP***	This parameter specifies the destination IP address of the BFD session. Set this parameter based on the network plan. ● If SBFD is used and the source IP address is configured on an Ethernet port, the source and destination IP addresses must be on the same network segment. ● The destination IP address of each BFD session must be unique.
Min TX Interval	**BFDSESSION**.***MINTI***	Set this parameter to **100**.
Min RX Interval	**BFDSESSION**.***MINRI***	Set this parameter to **100**.
Detection Multiplier	**BFDSESSION**.***DM***	Set this parameter to **3**.
Session Catalog	**BFDSESSION**.***CATLOG***	● If this parameter is set to **MAINTENANCE**, the BFD session is only used for performing a connectivity check. ● If this parameter is set to **RELIABILITY**, the BFD session is used to trigger route interlock. Route interlock enables the standby route to take over once the active route becomes faulty, and prevents service interruptions caused by route failures.
DSCP	**BFDSESSION**.***DSCP***	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
BFD Authentication Switch	**BFDSESSION**.***BFDAUTHSW***	● Enable BFD authentication if high security is required for BFD control packets. ● Otherwise, disable BFD authentication.
BFD Authentication Algorithm	**BFDSESSION**.***BFDAUTHTYPE***	Set this parameter based on the network plan. The following algorithms are supported: ● MD5(MD5) ● Meticulous MD5(MeMD5) ● SHA1(SHA1) ● Meticulous SHA1(MeSHA1) The base station and the peer device must negotiate to use the same algorithm.
BFD Authentication Key Chain ID	**BFDSESSION**.***KEYCHAINID***	Set this parameter to the same value as ***KEYCHAINID***. Only one key chain is currently supported.
BFD Authentication Key Chain ID	**BFDKEYCHAIN**.***KEYCHAINID***	Only one key chain is currently supported.
Key Chain Description	**BFDKEYCHAIN**.***KEYCHAINDESC***	Only one key chain is currently supported.
KEY Identity	**BFDKEY**.***KEYID***	The base station and the peer device must negotiate to use the same algorithm.
KEY String	**BFDKEY**.***KEY***	The base station and the peer device must negotiate to use the same algorithm.

6.1.3.3.2 Using MML Commands

Before using MML commands, refer to **6.1.2 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

- Activation

  Run the **ADD BFDSESSION** command with **My Discriminator Generation Mode** set to **STATIC_CONFIG**, **Hop Type** set to **SINGLE_HOP**, and **Session Catalog** set to **RELIABILITY** to add a BFD session.

  NOTE

  Set **Protocol Version** to the BFD session protocol version supported by the peer device, which can be either **DRAFT4** or **STANDARD**.

- Deactivation

  Run the **RMV BFDSESSION** command to remove a BFD session.

- Reconfiguration

  Run the **MOD BFDSESSION** command to reconfigure the **MINTI**, **MINRI**, and **DM** parameters, if required.

  – If BFD packets are sent at a too high frequency, excessive bandwidth is required, which will affect the performance of the peer device. In this case, increase the value of the **MINTI** parameter (negotiation with the operator is required).

  – If BFD packets are sent at a too low frequency, detection results are less accurate. In this case, decrease the value of the **MINTI** parameter (negotiation with the operator is required).

  – If fault detection sensitivity is too high at the peer end, this will result in a ping-pong effect. In this case, increase the value of the **MINRI** and **DM** parameters (negotiation with the operator is required).

  – If fault detection sensitivity is too low at the peer end, fault detection is difficult and will require too much time. In this case, decrease the value of the **MINRI** and **DM** parameters (negotiation with the operator is required).

Activation Command Examples

```
//Adding a BFD authentication key chain on the base station (BFD Authentication Key Chain ID = 0; Key Chain Description = bfdkeychain)
ADD BFDKEYCHAIN: KEYCHAINID=0, KEYCHAINDESC="bfdkeychain";
//Adding a key to the BFD key chain on the base station (BFD Authentication Key Chain ID = 0; KEY Identity = 0; KEY String = 123)
ADD BFDKEY: KEYCHAINID=0, KEYID=0, KEY="*****";
//Adding a BFD session on the base station (Cabinet No. = 0; Subrack No. = 0; Slot No. = 6; Session ID = 0; Source IP = 192.168.5.5; Destination IP = 192.168.5.6; My Discriminator Generation Mode = STATIC_CONFIG; Hop Type = SINGLE_HOP; Session Catalog = RELIABILITY; DSCP = 0; Protocol Version = STANDARD; BFD Authentication Switch = ON; BFD Authentication Algorithm = MeSHA1; BFD Authentication Key Chain ID = 0)
ADD BFDSESSION: CN=0, SRN=0, SN=6, BFDSN=0, SRCIP="192.168.5.5", DSTIP="192.168.5.6", MYDSCRMGENMODE=STATIC_CONFIG, HT=SINGLE_HOP, CATLOG=RELIABILITY, DSCP=0, VER=STANDARD, BFDAUTHSW=ON,  BFDAUTHTYPE=MeSHA1, KEYCHAINID=0;
```

Deactivation Command Examples

```
//Removing a BFD session from the base station (Cabinet No. = 0; Subrack No. = 0; Slot No. = 0; Session ID = 0)
RMV BFDSESSION: CN=0, SRN=0, SN=0, BFDSN=0;
```

6.1.3.3.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.1.3.4 Activation Verification

The procedures for verifying SBFD and MBFD are the same.

Step 1 Run the **DSP BFDSESSION** (in the old model)/**DSP BFD** (in the new model) command on the base station. If **Session State** is **Up**, the function has taken effect.

----End

6.1.3.5 Network Monitoring

If ALM-25899 BFD Session Fault is reported after BFD is deployed, check whether the deployment is correct according to the alarm help.

6.2 GTP-U Echo

6.2.1 Network Analysis

6.2.1.1 Benefits

None

6.2.1.2 Impacts

The impacts between this function and other functions are described in **6.2.2.2.3 Function Impacts**.

After static GTP-U echo is enabled, if the user-plane path detection result shows that the path is faulty, no new bearer will be set up on the path. As a result, the access success rate of bearers may decrease and the access failure rate of bearers due to transport-layer faults may increase.

6.2.2 Requirements

6.2.2.1 Licenses

None

6.2.2.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.2.2.2.1 Prerequisite Functions

None

6.2.2.2.2 Mutually Exclusive Functions

None

6.2.2.2.3 Function Impacts

None

6.2.2.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards support this function. To learn which main control boards are NR-capable, see technical specifications of the BBU in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

This function does not depend on RF modules.

6.2.2.4 Networking

None

6.2.2.5 Others

The link must already be created to use link-level GTP-U echo.

6.2.3 Operation and Maintenance

6.2.3.1 When to Use

You are advised to enable GTP-U echo if the base station needs to check the GTP-U path disconnection duration as well as the number of times that GTP-U paths are disconnected.

6.2.3.2 Data Configuration

6.2.3.2.1 Data Preparation

The following table describes the key parameters that must be set in a **GTPU** MO to configure global GTP-U parameters.

Parameter Name	Parameter ID	Setting Notes
ECHO Frame Timeout	**GTPU**.***TIMEOUTTH***	Set this parameter based on the network plan.
ECHO Frame Timeout Count	**GTPU**.***TIMEOUTCNT***	Set this parameter based on the network plan.
DSCP	**GTPU**.***DSCP***	Set this parameter based on the network plan.
Static Check Switch	**GTPU**.***STATICCHK***	It is recommended that this parameter be set to **ENABLE**.

The following table describes the key parameters that must be set in an **EPGROUP** MO to configure the endpoint group static GTP-U echo switch in endpoint configuration mode.

Parameter Name	Parameter ID	Setting Notes
Static Check Mode	**GEPMODELPARA**.***STATICCHKMODE***	Set this parameter to **EPSTATICCHK**.
Static Check Switch	**EPGROUP**.***STATICCHK***	Retain the default value.

The following table describes the key parameters that must be set in a **USERPLANEPEER** MO to configure the link-level static GTP-U echo switch in endpoint configuration mode.

Parameter Name	Parameter ID	Setting Notes
Static Check Mode	**GEPMODELPARA**.***STATICCHKMODE***	Set this parameter to **UPPEERSTATICCHK**.
Static Check Switch	**USERPLANEPEER**.***STATICCHK***	Retain the default value.

6.2.3.2.2 Using MML Commands

Before using MML commands, refer to **6.2.2 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Step 1 If the global and link-level switch settings are to be used, run the **SET GEPMODELPARA** command with **STATICCHKMODE** set to **UPPEERSTATICCHK**.

1. Run the **MOD GTPU** command with **Static Check Switch** set to **ENABLE** to turn on the global static GTP-U echo switch.
2. Run the **ADD USERPLANEPEER** or **MOD USERPLANEPEER** command with **Static Check Switch** set to **FOLLOW_GLOBAL**, **ENABLE**, or **DISABLE** to set the link-level static GTP-U echo switch in endpoint configuration mode.

Step 2 If the endpoint group switch setting is to be used, run the **SET GEPMODELPARA** command with **Static Check Mode** set to **EPSTATICCHK**. Run the **ADD EPGROUP** or **MOD EPGROUP** command with **Static Check Switch** set to **ENABLE** or **DISABLE**.

----End

Activation Command Examples

Enabling GTP-U echo using the global and link-level switch settings

```
SET GEPMODELPARA: STATICCHKMODE=UPPEERSTATICCHK;
MOD GTPU: STATICCHK=ENABLE;
MOD USERPLANEPEER: UPPEERID=0, STATICCHK=FOLLOW_GLOBAL;
```

Enabling GTP-U echo using the endpoint group switch setting

```
SET GEPMODELPARA: STATICCHKMODE=EPSTATICCHK;
MOD EPGROUP: EPGROUPID=0, STATICCHK=ENABLE;
```

6.2.3.2.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.2.3.3 Activation Verification

The activation verification procedure is as follows:

Step 1 Run the **DSP GTPUECHO** command. Perform operations based on the value of **Static GTP-U ECHO test switch** in the command output.

Status	Operation
Static GTP-U echo switch has been turned on	Ensure that the cell has been activated.

Status	Operation
Static GTP-U echo switch has not been turned on	Ensure that bearers have been set up for UEs in the cell. This can be achieved by using UEs to access the cell and injecting packets in the uplink or downlink.

The base station can send GTP-U echo control packets in either of the preceding scenarios.

Step 2 Create a GTP-U echo monitoring task as follows:

1. On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**.
2. On the navigation tree of the **Signaling Trace Management** window, choose **Base Station Device and Transport** > **Transport Trace** > **GTPU Trace**.
3. In the displayed **GTPU Trace** window, set parameters of the GTP-U echo monitoring task to complete the task creation.

Step 3 Start the monitoring task, and check the results. If echo request and response messages can be viewed in real time, GTP-U echo has been successfully activated.

----End

6.2.3.4 Network Monitoring

None

6.3 LLDP

6.3.1 Network Analysis

6.3.1.1 Benefits

The Link Layer Discovery Protocol (LLDP) provides a standard link-layer discovery method. It sends local device information to directly connected neighbors, helping the network management system query and determine the communication quality of links.

- This helps network administrators learn about the network topology, including the number of devices on the entire network, number of interfaces on each device, and connections between devices.
- This helps network administrators diagnose network faults. Based on the LLDP interaction information, network administrators can quickly obtain the link communication status and locate the link fault point.

6.3.1.2 Impacts

The impacts between this function and other functions are described in **6.3.2.2.3 Function Impacts**.

LLDP encapsulates the management address, device ID, and interface information of a device into a Link Layer Discovery Protocol data unit (LLDPDU) and advertises the LLDPDU to neighbors, which consumes a small amount of transmission bandwidth.

6.3.2 Requirements

6.3.2.1 Licenses

None

6.3.2.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.3.2.2.1 Prerequisite Functions

None

6.3.2.2.2 Mutually Exclusive Functions

None

6.3.2.2.3 Function Impacts

None

6.3.2.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards support this function. To learn which main control boards are NR-capable, see technical specifications of the BBU in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

This function does not depend on RF modules.

6.3.2.4 Networking

LLDP has been enabled on the interconnected equipment.

6.3.2.5 Others

None

6.3.3 Operation and Maintenance

6.3.3.1 Data Configuration

6.3.3.1.1 Data Preparation (in the New Model)

Adding LLDP global configuration:

The following table lists the key parameters that must be set in the **LLDPGLOBAL** MO to specify LLDP global parameters.

Parameter Name	Parameter ID	Setting Notes
LLDPDU Tx Interval	**LLDPGLOBAL**.***TXINTVAL***	30s
LLDP Hold Multiplier	**LLDPGLOBAL**.***HOLDMULTI***	4
Reinit Delay	**LLDPGLOBAL**.***REINITDELAY***	2s
Transmission Delay	**LLDPGLOBAL**.***DELAY***	2s
LLDP Notify Switch	**LLDPGLOBAL**.***NOTIFYSW***	When the NE topology drawing function is enabled on the FMA, set this parameter to **ENABLE(ENABLE)**.
Notify Interval	**LLDPGLOBAL**.***NOTIFYINTERVAL***	5s
LLDP Port Configuration Mode	**LLDPGLOBAL**.***PORTCFGMODE***	You are advised to set this parameter to **MANUAL(Manual)**.

Parameter Name	Parameter ID	Setting Notes
Base Station Microwave Collaboration Switch	**LLDPGLOBAL**.***BTSMWCOSW***	It is recommended that the **BKP_PSAV_EVALUATE_DATA_SW** option be selected if the microwave device needs to subscribe to backup power saving data of the base station. After the data is successfully subscribed to, the base station transmits backup power saving data to the microwave device based on LLDP. The subscription can be successful only if the microwave device is a Huawei model directly connected to the base station and the corresponding switch on the microwave device is turned on.

Adding an LLDP local entity:

The following table describes the key parameters that must be set in the **LLDP** MO to specify the LLDP local entity parameters.

Parameter Name	Parameter ID	Setting Notes
LLDP ID	**LLDP**.***LLDPID***	Set this parameter based on the network plan.
Ethernet Port ID	**LLDP**.***ETHPORTID***	Set this parameter based on the network plan.
Binding VLAN	**LLDP**.***BNDVLAN***	Set this parameter based on the network plan.
VLAN ID	**LLDP**.***VLANID***	Set this parameter based on the network plan.
VLAN Priority	**LLDP**.***VLANPRI***	Set this parameter based on the network plan.

6.3.3.1.2 Data Preparation (in the Old Model)

Adding LLDP global configuration:

The following table lists the key parameters that must be set in the **LLDPGLOBAL** MO to specify LLDP global parameters.

Parameter Name	Parameter ID	Setting Notes
LLDPDU Tx Interval	**LLDPGLOBAL**.***TXINTVAL***	30s
LLDP Hold Multiplier	**LLDPGLOBAL**.***HOLDMULTI***	4
Reinit Delay	**LLDPGLOBAL**.***REINITDELAY***	2s
Transmission Delay	**LLDPGLOBAL**.***DELAY***	2s
LLDP Notify Switch	**LLDPGLOBAL**.***NOTIFYSW***	When the NE topology drawing function is enabled on the FMA, set this parameter to **ENABLE(ENABLE)**.
Notify Interval	**LLDPGLOBAL**.***NOTIFYINTERVAL***	5s
LLDP Port Configuration Mode	**LLDPGLOBAL**.***PORTCFGMODE***	You are advised to set this parameter to **MANUAL(Manual)**.
Base Station Microwave Collaboration Switch	**LLDPGLOBAL**.***BTSMWCOSW***	It is recommended that the **BKP_PSAV_EVALUATE_DATA_SW** option be selected if the microwave device needs to subscribe to backup power saving data of the base station. After the data is successfully subscribed to, the base station transmits backup power saving data to the microwave device based on LLDP. The subscription can be successful only if the microwave device is a Huawei model directly connected to the base station and the corresponding switch on the microwave device is turned on.

Adding an LLDP local entity:

The following table describes the key parameters that must be set in the **LLDPLOCAL** MO to specify the LLDP parameters.

Parameter Name	Parameter ID	Setting Notes
Binding VLAN	**LLDPLOCAL.*BNDVLAN***	Set this parameter based on the network plan.
VLAN ID	**LLDPLOCAL.*VLANID***	Set this parameter based on the network plan.
VLAN Priority	**LLDPLOCAL.*VLANPRI***	Set this parameter based on the network plan.

6.3.3.1.3 Using MML Commands (in the New Model)

Before using MML commands, refer to **6.3.2 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

LLDP can be enabled on the base station side using either of the following methods:

- Manually configuring LLDP local port information

  Run the **ADD LLDP** command to add an LLDP local port. To configure multiple ports, run this command for multiple times.
- Automatically configuring LLDP local port information in batches

  Run the **SET LLDPGLOBALINFO** command with **LLDP Port Configuration Mode** set to **AUTO(Auto)**.

NOTE

- When **LLDPGLOBAL.*PORTCFGMODE*** is set to **AUTO(Auto)**, the system does not support manual configuration of LLDP port information for a port by running the **ADD LLDPPORT** (old model)/**ADD LLDP** (new model) command.
- If the LLDP port information of a port has been manually configured using the **ADD LLDPPORT** (old model)/**ADD LLDP** (new model) command, the system does not allow setting **LLDPGLOBAL.*PORTCFGMODE*** to **AUTO(Auto)**.
- When the LLDP port information is automatically configured, the value of **Binding VLAN** for the LLDP port is always **NO(No)**. To bind the LLDP port to a VLAN, you need to manually configure the LLDP port.

LLDP can be disabled on the base station side using either of the following methods: In manual configuration mode, LLDP can be disabled only in manual mode. In automatic configuration mode, LLDP can be disabled only in automatic mode.

- Manually removing LLDP local port information

  Run the **RMV LLDP** command to remove an LLDP local port.
- Automatically removing LLDP local port information in batches

  Run the **SET LLDPGLOBALINFO** command with **LLDP Port Configuration Mode** set to **MANUAL(Manual)**.

Activation Command Examples

Manually configuring LLDP local port information

```
//Setting LLDP Port Configuration Mode to MANUAL(Manual)
SET LLDPGLOBALINFO: NOTIFYSW=ENABLE, PORTCFGMODE=MANUAL;
//Adding LLDP port configuration information to port 0 in slot 6, subrack 0, and cabinet 0 without binding the port to a VLAN
ADD ETHPORT: CN=0, SRN=0, SN=6, SBT=BASE_BOARD, PN=1, PORTID=1, PA=FIBER, SPEED=AUTO, DUPLEX=AUTO;
ADD LLDP: LLDPID=0, ETHPORTID=0, BNDVLAN=NO;
```

Automatically configuring LLDP port information for all ports

```
SET LLDPGLOBALINFO: NOTIFYSW=ENABLE, PORTCFGMODE=AUTO;
```

(Optional) Selecting the **BKP_PSAV_EVALUATE_DATA_SW** option of the **BTSMWCOSW** parameter when the microwave device needs to subscribe to backup power saving data of the base station

```
SET LLDPGLOBALINFO: BTSMWCOSW=BKP_PSAV_EVALUATE_DATA_SW-1;
```

Deactivation Command Examples

Deselecting the **BKP_PSAV_EVALUATE_DATA_SW** option of the **BTSMWCOSW** parameter

```
SET LLDPGLOBALINFO: BTSMWCOSW=BKP_PSAV_EVALUATE_DATA_SW-0;
```

Manually removing LLDP local port information from the port

```
//Removing the LLDP port information from port 0 in slot 6, subrack 0, and cabinet 0
RMV LLDP: LLDPID=0;
```

Automatically removing LLDP port information from all ports

```
SET LLDPGLOBALINFO: PORTCFGMODE=MANUAL;
```

6.3.3.1.4 Using MML Commands (in the Old Model)

Before using MML commands, refer to **6.3.2 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

LLDP can be enabled on the base station side using either of the following methods:

- Manually configuring LLDP local port information

  Run the **ADD LLDPPORT** command to add an LLDP local port. To configure multiple ports, run this command for multiple times.
- Automatically configuring LLDP local port information in batches

  Run the **SET LLDPGLOBALINFO** command with **LLDP Port Configuration Mode** set to **AUTO(Auto)**.

NOTE

- When **LLDPGLOBAL**.***PORTCFGMODE*** is set to **AUTO(Auto)**, the system does not support manual configuration of LLDP port information for a port by running the **ADD LLDPPORT** command.
- If the LLDP port information of a port has been manually configured using the **ADD LLDPPORT** command, the system does not allow setting **LLDPGLOBAL**.***PORTCFGMODE*** to **AUTO(Auto)**.
- When the LLDP port information is automatically configured, the value of **Binding VLAN** for the LLDP port is always **NO(No)**. To bind the LLDP port to a VLAN, you need to manually configure the LLDP port.

LLDP can be disabled on the base station side using either of the following methods: In manual configuration mode, LLDP can be disabled only in manual mode. In automatic configuration mode, LLDP can be disabled only in automatic mode.

- Manually removing LLDP local port information

  Run the **RMV LLDPPORT** command to remove an LLDP local port.
- Automatically removing LLDP local port information in batches

  Run the **SET LLDPGLOBALINFO** command with **LLDP Port Configuration Mode** set to **MANUAL(Manual)**.

Activation Command Examples

Manually configuring LLDP local port information

```
//Setting LLDP Port Configuration Mode to MANUAL(Manual)
SET LLDPGLOBALINFO: NOTIFYSW=ENABLE, PORTCFGMODE=MANUAL;
//Adding LLDP port configuration information to port 0 in slot 6, subrack 0, and cabinet 0 without binding the port to a VLAN
ADD LLDPPORT: CN=0, SRN=0, SN=6, SBT=ETH_COVERBOARD, PN=0, BNDVLAN=NO;
```

Automatically configuring LLDP port information for all ports

```
SET LLDPGLOBALINFO: NOTIFYSW=ENABLE, PORTCFGMODE=AUTO;
```

(Optional) Selecting the **BKP_PSAV_EVALUATE_DATA_SW** option of the **BTSMWCOSW** parameter when the microwave device needs to subscribe to backup power saving data of the base station

```
SET LLDPGLOBALINFO: BTSMWCOSW=BKP_PSAV_EVALUATE_DATA_SW-1;
```

Deactivation Command Examples

Deselecting the **BKP_PSAV_EVALUATE_DATA_SW** option of the **BTSMWCOSW** parameter


```
SET LLDPGLOBALINFO: BTSMWCOSW=BKP_PSAV_EVALUATE_DATA_SW-0;
```

Manually removing LLDP local port information from the port

```
//Removing the LLDP port information from port 0 in slot 6, subrack 0, and cabinet 0
RMV LLDPPORT: CN=0, SRN=0, SN=6, SBT=ETH_COVERBOARD, PN=0;
```

Automatically removing LLDP port information from all ports

```
SET LLDPGLOBALINFO: PORTCFGMODE=MANUAL;
```

6.3.3.1.5 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.3.3.2 Activation Verification

Step 1 Run the **DSP LLDPPORT** (old model)/**DSP LLDP** (new model) command to query information about the LLDP local port.

Step 2 Run the **DSP LLDPNEIGHBOR** command to query LLDP remote information.

Expected result: If the LLDP local port information and LLDP remote information can be queried successfully, the LLDP function is successfully enabled.

----End

6.3.3.3 Network Monitoring

None

7 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

Step 1 Open the EXCEL file of parameter reference.

Step 2 On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

Step 1 Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

8 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

9 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

10 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- *IPsec*
- *IPv4 Transmission*
- *IPv6 Transmission*
- *Transmission Resource Management*
- *X2 and S1 Self-Management in NSA Networking*
- *NG and Xn Self-Management*
- *X2 Self-Management in SA Networking*
- *Common Transmission*
- *License Control Item Lists*
- *Technical Specifications* in *3900 & 5900 Series Base Station Product Documentation*